High-Performance Computer Architecture Review
Chapters 1–8 with Integrated Cache Simulation Concepts

ECE 4100/6100

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Scope and Topic Map

How to Use This Text

Chapters 1 through 8 provide topic depth with definitions, equations, and worked examples. Chapter 5 provides integrated synthesis and cache-simulation context. Chapter 6 develops dynamic instruction-level parallelism mechanisms such as superscalar scheduling and out-of-order execution. Chapter 7 develops compiler-scheduled instruction-level parallelism through very long instruction word (VLIW) execution and software pipelining. Chapter 8 develops multiprocessor organization, coherence, and parallel programming models.

Cross-Chapter Reading Paths

1. Readers who want the performance-and-memory foundation should begin with Chapters 1 through 3 and then read the first part of Chapter 4.
2. Readers focused on control hazards, prediction, and data hazards should concentrate on the middle and later sections of Chapter 4.
3. Readers focused on instruction-level parallelism should read Chapter 6 for dynamic ILP and Chapter 7 for compiler-scheduled ILP.
4. Readers focused on simulator-style interpretation and consolidated worked checks should use Chapter 5 alongside the detailed discussions in Chapters 3, 4, and 6.
5. Readers focused on thread-level parallelism, coherence, and shared-memory multiprocessors should continue from Chapters 1 and 2 into Chapter 8.

Topic Coverage Map

Chapter 1: Introduction Coverage includes speculation with and without prediction, grain sizes of parallelism (bit, instruction, thread, and inter-program), Flynn taxonomy, single instruction, multiple data (SIMD) versus multiple instruction, multiple data (MIMD), pipelining as temporal parallelism, and the distinction between packed-word SIMD and pipelined vector operations.

Chapter 2: Performance Evaluation Coverage includes throughput versus runtime, the central processing unit (CPU)-time equation in terms of instruction count (IC), cycles per instruction (CPI), and cycle time, reduced instruction set computer (RISC) versus complex instruction set computer (CISC) tradeoffs, Amdahl's and Gustafson's laws, and correct use of arithmetic versus harmonic means.

Chapter 3: Cache Memories Coverage includes static random-access memory (SRAM) versus dynamic random-access memory (DRAM) and bit-line delay, locality, hierarchy levels, virtual-memory terminology and its analogy to caching, average access time (AAT), the three C's of misses, cache geometry and construction, replacement and write policies, insertion-policy choices such as MIP and LIP, prefetching strategies including Markov and hybrid schemes, techniques to improve miss rate, miss penalty, and hit time, simulator-style hierarchy statistics, and compiler/data-layout effects.

Chapter 4: Pipelining Coverage includes instruction fetch / instruction decode / execute / memory / write back (IF/ID/EX/MEM/WB) behavior, speedup equations, cycles per instruction (CPI) stall modeling, the principle that program behavior plus hazards produces stalls, structural and control hazards, delay and squash-slot ideas, branch target buffer (BTB) behavior with and without tags, return-address stack (RAS) behavior, direction-prediction heuristics, and dynamic predictors including one-bit and Smith-counter schemes, GShare, GSelect, Yeh/Patt, perceptrons, hybrid tournament selection, and delayed predictor-update timing.

Chapter 5: Integrated Synthesis and Cache Simulation Concepts Coverage includes integrated synthesis, worked pipeline calculations, hierarchy semantics, prefetching, and interpretation of simulator-style cache, branch, and average access time (AAT) statistics.

Chapter 6: Dynamic ILP, Superscalar Scheduling, and Out-of-Order Execution Coverage includes cycles per instruction (CPI) versus instructions per cycle (IPC) interpretation, instruction-level parallelism (ILP), historical superscalar context, fire/complete ordering models, scoreboarding and busy-bit scheduling, function-unit latency considerations, Tomasulo-style reservation stations, physical register files, result buses, register renaming intuition, precise versus imprecise exceptions, reorder buffer (ROB) and future-file organization, checkpoint/repair recovery, simulator-style fetched/retired and queue-pressure counter interpretation, and the security consequences of speculative out-of-order execution.

Chapter 7: Compiler-Scheduled ILP and Software Pipelining Coverage includes compiler-scheduled very long instruction word (VLIW) execution, dependence-graph and list-scheduling ideas, zero-no-op multi-op encoding with tail and pause fields, software pipelining and iterative modulo scheduling, and memory disambiguation.

Chapter 8: Multiprocessors, Interconnects, and Cache Coherence Coverage includes shared-memory versus message-passing organization, standalone multiprocessors versus accelerator systems, buses versus point-to-point links, the cache-coherence problem and its formal properties, shared-cache versus directory-based versus snooping solutions, MSI coherence states and transactions, transient coherence states, and shared-memory versus message-passing programming examples.

Chapter 1

Introduction

1.1 What Computer Architecture Is

Computer architecture is the discipline of designing computing systems that execute real workloads efficiently under practical constraints. In this text, architecture means deciding how a processor and memory system should be organized so that useful work is completed quickly, reliably, and within power and cost limits. It is important to separate this from neighboring fields. Architecture is not transistor fabrication, circuit layout, or process chemistry, even though those influence what architects can build.

Every architectural decision is a tradeoff among competing goals. A design that minimizes latency for one task may reduce throughput for a server-style workload. A design that aggressively improves performance may consume more energy or require expensive cooling. A design that is elegant on paper may be too costly in area or too difficult to verify at scale. For this reason, architecture is best treated as constrained optimization: choose structures and policies that best match the target application while respecting implementation limits.

1.2 Core Methods: Parallelism, Speculation, and Prediction

Most performance techniques in modern processors can be traced to three closely linked ideas: *parallelism*, *speculation*, and *prediction*. *Parallelism* increases performance by doing multiple useful operations at the same time. *Speculation* increases performance by doing work early, before the machine is fully certain that this work will be needed. *Prediction* estimates uncertain future behavior (for example branch direction or branch target) so speculative work is directed toward the most likely path. Prediction is not a third independent speedup method; it is a tool that makes speculation more selective and efficient. One useful informal nickname for this is "2.5 tricks" rather than three completely separate tricks. Parallelism is one direct source of speedup. Speculation is the second. Prediction is the "half" because it usually matters only by making speculation more effective.

Classifying common optimizations This distinction is useful because many architectural features combine these ideas differently. Packed-word SIMD instructions and multicore processors are primarily forms of parallelism. A cache, a victim cache, or a write buffer relies largely on speculation without prediction: the machine keeps or moves data before it can prove the data will be needed immediately, trusting locality to make that early work useful. Branch prediction is speculation with prediction. Out-of-order superscalar execution combines both parallelism and speculation because

the machine overlaps independent work while also allowing some work to proceed before the final architectural outcome is known.

1.2.1 Speculation Without Prediction

Speculation does not require a learned predictor. A processor can speculate by optimistic overlap: continue execution under the assumption that an operation will complete normally, and recover only if that assumption fails. This is useful because waiting for every operation to retire before moving on would underutilize the pipeline.

Example: write buffer A write buffer is a standard example. When the core executes a store, it can enqueue the write and continue with subsequent instructions before that store reaches the lower memory hierarchy. If the store later encounters a fault condition, architectural recovery and exception handling restore correctness. The processor did not predict branch direction or data values in this case; it simply advanced optimistically to avoid stalling.

1.2.2 Speculation With Prediction

When prediction is added, speculation becomes more selective and usually more efficient. The processor estimates likely future behavior, executes along the estimated path, and commits the work only if the estimate is correct.

Example: branch prediction Branch prediction is the canonical case. If the predictor guesses that a branch is taken, instruction fetch and decode proceed down the predicted target path immediately. A correct guess keeps the front end busy and reduces control-hazard stalls. An incorrect guess forces the machine to squash the speculative instructions and restart on the correct path, which introduces a misprediction penalty. The basic tradeoff is straightforward: high prediction accuracy yields high performance; low accuracy can erase the benefit of speculation.

1.3 Parallelism by Grain Size

Parallelism is best understood by *grain size*, meaning the amount of work grouped into one concurrent unit. At the smallest scale, bit-level parallelism appears in wide datapaths and arithmetic logic that operate on many bit positions at once. A 64-bit adder, for example, processes far more state per cycle than an 8-bit design because the hardware width itself exposes more fine-grained parallel work.

At *instruction-level parallelism (ILP)*, the machine overlaps independent instructions within a thread. Superscalar issue, out-of-order scheduling, and pipelining all exploit ILP. Consider two arithmetic instructions: `R1=R2+R3` and `R4=R5+R6`. Because neither depends on the other, a dual-issue machine can dispatch them together. In contrast, `R1=R2+R3` followed by `R4=R1+R6` introduces a data dependence that limits overlap. This illustrates the central ILP limitation: dependencies, control hazards, and hardware resource conflicts bound achievable parallel speedup.

At *thread-level parallelism (TLP)*, multiple threads execute concurrently on multithreaded or multicore hardware. TLP can provide substantial throughput gains, but only if the workload contains enough independent work and synchronization overhead remains controlled. Inter-program parallelism is the coarsest common level, where independent processes run simultaneously under an operating system scheduler. This is the dominant form in desktop multitasking and cloud services, where many programs share the same physical platform.

1.4 Flynn's Taxonomy

Flynn's taxonomy classifies parallel machines by counting instruction streams and data streams. In *single instruction, single data (SISD)*, one instruction stream acts on one data stream, matching the traditional scalar uniprocessor model. In *single instruction, multiple data (SIMD)*, one instruction stream is applied across many data elements in lockstep, which is highly effective for dense data-parallel computations such as vector math, image processing, and many graphics processing unit (GPU) kernels. The data values may differ across lanes, but the issued operation is the same across the whole SIMD group.

In *multiple instruction, multiple data (MIMD)*, multiple processing elements execute different instruction streams on different data streams. This is the dominant model for modern multicore central processing units (CPUs), distributed servers, and cluster systems because it supports general-purpose, irregular workloads. *multiple instruction, single data (MISD)* exists mostly in specialized contexts, such as certain fault-tolerant designs, and is uncommon in mainstream systems.

1.4.1 SIMD vs. MIMD

SIMD and MIMD solve different problems. SIMD is efficient when the same operation must be applied repeatedly to large collections of data. Its lockstep execution naturally provides synchronization across lanes, which reduces control overhead. If different cores or lanes truly follow different code paths at the same time, the machine is no longer acting like classical SIMD for that region; that behavior belongs to MIMD or to a SIMT-style execution model with divergence handling. MIMD is more flexible because each processor can follow its own control path, but this flexibility introduces synchronization and communication costs that must be managed explicitly by hardware and software.

Modern GPUs often use a hybrid model sometimes called *single instruction, multiple threads (SIMT)*. SIMT provides a SIMD-like execution efficiency for the hardware while allowing a MIMD-like programming model where each thread appears to have its own scalar program counter. The hardware manages the divergence and convergence of these threads, grouping those on the same path into SIMD-style execution units.

Representative workload mappings Simple examples help fix the taxonomy. A small embedded controller such as a thermostat is usually best viewed as SISD. A dense image transform that applies the same arithmetic to many pixels is a classic SIMD workload. A cloud-service cluster is naturally MIMD because many processors or machines execute different control paths for different requests. Some tasks, such as brute-force hash search, can be built in several ways: a lockstep array of identical workers looks SIMD-like, while a farm of largely independent workers looks MIMD. The important step is to identify whether the machine is reusing one instruction stream across many data elements or allowing many instruction streams to proceed independently.

1.4.2 Where pipelining fits

Pipelining is orthogonal to Flynn's taxonomy. It is a temporal form of parallelism in which different stages operate on different instructions concurrently. Because pipeline structure is an internal execution organization, pipelining can appear inside SISD cores, SIMD engines, and MIMD processors alike.

1.5 Vectors: Packed-Word SIMD vs. Pipelined Vector Units

Packed-word SIMD and classic vector pipelines both exploit data parallelism, but they do so with different execution structures. Packed-word SIMD divides a fixed-width register into lanes (for example, four 32-bit lanes in 128 bits) and applies one instruction across all lanes in lockstep. An operation such as `ADD_PACKED` performs multiple elementwise additions per instruction, and lane boundaries isolate carries so one lane's overflow does not spill into neighboring lanes. This style underlies CPU SIMD extensions such as streaming SIMD extensions (SSE), advanced vector extensions (AVX), and Arm NEON.

Vector pipelines, in contrast, stream long vectors through deep arithmetic pipelines. A floating-point multiply can be viewed conceptually as operand preparation, multiply, and result normalization/writeback stages. If a pipeline has k stages and processes n elements, a serial implementation needs nk cycles, while a pipelined implementation needs approximately $k + n - 1$ cycles after fill effects are accounted for (Eq. (1.1)):

$$\text{serial cycles} = nk, \qquad \text{pipeline cycles} \approx k + n - 1. \tag{1.1}$$

The result is substantial throughput gain for long vectors even though single-element latency is not eliminated.

Worked vector-throughput example Suppose a vector arithmetic pipeline has 6 stages and the program processes 100 elements. A non-pipelined implementation would spend

$$nk = 100 \times 6 = 600 \text{ cycles} \tag{1.2}$$

to process the vector. The pipelined form instead needs approximately

$$k + n - 1 = 6 + 100 - 1 = 105 \text{ cycles.} \tag{1.3}$$

The difference is the standard vector-pipeline gain: the pipeline still pays 6-stage latency for the first element, but after fill it can accept and complete additional elements at a much higher steady-state rate.

Chaining Vector chaining improves this further. If a second vector operation consumes results from a first operation, the hardware can forward early produced elements directly into the consumer pipeline instead of waiting for the full vector to complete. Chaining reduces producer-consumer delay and improves sustained throughput in vectorized loops.

1.6 Common Chapter 1 Pitfalls

Students often conflate speculation with prediction. The correct view is that prediction is one way to make speculation more effective, not the definition of speculation itself. Another recurring error is to assume that additional cores guarantee proportional speedup; real speedup is limited by dependencies, communication, and serial regions of code. It is also common to treat SIMD and MIMD as interchangeable, even though they target different workload structures and have different programming implications.

A final source of confusion is the difference between packed-word SIMD and long-vector pipelining. Both process multiple data elements, but packed SIMD exploits lane-level parallel arithmetic within instruction-width registers, while vector pipelines exploit staged streaming over long element sequences. Related to this, pipelining primarily improves throughput rather than the latency of a single operation.

1.7 Chapter 1 Summary

Chapter 1 introduces computer architecture as disciplined performance engineering under practical constraints. The chapter frames nearly all speedup mechanisms as forms of parallelism or speculation, then organizes parallelism by grain size from bits through instructions, threads, and entire programs. Flynn's taxonomy provides a clean language for distinguishing SISD, SIMD, MISD, and MIMD machines, especially the practical differences between SIMD efficiency and MIMD flexibility. Finally, the chapter distinguishes packed-word SIMD from pipelined vector execution and shows why chaining is valuable in vector data paths. Together, these ideas form the conceptual base used throughout the rest of the book.

1.8 Practice Questions

The following questions reinforce the main ideas of Chapter 1 and are intended for self-checking after the summary.

1. For each of the following techniques, state whether it primarily exploits *parallelism*, *speculation without prediction*, *speculation with prediction*, or more than one of these ideas: packed-word SIMD, a cache, a branch predictor, and out-of-order execution with branch prediction.

2. Classify each of the following workloads using Flynn's taxonomy, and give a one-sentence justification for each answer: a dense image convolution engine, a cloud-service cluster that handles independent requests, a small thermostat controller, and a brute-force hash search implemented as many independent workers.

3. Explain why pipelining is orthogonal to Flynn's taxonomy. Give one example of a pipelined SISD design and one example of pipelining inside a wider SIMD or MIMD system.

4. A vector arithmetic pipeline has 6 stages and processes a vector of 100 elements. Compute the approximate cycle count for a serial implementation and for a pipelined implementation. Then explain how chaining could reduce producer-consumer delay for two dependent vector operations.

5. Distinguish packed-word SIMD from classic pipelined vector execution. In particular, explain what is fixed by register width in packed SIMD and what is streamed through the pipeline in classic vector execution.

Chapter 2

Performance Evaluation

2.1 Why Performance Evaluation Matters

Architecture decisions are only meaningful if they are evaluated with the right metric. A design that improves one metric can degrade another, and a design that looks fast on one workload can be slow on the workload that actually matters. The goal of this chapter is therefore methodological: define performance correctly, choose representative workloads, and use mathematically correct summaries when comparing alternatives.

2.2 Throughput vs. Runtime

Two primary performance metrics appear repeatedly in computer architecture. *Runtime* (also called response time or turnaround time) measures how long one job takes from start to completion. *Throughput* measures how many jobs complete per unit time. These metrics are related but not interchangeable.

For a single-user interactive application, runtime is usually the dominant concern: the user wants one operation to finish quickly. For servers and shared systems, throughput is often dominant: the owner cares about how many requests the system can process per second. The same machine can be better under one metric and worse under the other, so performance claims must always specify which metric is being optimized.

This distinction also explains why pipelining can be valuable even when individual operation latency does not fall dramatically. A deeper pipeline can increase the number of completed operations per unit time, improving throughput, while leaving single-operation latency unchanged or slightly worse.

2.3 CPU Runtime Equation

The fundamental central processing unit (CPU) time equation used throughout performance analysis is (Eq. (2.1)):

$$\text{CPU time} = \text{Instruction Count (IC)} \times \text{Cycles per Instruction (CPI)} \times \text{Cycle Time (CT)}. \tag{2.1}$$

Using clock frequency $f = 1/\text{CT}$, the same equation is (Eq. (2.2)):

$$\text{CPU time} = \frac{\text{IC} \times \text{CPI}}{f}. \tag{2.2}$$

This decomposition is useful because each factor is controlled by different mechanisms. *Instruction count* is influenced by algorithm choice, compiler quality, and instruction set architecture (ISA) expressiveness. *Cycles per instruction (CPI)* is influenced by microarchitecture, hazard behavior, memory hierarchy behavior, and available parallelism. *Cycle time* is influenced by circuit and pipeline timing constraints.

For instruction mixes, total cycles are computed directly (Eq. (2.3)):

$$\text{Total cycles} = \sum_i (\text{IC}_i \times \text{CPI}_i), \qquad \text{CPI}_{avg} = \frac{\text{Total cycles}}{\text{Total instructions}}. \tag{2.3}$$

Worked example Consider a workload that executes 76 loads at 6 cycles each, 30 stores at 4 cycles each, 300 adds at 1 cycle each, and 58 multiplies at 5 cycles each. Then total cycles are

$$76 \cdot 6 + 30 \cdot 4 + 300 \cdot 1 + 58 \cdot 5 = 1166. \tag{2.4}$$

At 2.2 MHz, runtime is

$$\frac{1166}{2.2 \times 10^6} \approx 5.30 \times 10^{-4}\text{s} = 0.53\text{ms}. \tag{2.5}$$

If multiply latency is reduced from 5 cycles to 3 cycles, new total cycles become

$$76 \cdot 6 + 30 \cdot 4 + 300 \cdot 1 + 58 \cdot 3 = 1050, \tag{2.6}$$

which yields a speedup of

$$\frac{1166}{1050} \approx 1.11. \tag{2.7}$$

The example illustrates a recurring theme: accelerating only one instruction class can help, but total improvement depends on how often that class occurs.

Worked machine-comparison example Now consider three machines that all execute the same source-level program. Machine J runs at 5.1 GHz with $CPI = 3.0$. Machine K runs at 5.5 GHz with $CPI = 3.7$ and the same dynamic instruction count as J. Machine G runs at 6.1 GHz with $CPI = 3.2$ but requires 1.5× as many dynamic instructions. Using Eq. (2.2), the J-versus-K comparison is

$$\frac{T_K}{T_J} = \frac{3.7/5.5}{3.0/5.1} \approx 1.146, \tag{2.8}$$

so K is slower even though its clock is faster. For J versus G, runtimes are proportional to

$$T_J \propto \frac{1.0 \times 3.0}{5.1}, \qquad T_G \propto \frac{1.5 \times 3.2}{6.1}. \tag{2.9}$$

Numerically,

$$T_J \propto 0.588, \qquad T_G \propto 0.787. \tag{2.10}$$

So the faster clock does not rescue Machine G: the larger dynamic instruction count dominates and Machine J is faster overall. This is the central Chapter 2 discipline: compare complete CPU time, not any single factor in isolation.

2.4 RISC vs. CISC Through the Performance Equation

Reduced instruction set computer (RISC) and *complex instruction set computer (CISC)* are best understood as design tradeoffs, not as a simple "better/worse" ranking. Historically, CISC designs emphasized richer, denser instructions to reduce dynamic instruction count and code size. RISC designs emphasized simpler instructions and more regular pipelines, often improving CPI and enabling higher clock rates. In CPU-time terms, this is a direct trade (Eq. (2.11)):

$$\text{IC} \downarrow \text{may come with} \text{CPI} \uparrow \text{or} \text{CT} \uparrow, \quad \text{or vice versa.} \tag{2.11}$$

In modern systems the boundary is less rigid than textbook definitions suggest, and x86 implementations, for example, decode complex instructions into simpler internal micro-operations and then execute them with highly RISC-like back ends. Arm designs, traditionally associated with RISC, also include increasingly rich instruction features to improve efficiency for modern workloads. The practical conclusion is that ISA style should be evaluated by end-to-end performance, power, and software ecosystem impact rather than by labels alone.

2.5 Amdahl's Law

Amdahl's Law quantifies the speedup limit when only part of a workload is accelerated. If fraction f of execution time benefits from a speedup factor s, total speedup is

$$S_{Amdahl} = \frac{1}{(1-f) + \frac{f}{s}}. \tag{2.12}$$

The limit as $s \to \infty$ is

$$S_{\text{upper}} = \frac{1}{1-f}. \tag{2.13}$$

This bound is the key insight: the unaccelerated fraction eventually dominates.

Worked example Consider a case in which 80% of runtime is parallelizable and the accelerated part can use 100,000 processors ideally ($s = 100{,}000$). Then

$$S = \frac{1}{0.2 + \frac{0.8}{100000}} \approx 5.0. \tag{2.14}$$

Even enormous parallel resources cannot overcome a significant serial fraction. This is why architectural proposals must report both the accelerated fraction and the achieved acceleration factor.

Maximum-speedup interpretation When a problem states that a new mechanism speeds up "25% of a benchmark program," the 25% must be interpreted as a fraction of the *original execution time* of the whole benchmark unless stated otherwise. If that accelerated fraction were made infinitely fast, the maximum possible speedup would be

$$S_{\max} = \frac{1}{1-0.25} = \frac{4}{3} \approx 1.33. \tag{2.15}$$

This is the limiting-case form of Amdahl's bound: once the improved portion becomes arbitrarily fast, the remaining 75% of the original runtime dominates completely.

2.6 Gustafson's Law and Why It Does Not Contradict Amdahl

Gustafson's Law addresses a different question: instead of fixing problem size, it assumes the workload scales with available processors. Let α be the sequential fraction measured on an N-processor run. Scaled speedup is

$$S_{Gustafson} = \alpha + (1 - \alpha)N. \tag{2.16}$$

This often grows nearly linearly with N when α is small.

Amdahl and Gustafson do not contradict each other because their assumptions differ. Amdahl is *strong-scaling* analysis (fixed total work). Gustafson is *weak-scaling* analysis (larger total work as resources grow). Both are valid, but they answer different engineering questions.

Example With $\alpha = 0.15$ and $N = 16$,

$$S_{Gustafson} = 0.15 + 0.85 \cdot 16 = 13.75. \tag{2.17}$$

This can exceed Amdahl's fixed-size bound because the scaled workload includes more total useful work.

2.7 Combining Performance Numbers Correctly

Single-number summaries are useful only when the averaging method matches the quantity being averaged. For times (seconds per task), *arithmetic mean* is appropriate (Eq. (2.18)):

$$\bar{T}_{arith} = \frac{1}{n}\sum_{i=1}^{n} T_i. \tag{2.18}$$

For rates (tasks per second), use *harmonic mean* (Eq. (2.19)):

$$\bar{R}_{harm} = \frac{n}{\sum_{i=1}^{n} \frac{1}{R_i}}. \tag{2.19}$$

Using arithmetic mean on rates can select the wrong system. Suppose machine A runs two workloads at 4 and 4 transactions/s, while machine B runs them at 2 and 7 transactions/s. Arithmetic means are 4.0 and 4.5, which appears to favor B. But for one transaction of each workload, total throughput is determined by total work divided by total time (Eq. (2.20)):

$$R_A = \frac{2}{\frac{1}{4} + \frac{1}{4}} = 4, \qquad R_B = \frac{2}{\frac{1}{2} + \frac{1}{7}} = \frac{28}{9} \approx 3.11. \tag{2.20}$$

Under equal-work comparison, A is better. This is exactly why rates should be combined harmonically, not arithmetically.

Weighted workload mixes Real benchmark suites are often not uniform. If workload i contributes fraction w_i of the jobs or requests, with $\sum_i w_i = 1$, then the weighted average runtime is

$$\bar{T}_{weighted} = \sum_i w_i T_i. \tag{2.21}$$

For rates under the corresponding fixed-work interpretation, the comparable aggregate rate is

$$R_{eq} = \frac{1}{\sum_i \frac{w_i}{R_i}}. \tag{2.22}$$

These weighted forms are the correct way to combine benchmark results when workloads do not occur equally often.

Worked weighted-runtime example Suppose three workloads occur with weights 0.50, 0.30, and 0.20, and one machine runs them in 4 s, 9 s, and 6 s, respectively. Then the weighted average runtime is

$$\bar{T}_{weighted} = 0.50(4) + 0.30(9) + 0.20(6) = 5.9 \text{ s}. \tag{2.23}$$

The same arithmetic form should *not* be applied blindly to rates. If the reported numbers were throughputs, the fixed-work interpretation would instead require the weighted harmonic form in Eq. (2.22).

2.8 Workload Dependence and Benchmarking

Performance numbers are workload-dependent. Miss ratio, CPI, and even effective clock behavior under thermal constraints depend on the executed program mix. A claim such as "processor X is faster" is incomplete unless it states the benchmark set, input sizes, and aggregation method.

Representative benchmarking therefore requires matching the measured workloads to expected deployment behavior. A web service, a game engine, and a scientific simulation stress very different execution paths and memory behaviors. Chapter 2 methodology demands that architects evaluate on relevant traces and report assumptions explicitly; otherwise conclusions will not transfer to real systems.

2.9 Cost-Aware Performance Comparison

Raw speedup is not always the only decision metric. If machine m costs $Cost_m$ and delivers speedup S_m relative to a chosen baseline, two useful value measures are

$$\text{speedup per dollar} = \frac{S_m}{Cost_m}, \qquad \text{cost per unit speedup} = \frac{Cost_m}{S_m}. \tag{2.24}$$

The first form is convenient when maximizing performance for a fixed budget. The second form is convenient when minimizing cost for a required performance gain. Both require the same discipline as every other Chapter 2 metric: use one baseline, one workload definition, and one aggregation method throughout the comparison.

Worked value-comparison example Suppose Server A costs \$4,000 and delivers speedup 1.25, while Server B costs \$5,400 and delivers speedup 1.55. Their speedup per dollar values are

$$\frac{1.25}{4000} = 3.125 \times 10^{-4}, \qquad \frac{1.55}{5400} \approx 2.87 \times 10^{-4}, \tag{2.25}$$

so Server A gives slightly more performance per dollar. Their cost per unit speedup values are

$$\frac{4000}{1.25} = 3200, \qquad \frac{5400}{1.55} \approx 3483.87, \tag{2.26}$$

which leads to the same conclusion under the reciprocal metric. This example is intentionally simple: the point is to show that cost-aware comparisons are still just ratio comparisons once a common baseline is fixed.

2.10 Chapter 2 Summary

Chapter 2 provides the framework for disciplined performance reasoning. Runtime and throughput are distinct objectives and must not be mixed casually. CPU time decomposes into IC, CPI, and cycle time, which allows precise reasoning about optimization impact. RISC and CISC are best evaluated as tradeoff points in Eq. (2.1) rather than as absolute categories. Amdahl's Law gives hard limits for fixed-size speedup, while Gustafson's Law explains scaled-workload growth without contradiction. Finally, performance summaries require correct statistics: arithmetic means for time, harmonic means for rate, and weighted forms when the workload mix is not uniform. These tools form the quantitative core used in later chapters.

2.11 Practice Questions

The following questions reinforce the main ideas of Chapter 2 and are intended for self-checking after the summary.

1. Machine A executes a workload with $IC = 1.2 \times 10^9$, $CPI = 1.5$, and clock rate 2.4 GHz. Machine B executes the same workload with $IC = 0.9 \times 10^9$, $CPI = 1.9$, and clock rate 3.0 GHz. Compute the runtime of both machines and the speedup of the faster machine over the slower one.

2. Two machines execute the same program. Machine J runs at 5.1 GHz with $CPI = 3.0$. Machine K runs at 5.5 GHz with $CPI = 3.7$. Compute the speedup of K over J. Then consider Machine G, which runs at 6.1 GHz with $CPI = 3.2$ but requires 1.5× as many dynamic instructions as J for the same work. Compute the speedup of G over J.

3. A benchmark mix contains three workloads with weights 0.50, 0.30, and 0.20. On one machine, their runtimes are 4 s, 9 s, and 6 s. Compute the weighted average runtime. Then explain why the same weighted arithmetic mean would be inappropriate if the reported quantities were throughputs instead of runtimes.

4. A proposed optimization accelerates the branch-recovery portion of a program, which currently accounts for 30% of total runtime, by a factor of 4. Use Amdahl's Law to compute the overall speedup. Then compute the maximum possible speedup if that portion were made infinitely fast.

5. A program spends 20% of its time in vectorizable code. Suppose an added accelerator makes only that portion run four times faster. Compute the overall speedup. Then explain why adding more parallel hardware gives diminishing returns when the accelerated fraction is fixed.

6. A parallel machine runs a scaled problem with serial fraction 0.10 on 16 processors. Compute the speedup predicted by Gustafson's Law. Then explain why this result does not contradict Amdahl's Law.

7. For each of the following, state whether arithmetic mean, harmonic mean, or a weighted variant is the correct aggregation method, and explain why: runtimes for an equal-work benchmark suite, throughputs for an equal-work benchmark suite, and runtimes for a production workload mix with known fractions.

8. Two candidate servers cost $4,000 and $5,400 and deliver speedups of 1.25 and 1.55 relative to the same baseline. Compute speedup per dollar and cost per unit speedup for both systems. State which system is preferable under each metric.

Chapter 3

Cache Memories

3.1 Memory Technology and the Hierarchy

Modern cache design begins with a technology reality: fast memory is expensive and dense memory is slower. *Static random-access memory (SRAM)* cells use more transistors per bit, provide lower access latency, and do not require refresh, so SRAM is the natural choice for on-chip caches. *Dynamic random-access memory (DRAM)* cells are denser and cheaper per bit but slower and refresh-driven, so DRAM is the natural choice for main memory capacity. This is why systems are built as hierarchies rather than as one uniform memory.

A useful mental model is that each level trades speed for size as distance from the core increases. L1 and often L2 are SRAM-based and optimized for very low hit latency. Lower levels and main memory prioritize capacity and cost. Architecture work in Chapter 3 is mostly about exploiting this hierarchy so that accesses are absorbed by fast levels as often as possible.

At the circuit level, one practical limiter is wire delay. Longer bit lines have higher effective capacitance and resistance, so sense amplifiers need more time to resolve values. This is one reason very large single-level caches become slow: making arrays taller increases bit-line delay even if logic depth is unchanged.

3.2 Typical Hierarchy Levels

In mainstream systems, the first level is a small L1 cache optimized for minimum hit time. A larger L2 cache is usually next, trading some latency for much lower miss ratio. Below L2 sits physical memory (DRAM), which is much larger but much slower. Beneath DRAM is virtual-memory backing store (for example solid-state drive (SSD)), managed by the operating system through paging.

From the central processing unit (CPU) perspective, each lower level is accessed only when the upper level misses. This staged access is exactly why average-access-time analysis composes recursively across levels.

3.3 Locality: Why Caches Work

Cache behavior is dominated by *locality*. *Temporal locality* means that if data is used now, it is likely to be used again soon. *Spatial locality* means that if one address is accessed, nearby addresses are likely to follow. *Sequential locality* is a common special case of spatial locality in which accesses proceed in order, such as instruction fetch or array traversal.

These properties justify block-based transfer. Instead of moving one byte at a time from DRAM, the system moves an entire cache block, betting that adjacent bytes will be useful soon. The same properties also justify replacement policies that retain recently used data and evict older candidates.

3.4 Cache Geometry and Address Partitioning

To reason cleanly about cache structure, use the triplet (C, B, S) (Eq. (3.1)):

$$\text{cache capacity} = 2^C \text{ bytes}, \qquad \text{block size} = 2^B \text{ bytes}, \qquad \text{associativity} = 2^S \text{ ways}. \tag{3.1}$$

Address fields in (C, B, S) notation In this notation, S is **not** an extra address field beyond tag/index/offset. Instead, S changes the number of sets and therefore the index width.

$$\text{sets} = \frac{2^C}{2^B \cdot 2^S} = 2^{C-B-S}. \tag{3.2}$$

Total line count follows directly from sets and ways (Eq. (3.3)):

$$\text{lines} = \text{sets} \cdot \text{ways} = 2^{C-B-S} \cdot 2^S = 2^{C-B}. \tag{3.3}$$

For direct-mapped caches ($S = 0$), ways $= 1$ so lines $=$ sets. For set-associative caches ($S > 0$), lines $>$ sets because there are multiple ways per set.

$$|\text{Offset}| = B, \qquad |\text{Index}| = \log_2(\text{sets}) = C - B - S. \tag{3.4}$$

Because index plus offset consume $(C - S)$ low-order bits (Eq. (3.5)):

$$|\text{Index}| + |\text{Offset}| = (C - B - S) + B = C - S. \tag{3.5}$$

For an address width of AW bits (Eq. (3.6)):

$$|\text{Tag}| = AW - (|\text{Index}| + |\text{Offset}|) = AW - (C - S). \tag{3.6}$$

So the split remains (Eq. (3.7)):

$$\text{Address} = [\text{Tag} \mid \text{Index} \mid \text{Offset}], \tag{3.7}$$

with

$$|\text{Offset}| = B, \qquad |\text{Index}| = C - B - S, \qquad |\text{Tag}| = AW - (C - S). \tag{3.8}$$

Where the S bits "go" Higher associativity means fewer sets, so index width decreases by S relative to a direct-mapped organization with the same C and B. That is why S appears in index/tag equations but is not carried as an address field.

What selects the way? The way is selected by hardware after tag comparison, not by an explicit address field:

1. Index selects one set.
2. Tags for all 2^S ways in that set are compared in parallel.
3. Match logic identifies the hitting way (if any).
4. Internal way-select control drives the data mux (conceptually an S-bit code since $\log_2(2^S) = S$).

Quick example: 32 KB, 32 B blocks, 4-way With $(C, B, S) = (15, 5, 2)$ (Eq. (3.9)):

$$|\text{Index}| = C - B - S = 15 - 5 - 2 = 8. \tag{3.9}$$

For the same C, B but direct-mapped ($S = 0$) (Eq. (3.10)):

$$|\text{Index}| = 15 - 5 - 0 = 10. \tag{3.10}$$

So moving to 4-way associativity reduces index width by 2 bits (equivalently reduces sets by 4×).

32-bit address specialization If $AW = 32$ (Eq. (3.11)):

$$|\text{Offset}| = B, \qquad |\text{Index}| = C - B - S, \qquad |\text{Tag}| = 32 - (C - S) = 32 - C + S. \tag{3.11}$$

Worked example (metadata sizing) Consider a 64-bit machine with $(C, B, S) = (13, 5, 0)$, an 8 KB direct-mapped cache with 32-byte blocks. The cache has $2^{13-5} = 256$ blocks and $2^{13-5-0} = 256$ sets. Tag width is $64 - (13 - 0) = 51$ bits. If each block stores one valid bit and one dirty bit, metadata per block is 53 bits. Total tag-store plus metadata is (Eq. (3.12)):

$$256 \times 53 = 13{,}568 \text{ bits}. \tag{3.12}$$

If B is reduced to 3 while C and S are fixed, the number of blocks rises to $2^{10} = 1024$, so tag overhead increases to $1024 \times 53 = 54{,}272$ bits. This illustrates an important Chapter 3 point: smaller blocks can increase tag overhead significantly.

Concrete Tag-Store Accounting The quickest reliable procedure is:

1. determine the number of lines from 2^{C-B},
2. determine tag width from the relevant physical or virtual address width,
3. add only the metadata bits the problem explicitly asks for, and
4. multiply per-line metadata by line count.

For example, suppose a cache is *write-back, write-allocate* and *physically indexed, physically tagged* with $(C, B, S) = (18, 7, 3)$, 64-bit virtual addresses, 4 KB pages, and 32 GB of physical memory. The physical address width is 35 bits because $32\,\text{GB} = 2^{35}$ bytes. The cache therefore has $2^{18-7} = 2^{11} = 2048$ lines, and the physical tag width is

$$|\text{Tag}| = 35 - (18 - 3) = 20 \text{ bits}. \tag{3.13}$$

Because the level is write-back, write-allocate, each line also needs one valid bit and one dirty bit, so per-line tag-store plus required metadata is 22 bits. Total tag-store size is therefore

$$2048 \times (20 + 1 + 1) = 45{,}056 \text{ bits}. \tag{3.14}$$

If the problem says to ignore replacement metadata such as LRU state, then it should not be added.

Address partitioning example Consider the same $(13, 5, 0)$ 64-bit cache. For the virtual address `0000_1234_ABCD_5678`: **Offset (5 bits):** the lowest 5 bits of `78` (`0111_1000`) are `11000`, which is `18` (decimal 24). **Index (8 bits):** bits 5 through 12 come from hex digits `5`, `6`, and `7`. In binary these digits are `0101`, `0110`, and `0111`; extracting bits 5–12 yields `1011_0011` (binary), which is `B3`. **Tag (51 bits):** the remaining upper bits are the tag. This direct bit-mapping is the fundamental operation performed by hardware on every memory access. It illustrates how physical cache structure determines exactly which address bits are used for indexing and comparison.

3.5 Direct-Mapped, Set-Associative, and Fully Associative

A *direct-mapped cache* has one way per set, so each block address maps to exactly one location. Lookup is simple and fast, but conflict misses can be severe. A *set-associative cache* allows each index to hold multiple candidate blocks; it reduces conflicts at the cost of parallel tag comparisons and a way-select multiplexer. A *fully associative cache* removes index constraints entirely and minimizes conflict misses, but hardware search complexity and power become substantial for large capacities.

Under (C, B, S) notation, direct-mapped means $S = 0$. Fully associative means one set, so $C - B - S = 0$ and therefore $S = C - B$. This distinction is critical, because confusing these two conditions leads to incorrect address math and wrong simulator behavior.

How each cache is constructed Direct-mapped construction uses one data line and one tag entry per index. The index decoder selects one row, then one tag compare and one word select complete the access. A 2^S-way set-associative cache replicates tag and data entries 2^S times per index; all tags in the selected set are compared in parallel, and a way-select mux picks the winning line. A fully associative cache has a single set containing all lines, so every access performs an associative search across the entire tag store and relies heavily on replacement metadata.

3.6 Hit, Miss, and Average Access Time

For a cache hit, access latency is the *hit time* HT. For a miss, the processor first spends hit-time logic discovering the miss, then pays *miss penalty* MP to fetch from the next level. The *average access time* is (Eq. (3.15)):

$$AAT = HT + MR \cdot MP, \tag{3.15}$$

where MR is *miss ratio*. In Chapter 3 analysis, Eq. (3.15) is the cache analogue of the CPU-time equation in Eq. (2.1).

If multiple workloads share one machine and workload i accounts for fraction w_i of the accesses, then weighted average access time is

$$AAT_{weighted} = \sum_i w_i \, AAT_i, \tag{3.16}$$

with $\sum_i w_i = 1$. This weighted form is the correct way to compare cache organizations against a non-uniform workload mix.

Hit time is not a single gate delay. It includes row decoding, word-line drive, bit-line sensing, tag comparison, way selection, and word/byte selection. Cache-geometry choices alter these terms. Taller arrays hurt bit-line delay, wider arrays increase mux and comparator complexity, and higher associativity increases tag-compare and selection cost.

Which change affects which AAT term? When a problem asks which of HT, MR, or MP changes, the most useful first-pass mapping is usually:

1. **data-layout or tiling changes** primarily move *miss ratio* because they change locality and conflict behavior,

2. **VIPT versus PIPT** primarily moves *hit time* because it changes whether TLB lookup can overlap cache indexing, and

3. **write buffers** primarily reduce effective *miss penalty* or visible lower-level service time on writes by decoupling cache progress from memory completion.

This classification is approximate rather than absolute, but it is the right first-pass answer for most design analyses.

3.7 The Three C's of Misses

Misses are classed as *compulsory*, *capacity*, or *conflict*. Compulsory misses are first-touch misses and are unavoidable unless data is brought in before first demand access. Capacity misses happen because the working set exceeds what the cache can hold over the reuse interval. Conflict misses happen when mapping restrictions force evictions even though total capacity would have been enough under freer placement.

In practice, this taxonomy provides a diagnosis strategy. If misses are mainly conflict-driven, associativity or victim structures can help. If misses are mainly capacity-driven, larger effective storage or better blocking can help. If misses are mainly compulsory, prefetching and layout changes are primary tools.

Sizing for a repeating working set For a repeating access pattern, the smallest cache that avoids capacity misses must hold every block that remains live across the reuse interval. If N_{live} distinct blocks must stay resident until reuse, then the cache needs at least N_{live} lines. That condition is necessary but not always sufficient in a direct-mapped or low-associativity cache, because several of those live blocks may still collide onto the same set. This is why trace analysis must separate "enough total lines" from "enough placement freedom."

3.8 Replacement Policies and Their Tradeoffs

When a set is full and a miss occurs, one resident block must be evicted. *Belady's optimal policy* is a theoretical lower bound that requires future knowledge and is not implementable directly. *Least recently used (LRU)* approximates this by evicting the least recently touched entry, and usually performs well under temporal locality. *First in, first out (FIFO)* ignores reuse and can evict hot lines. *Random* is simple and sometimes surprisingly competitive but unpredictable. *Least frequently used (LFU)* tracks frequency but can lag phase changes.

For high associativity, exact LRU state grows costly. Practical designs use approximations, tree-based encodings, or hybrid insertion policies. A notable policy distinction is insertion behavior: a line can be installed as *most recently used (MRU)* (default in many demand-miss paths) or near LRU (useful for speculative prefetch lines to reduce pollution). Related insertion-policy abbreviations are *most recently used insertion policy (MIP)* and *least recently used insertion policy (LIP).*

Simulating LRU with timestamps One exact way to simulate LRU is to attach a "last used" timestamp to each line in a set. On every hit, update the touched line's timestamp to the current access time. On a miss, fill an invalid line if one exists; otherwise evict the valid line with the smallest timestamp because that line was touched least recently. This method is conceptually simple and is often the most direct way to hand-simulate LRU on short traces.

Worked cache-trace decode Consider a cache with $(C, B, S) = (6, 4, 1)$. Then the cache has $2^{6-4} = 4$ lines, arranged as $2^{6-4-1} = 2$ sets with 2 ways per set. So the offset is 4 bits, the index is 1 bit, and the remaining upper bits form the tag. Now consider the access sequence `0x24`, `0x34`, `0x2C`, `0x44`. Because the offset is 4 bits, the block numbers are `0x2`, `0x3`, `0x2`, and `0x4`. Using the low-order bit of the block number as the set index, those accesses map to set 0, set 1, set 0, and set 0, with tags 1, 1, 1, and 2, respectively. The first two accesses are compulsory misses, the third is a hit on the already resident block for `0x24`, and the fourth is another miss because it introduces a different tag into set 0. This is the basic hand-simulation loop: decode block, split into set and tag, check the resident ways, then update replacement state.

Worked timestamp-LRU trace Consider a 2-way set with two accesses already resident:

Tag	Timestamp
A	4
B	7

If the next access is a hit on A at time 8, then A's timestamp becomes 8 while B remains at 7. If the following access is a miss at time 9 and no invalid line exists, then B is evicted because it has the smaller timestamp and is therefore the least recently used line. This is the mechanical rule to remember: every hit refreshes exactly one timestamp, and every full-set miss evicts the smallest timestamp.

Exact LRU with a bit matrix Another exact representation for an n-way set is an $n \times n$ recency matrix. Entry $M[i][j] = 1$ means way i is more recent than way j. When way k is accessed, set row k to 1, set column k to 0, and keep the diagonal in its fixed convention. After the update, the least-recently used way is the one whose off-diagonal row bits are all 0, because every other way is marked as newer than it.

Worked bit-matrix update For a 4-way set, suppose the current access is to way 2. Then row 2 is set to 1 against the other ways and column 2 is cleared:

	0	1	2	3
0	–	?	0	?
1	?	–	0	?
2	1	1	–	1
3	?	?	0	–

The question marks keep their previous values because the relative recency order among the untouched ways does not change. After several accesses, the LRU way is the row whose off-diagonal entries are all 0. That row is the one that every other way claims to be newer than.

3.9 Write Policies

Write behavior has two independent decisions. On a write hit, either update only the cache and defer memory update (*write-back*), or update both cache and memory immediately (*write-through*). On a write miss, either fetch the block then write it (*write-allocate*), or write to lower memory without installing in cache (*write-no-allocate*).

Write-back reduces memory traffic when repeated writes target the same line, but requires a dirty bit and careful eviction handling. Because dirty data can live in cache before eviction, abrupt power loss can drop recently written values that have not yet reached lower levels. Write-through simplifies coherence visibility and durability behavior because lower memory is updated on each write hit, and it does not require dirty metadata at that cache level, but it increases memory bandwidth pressure. The common pairings are write-back plus write-allocate for reuse-heavy data paths, and write-through plus write-no-allocate for streaming paths with little read-after-write reuse.

Reading write traces Write traces are easiest to interpret by separating three questions. First, does the access hit or miss in the current cache state? Second, does the policy allocate the block on a miss? Third, if a valid victim is evicted, is it dirty? In a write-back, write-allocate cache, a write miss usually installs the block and marks it dirty immediately. Each later dirty eviction contributes one writeback event. Dirty lines that remain in the cache at the end of the trace are not counted as writebacks unless the problem explicitly asks for a final flush.

Worked dirty-eviction trace Consider a write-back, write-allocate cache whose relevant set can hold only two lines. Suppose the set is initially empty and the access sequence is

$$W(A),\ W(B),\ W(A),\ R(C),$$

with C mapping to the same full set as A and B. The first write to A is a miss, so A is allocated and marked dirty. The first write to B does the same for B. The second write to A is a hit and leaves A dirty. When C is later accessed, one of the dirty lines A or B must be evicted; that eviction produces a writeback because the victim holds modified data that has not yet reached the next level.

Valid and dirty bits: meaning, need, and size impact Each cache line typically carries at least one *valid bit*. Valid= 1 means the tag/data entry contains a meaningful memory block; valid= 0 means the entry must not hit, even if tag bits happen to match after reset or replacement. A *dirty bit* is needed when that cache level uses *write-back*: dirty= 1 means the line was modified in cache and must be written to the next level before eviction; dirty= 0 means eviction can discard the line without write-back traffic. In a pure write-through level, dirty bits are not required because lower memory is updated on every write hit.

If line count is $N_{lines} = 2^{C-B}$, valid-bit storage is N_{lines} bits. Dirty-bit storage is another N_{lines} bits only when write-back is enabled at that level. Relative to data-array size 2^C bytes, overhead from these two bits alone is (Eq. (3.17)):

$$\text{overhead fraction} = \frac{n_v + n_d}{2^{B+3}}, \tag{3.17}$$

where $n_v = 1$ (valid always) and $n_d \in \{0, 1\}$ (dirty only for write-back). Example: with 32-byte blocks ($B = 5$), valid-only overhead is $1/256 \approx 0.39\%$, and valid+dirty overhead is $2/256 \approx 0.78\%$ (excluding tag, replacement, and coherence bits).

3.10 Reducing Miss Ratio

The first lever is **cache size (C)**. Increasing C reduces capacity misses because more of the working set fits in the hierarchy. However, larger caches increase hit time due to longer bit lines and decoder delays, and they consume more area and power.

The second lever is **associativity (S)**. Increasing S reduces conflict misses by providing more locations for a given index. The benefit is most significant when moving from direct-mapped to 2-way or 4-way, after which returns diminish. The cost includes increased hit time (due to the way-select multiplexer and parallel tag comparisons) and increased power consumption.

The third lever is **block size (B)**. Increasing B reduces compulsory misses by exploiting spatial locality (fetching more useful data per miss). However, if B becomes too large, it can increase conflict misses (by reducing the total number of blocks) and will always increase miss penalty (longer transfer time). So the statement "increasing block size cannot improve miss ratio" is false. With a fixed cache capacity, a larger block often improves miss ratio at first by capturing nearby useful data, but the improvement is not monotonic because fewer total lines remain available.

Hardware Prefetching Hardware prefetching targets compulsory and capacity effects by pulling data before demand. A +1 *prefetcher* fetches block $i+1$ when block i is accessed. A *stride prefetcher* detects constant deltas (for example $i, i+k, i+2k$) and fetches the next predicted address. A *Markov prefetcher* uses a correlation table to learn non-linear transitions; if address A is often followed by B then C, the predictor learns and issues those future blocks when A appears. A *hybrid prefetcher* combines a richer predictor such as Markov correlation with a simpler fallback such as next-block prefetching, so the machine can still make a reasonable guess when the correlation table has no useful entry. To prevent *cache pollution* (evicting useful data for speculative data that might not be used), many designs use a *prefetch buffer*. This is a small, separate storage area where prefetched lines sit until they are actually demanded by the CPU. If demanded, they are moved into the main cache.

Prefetch insertion policy matters Prefetched lines are often installed with a low-priority policy such as *least recently used insertion policy (LIP)* rather than as most-recently used lines. The goal is to avoid immediately evicting useful demand data for speculative lines that may never be touched. This helps in set-associative caches, but it does not eliminate the placement limitations of a direct-mapped cache. In a direct-mapped design, a +1 prefetch can still evict useful demand data if the prefetched block maps to the same line.

Trace-driven prefetch statistics Detailed simulator outputs often distinguish three prefetch counters. *Prefetches issued* count speculative requests launched by the prefetcher. A *prefetch hit* means a later demand access found a line that was present only because of an earlier prefetch. A *prefetch miss*, sometimes called a useless prefetch, means the prefetched line was evicted before any demand access used it. These counters matter because a prefetcher can lower demand misses and still be a poor design if it wastes bandwidth or pollutes the cache badly.

3.11 Reducing Miss Penalty

Miss penalty is shaped strongly by transfer mechanics. If a block has 2^B bytes and bus transfer width is W bytes, then the number of sub-block transfers is $N_{sb} = 2^B/W$. With first-transfer

startup latency T_1 and per-following transfer time T_f, block transfer time is (Eq. (3.18)):

$$MP_{full} = T_1 + (N_{sb} - 1)T_f. \tag{3.18}$$

Early restart returns data to the CPU as soon as the requested sub-block arrives rather than after full-block completion. If critical-word position is uniformly distributed and transfer order is fixed from sub-block 0 upward, average CPU-visible penalty becomes (Eq. (3.19)):

$$MP_{ER} \approx T_1 + \frac{N_{sb} - 1}{2} T_f. \tag{3.19}$$

Critical-word-first improves further by requesting the demanded sub-block first, then wrapping to complete the rest in the background. That benefit is only CPU-visible if the machine can resume execution as soon as the critical word arrives. Without early restart or an equivalent partial-consume mechanism, critical-word-first alone does not reduce the stall seen by the processor, because the core would still wait for the full block to complete before continuing.

Non-uniform critical-word distributions The uniform approximation in Eq. (3.19) is only one special case. If sub-block k is demanded with probability p_k and fixed-order transfer would deliver that word after k follow-up transfers, then the early-restart penalty is

$$MP_{ER} = \sum_{k=0}^{N_{sb}-1} p_k \left(T_1 + kT_f\right). \tag{3.20}$$

Under the same simple model, critical-word-first with early restart reduces the CPU-visible penalty to approximately T_1 because the demanded sub-block is requested first and the rest of the block finishes in the background.

Worked miss-penalty comparison Consider a 32-byte block on an 8-byte bus, so the miss repairs 4 words. Suppose each requested word transfer takes 70 ns and transfers are serialized. Baseline repair waits for all 4 words, so the visible miss penalty is

$$MP_{full} = 4 \times 70 = 280 \text{ ns}. \tag{3.21}$$

With early restart under a uniform critical-word position, the demanded word arrives after 1, 2, 3, or 4 transfers with equal probability, so the average visible penalty is

$$MP_{ER} = \frac{70 + 140 + 210 + 280}{4} = 175 \text{ ns}. \tag{3.22}$$

With critical-word-first plus early restart, the demanded word is requested first, so the processor-visible penalty returns to approximately one transfer time, 70 ns, while the rest of the block completes in the background.

Another major lever is adding lower cache levels. If L1 misses probe L2 before DRAM, L1 penalty becomes L2 average access time (Eq. (3.23)):

$$AAT_{L1} = HT_{L1} + MR_{L1} \cdot AAT_{L2}. \tag{3.23}$$

This recursive form extends naturally to deeper hierarchies.

Worked two-level AAT example Suppose $HT_{L1} = 1$ cycle, $MR_{L1} = 0.08$, $HT_{L2} = 8$ cycles, $MR_{L2,read} = 0.25$, and $DRAM_TIME = 60$ cycles. Then

$$AAT_{L2} = 8 + 0.25 \times 60 = 23 \text{ cycles.} \tag{3.24}$$

That lower-level average time becomes the miss penalty seen by L1, so

$$AAT_{L1} = 1 + 0.08 \times 23 = 2.84 \text{ cycles.} \tag{3.25}$$

The main reasoning pattern is recursive rather than additive-by-guess: first solve the lower level, then substitute that average behavior upward.

Inclusive policy: subset, not equality If L2 is *inclusive* of L1, then every line present in L1 must also be present in L2. The reverse is not required: L2 can contain many lines that are not currently in L1. So inclusivity means "L1 is a subset of L2," not "L1 and L2 contain exactly the same data."

Why many L2 ratios are read-only In a hierarchy where L1 uses write-back, write-allocate and L2 uses write-through, write-no-allocate, simulators often report separate L2 read and write traffic. Read hits and read misses form a meaningful reuse ratio because reads probe the resident L2 contents. Writes, by contrast, may simply reflect L1 writebacks or pass-through stores and may not allocate lines on a miss. For that reason, many reports define L2 hit ratio and miss ratio over *reads only* rather than over total L2 traffic.

Victim caches are also miss-penalty tools. A small fully associative buffer captures lines evicted from the primary cache. On a later conflict-driven re-reference, data can be swapped back quickly without going to DRAM. Conceptually, a victim cache provides a small amount of flexible associativity exactly where pressure appears.

Victim cache versus L2 A victim cache and an L2 do not solve exactly the same problem. A victim cache is usually the better value when misses are dominated by a small number of conflict patterns near the L1 boundary, because a few fully associative entries can undo repeated index collisions cheaply. A larger L2 is usually the better value when misses reflect broader capacity pressure or many different miss streams, because it filters a much wider set of misses rather than only recently evicted L1 lines. The right choice depends on whether the workload is mainly suffering from local conflicts or from a deeper shortage of effective capacity.

Write buffers reduce stalls by decoupling cache writes from lower-memory completion. They are especially useful when a following miss would otherwise wait behind pending writes. Correct implementation requires forwarding from the newest matching buffered write and careful ordering semantics.

3.12 Reducing Hit Time

Hit-time optimization frequently conflicts with miss-rate optimization. Smaller arrays and lower associativity usually improve hit latency. However, overly aggressive simplification can raise miss ratio enough to hurt total performance. In general, small caches and direct-mapped organizations are fast on the hit path, but they are often unacceptable unless miss ratio remains controlled by workload behavior or lower-level caches.

Address translation complicates this tradeoff. A naive *physically indexed, physically tagged* path can serialize translation lookaside buffer (TLB) lookup before cache access, increasing L1 hit latency. *Virtually indexed, physically tagged (VIPT)* caches overlap TLB translation with set indexing and preserve physical-tag correctness.

PIPT vs. VIPT timing view With virtual memory, the CPU generates a virtual address split into virtual page number (VPN) and page offset. In a *physically indexed, physically tagged (PIPT)* L1, both index and tag use physical address bits. That means the TLB must produce the physical address before cache indexing can begin. So a first-order timing model is (Eq. (3.26)):

$$HT_{L1,\text{PIPT}} \approx T_{TLB} + T_{L1,\text{index+array+tag}}. \tag{3.26}$$

In a *VIPT* L1, indexing starts immediately using low-order virtual bits while the TLB is translating in parallel. When the physical address arrives, the physical tag compare finalizes hit/miss. A first-order model is (Eq. (3.27)):

$$HT_{L1,\text{VIPT}} \approx \max(T_{TLB},\, T_{L1,\text{index+array}}) + T_{\text{phys_tag_compare}}. \tag{3.27}$$

The practical effect is that VIPT can hide much of TLB latency on the L1 hit path, while PIPT often exposes it directly.

Why the VIPT constraint exists For VIPT to work cleanly, index-plus-offset bits must fit within page-offset bits (Eq. (3.28)):

$$C - S \leq P, \tag{3.28}$$

where P is page-offset width in bits. If $C - S > P$, then some index bits come from VPN bits. Different virtual addresses that map to the same physical page (*synonyms*) can then index different sets, allowing duplicated physical data in L1. That creates alias-management problems: writes through one synonym may not immediately update the other cached copy, risking stale reads until extra coherence/synchronization machinery repairs it. This constraint often forces minimum associativity for a given L1 size and page size. For example, with $P = 12$ (4 KB pages) and $C = 15$ (32 KB L1), one needs $S \geq 3$, i.e., at least 8-way associativity. If $S = 2$ (4-way), then $C - S = 13 > 12$, so VIPT indexing would extend beyond page offset and synonym risk appears.

Concrete VIPT Associativity Check Suppose $C = 18$ and page size is 4 KB, so $P = 12$. To make the cache VIPT without indexing beyond the page offset, require

$$C - S \leq P \Rightarrow 18 - S \leq 12 \Rightarrow S \geq 6. \tag{3.29}$$

So the minimum acceptable associativity field is $S = 6$, which corresponds to $2^6 = 64$ ways. The arithmetic is simple; the main thing to remember is that the constraint is on *index plus offset*, which in (C, B, S) notation is exactly $C - S$.

Write buffers can also improve effective hit behavior for writes by decoupling write completion from the immediate critical path and by forwarding recent write data when needed. This prevents some write-after-read and read-after-write timing stalls from stretching apparent hit latency.

3.13 Virtual Memory Technology, Terminology, and Analogy to Caching

Virtual-memory terminology mirrors cache terminology closely. A virtual address is typically split into a *virtual page number (VPN)* and page offset. Translation maps VPN to a *physical frame number (PFN)* through a page table. A *translation lookaside buffer (TLB)* is a cache of recent translations. A missing translation in the TLB causes a page-table walk; if the page is not in DRAM, a page fault triggers operating system (OS) handling and secondary-storage transfer.

The analogy to caching is direct. VPN behaves like a tag, page offset behaves like block offset, and the TLB behaves like a small associative cache of mapping metadata. Likewise, DRAM behaves as a cache for backing store, while on-chip caches behave as caches for DRAM. This layered view is useful because it applies one conceptual framework to both hardware cache hierarchy and OS-managed virtual memory.

3.14 Compiler and Data-Layout Techniques

Software organization strongly affects cache performance. **Compiler layout of instructions** places frequently executed basic blocks close together in memory to improve I-cache hit rate and reduce fetch stalls. **Compiler-directed prefetching** issues non-binding prefetch instructions ahead of use, which is effective for predictable array accesses. **Data-layout choices**, including structure-of-arrays (SoA), field reordering, and padding, can reduce cache-line waste and false sharing.

Tiling (blocking) re-organizes loops to process data in smaller cache-resident regions. For example:

```
// Before tiling:
for (i=0; i<N; i++) for (j=0; j<N; j++) A[i][j] = B[j][i];
// After tiling (where b is the tile size):
for (ii=0; ii<N; ii+=b) for (jj=0; jj<N; jj+=b)
    for (i=ii; i<ii+b; i++) for (j=jj; j<jj+b; j++) A[i][j] = B[j][i];
```

Choosing the tile factor is a working-set problem. The largest profitable tile is usually the largest one whose active data footprint still fits in the effective cache space while avoiding severe mapping conflicts. For direct-mapped or low-associativity caches, this means a tile can fail even when the byte count appears to fit, because the active blocks may still alias onto the same few sets. The practical rule is therefore: first check footprint size, then check placement conflicts. This is why tiled matrix multiplication often shows much better cache behavior than a naive matrix multiply, even when both perform the same arithmetic work.

Concrete Tile-Factor Estimate Consider a cache with $(C, B, S) = (10, 6, 2)$, so total capacity is $2^{10} = 1024$ bytes, block size is 64 bytes, and associativity is 4-way. If each matrix element is 8 bytes and the tiled working set keeps TF rows of 32 elements active, then one tiled row occupies

$$32 \times 8 = 256 \text{ bytes.} \tag{3.30}$$

A footprint estimate for the tile is therefore $256 \times TF$ bytes. The largest tile that still fits in 1024 bytes is

$$256 \times TF \leq 1024 \Rightarrow TF \leq 4. \tag{3.31}$$

Here $TF = 4$ is also structurally reasonable because each row occupies four 64-byte blocks, and the 4-way organization provides enough placement freedom to avoid immediate within-tile self-conflict.

Loop fusion combines independent loops that touch related data to improve temporal locality. For example:

```
for (i=0; i<N; i++) a[i] = b[i] + 1; // loop 1
for (i=0; i<N; i++) c[i] = a[i] * 2; // loop 2
// Fused:
for (i=0; i<N; i++) { a[i] = b[i] + 1; c[i] = a[i] * 2; }
```

In practice, these compiler and data-layout techniques frequently determine whether hardware cache features help or saturate.

3.15 Chapter 3 Summary

Chapter 3 develops cache memory from first principles to practical optimization. The chapter links SRAM/DRAM technology limits to hierarchical design, formalizes cache geometry with (C, B, S) math, and uses AAT to connect structure to performance. Misses are interpreted through the three C's, while replacement and write policies define behavioral tradeoffs. Improvement strategies are organized by target term: miss ratio, miss penalty, or hit time. Prefetching, victim caches, write buffers, multi-level hierarchies, simulator-style hierarchy and prefetch counters, and VIPT constraints show how modern designs balance speed, capacity, and correctness. Finally, compiler and data-layout techniques complete the picture by showing that cache performance is a hardware-software co-design problem.

3.16 Practice Questions

The following questions reinforce the main ideas of Chapter 3 and are intended for self-checking after the summary.

1. A physically indexed, physically tagged cache has total capacity 32 KiB, block size 64 B, associativity 4-way, and 48-bit addresses. Compute the number of sets, the number of offset bits, the number of index bits, and the number of tag bits. Then compute the total tag-store size if each line stores a valid bit and a dirty bit in addition to the tag.

2. For each of the following changes, state whether it primarily targets *hit time*, *miss ratio*, or *miss penalty*: adding a victim cache, adding a write buffer, increasing associativity, increasing block size, and using critical-word-first with early restart.

3. Explain the difference between critical-word-first and early restart. Why does critical-word-first by itself not guarantee a reduction in processor-visible miss penalty?

4. A system uses 32-byte blocks, an 8-byte memory bus, and serialized memory transfers. A miss first incurs a 70 ns round trip for the requested word, and each additional word also requires 70 ns. The cache hit time is 10 ns and the hit rate is 15%. Compute the average access time under three cases: baseline repair with no early use of returned data, early restart with a uniform probability of needing any of the four words in the block, and critical-word-first with early restart.

5. Consider a $(C, B, S) = (6, 4, 2)$ cache with LRU replacement and the access stream `0x589e`, `0xa7a1`, `0x3767`, `0xcce2`, `0xa7aa`, `0xff4d`, `0x376c`, `0x5892`. For each access, determine the set, tag, hit or miss outcome, and final contents of the tag store after the full trace.

6. A 4-way set uses an exact LRU bit-matrix implementation. Starting from a state in which way 0 is least recently used and way 3 is most recently used, update the matrix for the hit sequence $2 \to 0 \to 3 \to 1 \to 2$. Which way is least recently used at the end?

7. A VIPT L1 cache has capacity 64 KiB, block size 64 B, and page size 4 KiB. Compute the minimum associativity required so that the set index fits entirely within the page offset.

8. A miss stream repeatedly alternates between two blocks that map to the same direct-mapped L1 set. Would a small victim cache or a simple +1 prefetcher be the more natural first fix? Explain why.

9. A tiled matrix kernel keeps three $b \times b$ tiles of double-precision data active at once. If the effective L1 capacity available to the tiles is 24 KiB, what is the largest integer tile size b that satisfies the footprint constraint $3b^2 \cdot 8 \leq 24$ KiB?

10. A two-level hierarchy has $HT_{L1} = 1$ cycle, $MR_{L1} = 0.08$, $HT_{L2} = 8$ cycles, $MR_{L2,read} = 0.25$, and $DRAM_TIME = 60$ cycles. Compute AAT_{L2} and then AAT_{L1}.

11. A write-back, write-allocate cache receives the sequence: write block A, write block B, write block A again, then access block C, which maps to the same full set as A and B and forces an eviction. Describe when dirty state is created and whether the eviction causes a writeback.

Chapter 4

Pipelining

4.1 Pipeline Motivation

Pipelining improves throughput by overlapping instruction execution stages in time. Instead of waiting for one instruction to complete all work before starting the next, the processor places stage registers between major datapath functions and allows multiple instructions to be in flight simultaneously. The key gain is not that one instruction becomes instantly shorter in latency, but that completed instructions per unit time rise after the pipeline fills.

4.2 Five-Stage Example Pipeline

The canonical integer pipeline has five stages (Eq. (4.1)):

$$IF \rightarrow ID \rightarrow EX \rightarrow MEM \rightarrow WB. \tag{4.1}$$

Instruction Fetch (IF) reads the instruction from the instruction memory hierarchy and advances the program counter (PC). Instruction Decode (ID) decodes opcode and operands and reads source registers. Execute (EX) performs ALU work and effective-address calculation. Memory (MEM) performs data-memory access for loads and stores. Write Back (WB) writes results to the architected register file.

Even instructions that do not logically need memory often still occupy the MEM stage in a basic in-order five-stage design so that all instructions remain aligned in a uniform stage progression. This simplifies control and preserves in-order retirement semantics.

4.3 Pipeline Speedup Calculation

Suppose an unpipelined design executes one instruction in N stage-equivalent delays, each of delay T_s. Its instruction time is approximately NT_s. A pipelined design with stage registers has clock (Eq. (4.2)):

$$T_{clk,pipe} \approx T_s + T_{reg}, \tag{4.2}$$

where T_{reg} is latch/flip-flop overhead. For a long stream, ideal CPI approaches 1, so time per instruction approaches $T_{clk,pipe}$ and ideal speedup becomes (Eq. (4.3)):

$$S_{ideal} \approx \frac{NT_s}{T_s + T_{reg}}. \tag{4.3}$$

If T_{reg} were zero, this approaches N. In practice, register overhead and stalls reduce realized speedup.

Finite-length form For M instructions in an N-stage pipeline (ignoring hazards), total cycles are $N + M - 1$, so (Eq. (4.4)):

$$T_{pipe} \approx (N + M - 1)T_{clk,pipe}, \qquad T_{unpiped} \approx MNT_s. \tag{4.4}$$

This shows fill and drain overhead is more visible for short instruction sequences.

Worked pipeline-speedup calculation Suppose an unpipelined implementation has effective $CPI = 5$, and a 5-stage pipelined version keeps the same clock period. The ideal maximum speedup is therefore 5. If 25% of instructions now experience one extra stall cycle, the pipelined CPI becomes

$$CPI_{pipe} = 1 + 0.25 \times 1 = 1.25. \tag{4.5}$$

The corresponding speedup relative to the original $CPI = 5$ design is then

$$\frac{5}{1.25} = 4, \tag{4.6}$$

which is below the ideal bound because hazards consume part of the pipelined throughput gain.

4.4 CPI Model and Stall Accounting

In steady state without hazards, ideal CPI is 1 (Eq. (4.7)):

$$CPI_{ideal} = 1. \tag{4.7}$$

With hazards, the practical model is (Eq. (4.8)):

$$CPI = 1 + \text{stalls}_{avg}, \tag{4.8}$$

where stalls_{avg} is average stall cycles per instruction. This form is central in Chapter 4: almost every pipeline optimization is either reducing stall frequency, reducing stall duration, or both.

4.5 Program Characteristics Plus Hazards Produce Stalls

Hardware hazards are potential conflicts. A stall occurs only when the running program triggers that conflict. This yields a guiding principle (Eq. (4.9)):

$$\text{Program characteristics} + \text{pipeline hazards} \Rightarrow \text{stalls}. \tag{4.9}$$

For example, a branch hazard matters only when branches occur in the dynamic instruction stream. A structural hazard from a shared memory port matters only when fetch and data access overlap in a conflicting cycle.

4.6 Structural Hazards

Structural hazards arise from insufficient hardware resources. A classic example is a single-ported memory system that must serve both IF and MEM stages. If one instruction needs data memory while another needs instruction fetch in the same cycle, one must wait, causing a stall.

Common avoidance strategies are straightforward architectural provisioning: split L1 into separate instruction and data caches, add ports, duplicate critical units, or schedule around known bottlenecks. In modern designs, many obvious structural hazards are eliminated in hardware because their stall cost is predictable and severe.

CPI impact example If 20% of instructions incur one extra stall cycle due only to a structural hazard, then the average CPI becomes

$$CPI = 1 + 0.2 \times 1 = 1.2. \tag{4.10}$$

This is the same accounting pattern used for other hazards: identify the frequency of the event and multiply by its average stall contribution.

4.7 Data Hazards

Data hazards occur when the pipeline changes the order of read/write accesses to operands so that the order differs from the order in the original sequential program.

Read After Write (RAW) A *read-after-write (RAW)* hazard occurs when an instruction refers to a result that has not yet been calculated or stored by a preceding instruction. This is the most common hazard in the classic 5-stage pipeline.

```
I1:  ADD R1, R2, R3    I2:  SUB R4, R1, R5
```
(4.11)

In this case, `I2` needs `R1` in its ID/EX stage, but `I1` does not write `R1` until its WB stage. Without intervention, `I2` would read a stale value.

Write After Read and Write After Write *Write-after-read (WAR)* and *write-after-write (WAW)* hazards do not typically occur in the simple 5-stage in-order pipeline because instructions read early (ID) and write late (WB) in a fixed order. However, they become critical in out-of-order execution or pipelines with multiple completion units of varying latencies.

4.7.1 Forwarding (Bypassing)

Forwarding allows a result to be used as an input to a subsequent instruction before it is written to the register file. If the result is already available in a pipeline register (such as EX/MEM or MEM/WB), hardware can forward it directly to the ALU inputs. This eliminates many RAW stalls.

4.7.2 Load-Use Stalls

Forwarding cannot solve every RAW hazard. If a load instruction is followed immediately by an instruction that uses the loaded value, the data is not available until the end of the MEM stage, which is too late for the EX stage of the following instruction.

```
I1:  LW R1, 0(R2)    I2:  ADD R3, R1, R4
```
(4.12)

This case requires a one-cycle load-use stall even with forwarding. Compilers attempt to schedule independent instructions into these "load-delay" slots to maintain performance.

4.7.3 Same-Cycle Register-File Write Then Read

One subtle pipeline detail is the timing of register-file writes relative to reads in the same cycle. If the register file writes in the early part of the cycle and decode reads in the later part, then an instruction in ID can sometimes see a value written back by an older instruction in WB in that same cycle. If that timing is not supported, the machine needs either an explicit WB-to-ID bypass path or an extra stall. This detail matters when reading timing diagrams because two pipelines with the same five visible stage names can still differ by one cycle on closely spaced dependences.

4.8 Control Hazards

Control hazards occur because the fetch stream does not immediately know the correct next PC for branches and jumps. The branch creates uncertainty in both target and direction (taken/not taken), and the pipeline may fetch wrong-path instructions that must later be squashed.

CPI impact example If branches occur every five instructions on average (branch frequency = 0.2) and each branch introduces one average stall cycle in a simple pipeline, then

$$CPI = 1 + 0.2 \times 1 = 1.2. \tag{4.13}$$

This simple calculation captures why even modest branch penalties have visible throughput cost.

4.8.1 Delay Slots

A *delay slot* ISA semantic says that one or more instructions after a branch execute before control transfer takes effect. The compiler tries to fill these slots with useful independent work. This can reduce explicit branch stalls when suitable instructions exist, but becomes increasingly difficult as pipelines deepen and dependence constraints tighten.

How software fills delay slots Delay-slot instructions execute whether the branch is taken or not. So a compiler may place an instruction into a delay slot only if doing so is safe on *both* outcomes. The usual search order is:

1. move an independent instruction from before the branch down into the slot,
2. if control-flow structure permits, move an instruction from a successor block only when it is safe for both paths, or
3. leave the slot as a `NOP` if no safe candidate exists.

This is why branch schedules often still leave some slots unfilled: correctness takes priority over slot utilization.

4.8.2 Squash Slots with Likely Bits

Some designs expose *squash-slot* behavior via branch-likely style semantics. The compiler places a candidate instruction near the branch and encodes likely behavior. Hardware keeps or squashes that instruction based on actual branch resolution and likely-bit policy. This improves performance when heuristics are good but complicates ISA semantics and compiler scheduling.

How squash-slot filling differs Squash slots are more permissive than delay slots because wrong-path instructions may be discarded. If the machine assumes a branch is likely taken, the compiler can place instructions from the predicted taken path into those slots. If the prediction is wrong, hardware squashes them. So the key distinction is:

- **delay slot:** instruction must be correct regardless of branch outcome,
- **squash slot:** instruction only needs to be correct on the predicted path because hardware may cancel it.

4.9 Branch Target Buffer (BTB)

The *branch target buffer (BTB)* predicts where a branch goes (target prediction) early enough to guide fetch.

4.9.1 BTB with Tags

A tagged BTB stores branch PC tags and associated targets. On fetch, the BTB is indexed and tag-compared; a valid tag hit yields a predicted target. Tagging eliminates *aliasing*, where two unrelated branches share an entry, ensuring higher prediction accuracy at the cost of significantly more hardware area and longer lookup time.

4.9.2 BTB without Tags

A tagless BTB behaves like a hashed direct lookup table from index to target metadata. It is fast and compact but prone to *aliasing*: unrelated branches may map to the same entry and pollute each other's predictions. This can still be acceptable if the table is sized well and collision rate is low for target workloads, trading accuracy for substantial area savings.

4.9.3 Hardware Return Address Stack

Function returns are indirect branches whose targets are difficult for generic BTB logic. A small hardware *return-address stack (RAS)* records call return PCs and pops on return, producing very high return-target accuracy in common call/return patterns. This is one of the most cost-effective branch-target optimizations.

Worked BTB/RAS trace Assume the BTB is indexed by $PC[7:4]$ and the RAS is initially empty. Now consider the short control-flow sequence

```
0x4000:  CALL 0x4A00
0x4A00:  BREQZ R1,0x4AF0 (taken)
0x4AF8:  RET
```

The call pushes return address `0x4004` onto the RAS and creates or refreshes a BTB entry for the call target `0x4A00`. The conditional branch updates its BTB entry with the taken target `0x4AF0`. The return then pops the top of the RAS and predicts target `0x4004`. The main bookkeeping distinction is that the BTB remembers branch and jump targets by PC, while the RAS handles the nested call-return discipline explicitly.

BTBs still matter for unconditional control transfers A BTB is not useful only for conditional branches. Even if every control transfer is unconditional, fetch still benefits from knowing the target early. Without a BTB or some equivalent early-target mechanism, the pipeline may need to wait until later decode or execute logic computes the jump target, creating avoidable fetch bubbles.

4.10 Whether-To Branch: Direction Prediction

Target prediction answers where a branch goes. *Direction prediction* answers whether it is taken. A simple static method is software heuristics, especially *backward taken / forward not taken (BTFNT)*: loop back-edges are often taken, many forward conditionals are often not taken. Compilers can also use profile-guided statistics to set likely-direction metadata (for example branch-likely annotations).

Dynamic predictors improve further by learning runtime behavior (for example saturating counters and history-based schemes). The essential Chapter 4 lesson is that deeper pipelines require higher direction accuracy because misprediction penalty grows with the number of in-flight wrong-path stages.

A very simple baseline is the *always-taken* predictor. It ignores PC and history entirely and predicts that every conditional branch will be taken. This is crude, but it remains a useful reference point because many loop back-edges are taken most of the time. More sophisticated predictors should therefore be judged not only against one another, but also against simple baselines such as backward taken / forward not taken and always taken.

4.11 Simple Dynamic Direction Predictors

The simplest dynamic predictors store a small amount of state per static branch or per BTB entry.

4.11.1 One-Bit Prediction

A *1-bit predictor* records only the last observed direction for a branch. In a compact BTB-style design, this may appear as a single prediction bit associated with the target entry. If the last outcome was taken, predict taken next time; if the last outcome was not taken, predict not taken. This adapts quickly, but it also flips too easily. For example, a loop-closing branch that is taken many times and then falls through once on loop exit will usually be mispredicted again on the next loop entry because the last outcome was not taken.

4.11.2 Smith Two-Bit Saturating Counters

A *Smith predictor* replaces the single bit with a small saturating counter, most commonly two bits. The four counter states are commonly interpreted as strongly not taken, weakly not taken, weakly taken, and strongly taken. Prediction comes from the high-order tendency of the counter, while update moves the counter only one step toward the actual outcome. This hysteresis avoids overreacting to one unusual branch result. In the loop example above, a two-bit counter usually mispredicts only on the loop exit rather than on both the exit and the next re-entry.

4.12 Global and Local History Predictors

History-based predictors use recent branch outcomes to select among several learned taken/not-taken tendencies rather than relying on one counter per static branch.

4.12.1 GShare

GShare uses a global history register (GHR) like other global-history predictors, but instead of concatenating PC bits and history bits it combines them with an exclusive-or (XOR). If the predictor uses n index bits, then a common form is

$$index = PC_{low}[n-1:0] \ \oplus \ GHR[n-1:0]. \tag{4.14}$$

The XOR can reduce some destructive aliasing patterns by spreading correlated PCs across the counter table more evenly than a plain concatenation. Conceptually, GShare and GSelect use the same ingredients; they differ mainly in how PC and history are mixed to form the table index.

4.12.2 GSelect

GSelect is a simple global-history predictor. It keeps a *global history register (GHR)* containing the most recent branch outcomes and combines that history with a small hash of the current PC to index a table of saturating counters. If the PC hash contributes p bits and the GHR contributes h bits, then the predictor selects one of 2^{p+h} counters. Prediction comes from the selected counter, usually by treating the upper half of the counter states as taken and the lower half as not taken. After the true outcome is known, the predictor updates that same counter and then shifts the actual outcome into the GHR. The key idea is that one branch can learn differently depending on the recent behavior of other branches.

Worked GSelect trace The following trace makes the lookup order explicit. Assume a 2-bit GHR whose rightmost bit is the newest outcome, a 1-bit PC hash equal to $PC[2]$, and 2-bit Smith counters where `00`/`01` predict not taken and `10`/`11` predict taken. Let every counter begin in state `10`, and let the initial GHR be `01`. For the four resolved branches below, prediction always uses the *current* PC hash and the *current* GHR. Only after that prediction is the selected counter updated and the actual outcome shifted into the GHR.

Time	PC	Act.	$PC[2]$	GHR_T	Ctr_T	Pred.	Ctr_{T+1}	GHR_{T+1}
T_0	`0x2017`	N	`1`	`01`	`10`	T	`01`	`10`
T_1	`0x2009`	T	`0`	`10`	`10`	T	`11`	`01`
T_2	`0x2017`	T	`1`	`01`	`01`	N	`10`	`11`
T_3	`0x2009`	N	`0`	`11`	`10`	T	`01`	`10`

After T_3, the final GHR is `10`. The visited counter at $(PC[2], GHR) = (0, \texttt{10})$ has been trained up to `11`, and the counter at $(0, \texttt{11})$ has been trained down to `01`. This is the mechanical rule to remember: read the counter indexed by the *old* GHR, make the prediction, update that same counter with the actual outcome, then shift the actual outcome into the GHR.

4.12.3 Yeh/Patt

Yeh/Patt separates branch history into two levels. A first table, indexed by the branch PC, stores a short per-branch history pattern. That recorded history then indexes a second table of saturating counters. In other words, GSelect uses one shared global history for every branch, while Yeh/Patt gives each branch its own history entry before consulting the prediction counters. Prediction therefore proceeds in two steps:

1. Use the branch PC to read the branch's recorded history.

2. Use that history to select the saturating counter that predicts taken or not taken.

Update proceeds in the same order in reverse: first train the selected counter toward the actual outcome, then shift the actual outcome into that branch's local history register. This structure is especially useful when different static branches follow different repetitive patterns.

Worked Yeh/Patt trace Now assume a 2-bit history table indexed by $PC[3:2]$ and a shared pattern table of 2-bit Smith counters. Let every history-table entry begin at `00`, and let every pattern-table counter begin at `01`. The sequence below uses two PCs, `0x3B98` and `0x3B9F`, whose hashes are `10` and `11`, respectively.

Time	PC	$PC[3:2]$	Hist_T	Ctr_T	Pred.	Act.	Ctr_{T+1}	Hist_{T+1}
T_0	`0x3B98`	`10`	`00`	`01`	N	N	`00`	`00`
T_1	`0x3B9F`	`11`	`00`	`00`	N	T	`01`	`01`
T_2	`0x3B98`	`10`	`00`	`01`	N	N	`00`	`00`
T_3	`0x3B9F`	`11`	`01`	`01`	N	N	`00`	`10`

This trace makes the two-level structure explicit. At T_1, branch `0x3B9F` still reads local history `00`, so it trains the shared pattern entry for history `00`, even though it is a different static branch from `0x3B98`. At T_3, however, the same branch now reads local history `01`, so it consults and updates a different pattern-table counter. The history table is branch-specific; the pattern table is shared by history value.

4.13 Perceptron Direction Prediction

Perceptron prediction replaces small counter tables with a simple linear classifier. For one static branch, let the history bits be encoded as

$$x_i = \begin{cases} +1, & \text{if history bit } i \text{ is taken} \\ -1, & \text{if history bit } i \text{ is not taken} \end{cases} \tag{4.15}$$

and let w_0 be a bias weight. The predictor forms

$$y = w_0 + \sum_{i=1}^{n} w_i x_i. \tag{4.16}$$

If $y \geq 0$, it predicts taken; otherwise it predicts not taken. After the real outcome is known, a common training rule is to update when the prediction was wrong or when the confidence $|y|$ is below a threshold θ. If the target value is $t = +1$ for taken and $t = -1$ for not taken, then the update is

$$w_0 \leftarrow w_0 + t, \qquad w_i \leftarrow w_i + t x_i. \tag{4.17}$$

The perceptron predictor can learn longer correlations than small counter tables, but it costs more arithmetic and storage per prediction.

Worked perceptron update Suppose the active perceptron has weights $[1, 2, -1, 0, 1]$, where the first value is the bias w_0, and suppose the current 4-bit global history is `1010`. Using taken $= +1$ and not taken $= -1$, that history becomes

$$x = [+1, -1, +1, -1].$$

The perceptron output is therefore

$$y = 1 + 2(+1) + (-1)(-1) + 0(+1) + 1(-1) = 3. \tag{4.18}$$

Because $y \geq 0$, the prediction is taken. If the true outcome is not taken, then the prediction is wrong and training must occur with target $t = -1$. The updated weights become

$$w_0 \leftarrow 1 - 1 = 0, \qquad [w_1, w_2, w_3, w_4] \leftarrow [2, -1, 0, 1] + [-1, +1, -1, +1] = [1, 0, -1, 2].$$

After the same outcome is shifted into the history register, the new GHR becomes `0100` if the newest bit is placed on the right.

4.14 Hybrid and Tournament Predictors

A *hybrid predictor* combines two or more component predictors and uses a chooser to decide which component to trust for a branch context. A common *tournament predictor* stores a small table of saturating counters indexed by branch PC or by a short hash of PC and history. When the chooser leans one way, the machine uses one predictor family; when it leans the other way, it uses the competing family.

The usual training rule is simple. If the component predictors disagree and one is correct while the other is wrong, the chooser is updated toward the correct component. If both are correct or both are wrong, the chooser often remains unchanged because that outcome did not distinguish which component was better for that context. This organization is useful when one predictor handles short repetitive patterns well while another captures longer correlations better. For example, a tournament design may arbitrate between a Yeh/Patt predictor and a perceptron predictor.

Predictor-quality metrics Two branch-prediction ratios appear repeatedly in simulator output. If N_{inst} is total dynamic instruction count, N_{branch} is dynamic branch count, and $N_{correct}$ is the number of correct branch predictions, then

$$f_{branch} = \frac{N_{branch}}{N_{inst}} \tag{4.19}$$

is the dynamic branch fraction and

$$Acc_{branch} = \frac{N_{correct}}{N_{branch}} = 1 - \frac{N_{mispred}}{N_{branch}} \tag{4.20}$$

is branch-prediction accuracy. These metrics answer different questions: branch fraction tells how often control-flow prediction matters, while prediction accuracy tells how often the predictor was correct when it mattered.

Prediction and update timing Prediction happens near fetch, but the true branch outcome becomes known only later when the branch resolves or retires. A pipelined predictor therefore cannot always train itself using only its *current* tables, because the history pattern, perceptron sum, or chooser state seen at update time may differ from the state that existed when the prediction was originally made. Practical hardware and simulators solve this by preserving a small prediction-time snapshot, such as the selected history entry or chooser index, until the branch is ready to update the predictor.

4.15 Timing-Diagram Interpretation

Timing diagrams are not just bookkeeping artifacts. They are often enough to infer what forwarding paths, register-file timing rules, and structural resources a pipeline must have. A reliable reading order is:

1. Identify true, anti-, and output dependences from the program text before looking at the diagram.
2. Mark every repeated stage occupancy in the diagram as a stall candidate.
3. For each stall candidate, decide whether the cause is data, structural, or control.
4. Infer which hardware feature would be required to remove that stall.

Common structural cues If one instruction is fetching while another is accessing data memory in the same cycle without conflict, then the design must have either separate instruction and data paths or enough memory-porting to support both operations. If fetch pauses exactly when a load or store reaches MEM, a shared memory path is the likely cause.

Common forwarding cues If a dependent ALU instruction reaches EX before the producer reaches WB, some bypass path must exist. If a load-use pair still suffers one bubble but longer ALU-to-ALU chains do not, the machine likely has EX/MEM-style forwarding but still cannot make load data available early enough for the immediately following EX stage. If a consumer in ID proceeds only when the producer is in WB, the design may be relying on WB-to-ID forwarding or on same-cycle register-file write-then-read timing rather than on an earlier bypass path.

Register-file timing cue Suppose a producer writes back in cycle k and a dependent consumer reads registers in ID in that same cycle k. If the consumer does not stall further, then the machine must support either same-cycle write-before-read behavior or an equivalent WB-stage bypass into decode. If the consumer waits one more cycle, then neither mechanism is present.

Control-hazard cue Wrong-path instructions that appear briefly and are then crossed out or squashed indicate speculative fetch. A gap after a branch with no wrong-path work suggests conservative branch resolution rather than aggressive prediction. Likely-bit or squash-slot behavior is visible when an instruction after the branch is fetched speculatively but only conditionally allowed to complete.

A fuller checklist remains in Appendix A, but these rules are the minimum set needed to infer pipeline structure from the kinds of timing traces used throughout this text.

4.16 Chapter 4 Summary

Chapter 4 frames pipelining as throughput engineering with explicit hazard management. The five-stage model clarifies stage roles (IF, ID, EX, MEM, WB), while speedup equations show why ideal gains are reduced by latch overhead and stalls. CPI is modeled as $1 + \text{stalls}_{avg}$, with stalls explained by the interaction of program behavior and hazard structure. Structural hazards are reduced through resource provisioning; control hazards are addressed through delay/squash semantics and prediction structures. BTBs predict branch targets, RAS handles returns efficiently, and direction prediction determines whether control transfer occurs, with software heuristics like backward taken/forward not taken and simple baselines like always taken providing reference points. Simple dynamic predictors such as one-bit prediction and Smith two-bit counters add robustness with very little hardware, while history-based predictors such as GShare, GSelect, and Yeh/Patt, neural-style schemes such as perceptrons, and hybrid tournament predictors refine that prediction further by learning correlations across prior branch outcomes. Because prediction and training happen at different times in the pipeline, practical predictors must also preserve enough prediction-time context to update correctly later. Timing diagrams then tie the whole chapter together by exposing which hazards, forwarding paths, and same-cycle register-file behaviors a concrete pipeline must implement. Together, these ideas define practical pipeline performance analysis and design.

4.17 Practice Questions

The following questions reinforce the main ideas of Chapter 4 and are intended for self-checking after the summary.

1. An unpipelined design has $CPI = 5$. A new implementation uses five pipeline stages. What is the maximum possible speedup if the pipelined clock period is unchanged? Then compute the new CPI if 25% of instructions suffer a single-cycle stall.

2. A scalar pipeline has base $CPI = 1$. Structural hazards add an average of 0.05×1 cycles per instruction, load-use hazards add 0.12×1, and branch mispredictions add 0.15×2. Compute the final CPI.

3. For the sequence below, list the RAW, WAR, and WAW dependences. Then identify which forwarding path is most important for reducing stalls.

```
LW R1, 0(R2)
ADD R3, R1, R4
SUB R5, R3, R6
SW R5, 0(R7)
```

4. A timing diagram shows that instruction fetch continues in the same cycle that another instruction accesses data memory, that a dependent ALU consumer reaches EX before the producer reaches WB, and that a decode-stage consumer can read a value in the same cycle an older instruction writes it back. What does each of these observations imply about the hardware organization?

5. Explain the distinct roles of the branch target buffer (BTB), return-address stack (RAS), and direction predictor. Why does a BTB remain useful even when the branch direction is trivial?

6. A BTB is indexed by `PC[7:4]`. The return-address stack is initially empty. Simulate this short instruction stream: `0x4000: CALL 0x4A00`, `0x4A00: BREQZ R1,0x4AF0` (taken), `0x4AF8: RET`, `0x4004: ADD`. State the BTB effect of each control instruction and the final contents of the top of the RAS.

7. A GSelect predictor uses a 2-bit global history register and indexes the predictor table with `PC[2]` and the current GHR. The row for `PC[2]=0` is `[01,10,11,00]` for GHR `00,01,10,11`, the row for `PC[2]=1` is `[10,01,00,11]`, and the initial GHR is `01`. Simulate the following two branches: `(0x2004,T)` and `(0x2008,N)`. For each, give the prediction, the updated counter, and the new GHR.

8. A Yeh/Patt predictor uses `PC[3:2]` to index a history table, and each history register is 2 bits wide. Explain how the history-table lookup and the pattern-table lookup differ from the indexing rule used in GSelect. Then state one situation in which Yeh/Patt can distinguish branches that simple global-history indexing tends to alias together.

9. A perceptron predictor uses $G = 4$ bits of global history. The active perceptron has weights $[1, 2, -1, 0, 1]$, the current GHR is `1010`, and the next actual branch outcome is not taken. Compute the output y, state the prediction, determine whether training occurs, and give the new GHR after update. You may assume the project-style threshold rule $\theta = \lfloor 1.93G + 14 \rfloor$.

Chapter 5

Integrated Synthesis and Cache Simulation Concepts

5.1 Chapter Purpose

This chapter draws together the main ideas from Chapters 1–4. The goal is not to repeat those chapters in full, but to show how performance metrics, hazards, memory behavior, and simulation-oriented abstractions fit into a single analysis framework.

5.2 Integrated Themes from Chapters 1 and 2

The first two chapters establish the measurement framework used throughout the rest of the book. An architectural change should first be classified by what it is trying to improve: parallelism, speculation, or some combination of the two. Its performance impact must then be evaluated with the correct metric, usually runtime for one job or throughput for a stream of jobs. The central CPU-time equation,

$$\text{CPU time} = \text{IC} \times \text{CPI} \times \text{CT},$$

turns that reasoning into a concrete tradeoff among instruction count, pipeline efficiency, and clock period. Within that framework, Amdahl's law bounds fixed-size speedup, Gustafson's law explains scaled-size behavior, and the choice between arithmetic and harmonic means determines whether a summary is mathematically meaningful.

Worked check Consider an optimization that accelerates 30% of runtime by 2×. The total speedup is

$$S = \frac{1}{(1-0.3) + \frac{0.3}{2}} = \frac{1}{0.85} \approx 1.176. \tag{5.1}$$

The unaccelerated fraction remains in the denominator and bounds overall speedup.

Worked check: maximum possible speedup Consider a mechanism that accelerates 25% of the original benchmark runtime and makes that portion arbitrarily fast. The maximum total speedup is

$$S_{\max} = \frac{1}{1-0.25} = \frac{4}{3} \approx 1.33. \tag{5.2}$$

This is the standard way to interpret the *maximum* speedup of a partial optimization.

5.3 Integrated Themes from Chapter 3

Chapter 3 adds the memory-hierarchy side of the performance model. That means translating a cache specification such as (C, B, S) into sets, ways, and tag/index/offset fields; distinguishing compulsory, conflict, and capacity misses; and understanding how replacement and write policies alter observed traffic. It also means using

$$AAT = HT + MR \cdot MP$$

as a compact summary of hierarchy behavior while remembering that cache size, block size, and associativity can influence all three terms in different ways.

Worked check Consider a cache with $HT = 1.8$ ns, $MR = 0.06$, and $MP = 28$ ns (Eq. (5.3)):

$$AAT = 1.8 + 0.06 \cdot 28 = 3.48 \text{ ns}. \tag{5.3}$$

Worked check: repeated trace reasoning Consider a direct-mapped L1 with $(C, B, S) = (8, 4, 0)$, write-back/write-allocate, and the repeated access sequence

$$R\ 0x41F1,\ R\ 0xB54A,\ W\ 0x4204,\ W\ 0xCA3B,\ R\ 0x420E,\ W\ 0xB537. \tag{5.4}$$

With $B = 4$, the six accesses touch the five blocks $\{0x41F, 0xB54, 0x420, 0xCA3, 0xB53\}$ because $0x4204$ and $0x420E$ lie in the same block. The first iteration therefore has 5 compulsory misses. The key structural fact is that blocks $0xCA3$ and $0xB53$ map to the same direct-mapped set, so after warm-up the sequence continues to suffer two conflict misses per iteration. Over iterations 2–10, that produces 18 additional misses. Those 18 misses are the non-compulsory misses for the full run, and because the conflicting writes repeatedly evict dirty lines, the full sequence produces 19 writebacks if dirty lines are *not* flushed after the final access.

Victim cache and +1 prefetch contrast That same trace separates two common remedies cleanly. A small victim cache can eliminate the recurring two-block ping-pong at one set; after warm-up, the combined L1+victim-cache miss count can drop to zero because each conflicting block is recovered from the victim structure instead of DRAM. By contrast, a +1 prefetcher is not automatically a cure. In this trace it can prefetch block $0x420$ early enough to remove one first-iteration demand miss, but it does not remove the recurring conflict between $0xCA3$ and $0xB53$. This is the key design lesson of the trace: next-block prefetching and victim caching address different miss mechanisms.

5.4 Integrated Themes from Chapter 4

Chapter 4 connects pipeline structure to observed performance. Reading a timing diagram is therefore an exercise in inference: the reader is reconstructing hazards, bypass paths, memory behavior, and branch-handling policy from visible stage occupancy. The same chapter also introduces the progression from static branch handling to dynamic predictors such as always taken, one-bit and Smith-counter schemes, GShare, GSelect, Yeh/Patt, perceptrons, and hybrid tournament predictors.

Pipeline Speedup

Assume a 5-stage design where each stage has 250 ps logic delay and pipeline-register overhead is 30 ps. Unpipelined instruction time is $5 \times 250 = 1250$ ps. Pipelined clock is $250 + 30 = 280$ ps. Ideal long-run speedup is (Eq. (5.5)):

$$S \approx \frac{1250}{280} \approx 4.46. \tag{5.5}$$

This is below 5 because register overhead is non-zero.

CPI with Branch Stalls

If branch frequency is 18% and each branch contributes an average of 1.6 stall cycles (Eq. (5.6)):

$$CPI = 1 + 0.18 \times 1.6 = 1.288. \tag{5.6}$$

This illustrates how even moderate branch penalties raise CPI above the ideal value of 1.

Integrated CPI Accounting

Suppose a scalar core has base $CPI = 1$. Loads are 30% of all instructions, the L1 data-cache miss ratio on loads is 5%, and each such miss adds 20 cycles. Branches are 18% of all instructions, the misprediction rate is 10%, and each misprediction adds 2 cycles. Then the added CPI from load misses is

$$0.30 \times 0.05 \times 20 = 0.30, \tag{5.7}$$

while the added CPI from branch mispredictions is

$$0.18 \times 0.10 \times 2 = 0.036. \tag{5.8}$$

So the total CPI estimate is

$$CPI \approx 1 + 0.30 + 0.036 = 1.336. \tag{5.9}$$

This kind of combined accounting is the most useful way to compare bottlenecks: all stall mechanisms are translated into the common unit of cycles per retired instruction.

Software Branch Handling

When a branch has delay slots, every instruction moved into those slots must be safe regardless of whether the branch is taken. When a branch instead has squash slots under an "assume taken" policy, the compiler may place instructions from the predicted taken path there, because wrong-path entries will be canceled. The resulting placement rule is:

1. fill delay slots only with path-independent instructions,
2. fill squash slots from the predicted path when profitable, and
3. leave `NOP`s when correctness would otherwise be uncertain.

5.5 Cache Simulation Conceptual Model

The cache simulation framework models hierarchy behavior at the level of addresses, tags, and policy events rather than payload data values. A common parameterization is (Eq. (5.10)):

$$(C_1, B, S_1, C_2, S_2, \{\text{MIP,LIP}\}, \{\text{None,+1,Markov,Hybrid}\}, r). \tag{5.10}$$

In this notation, C controls capacity (2^C bytes), B controls block size (2^B bytes), and S controls associativity (2^S ways). Here *most recently used insertion policy (MIP)* and *least recently used insertion policy (LIP)* are common insertion-policy options.

5.6 Hierarchy Semantics in the Model

The hierarchy model treats L1 accesses separately from L2 internals, while L2 requests are viewed as arriving from an upper-level requester. This separation makes it easier to reason about statistics at each level without conflating local policy with downstream timing.

Write strategy also changes the meaning of those statistics. *Write-back, write-allocate (WBWA)* behavior in L1 emphasizes dirty-state tracking and deferred propagation, whereas *write-through, write-no-allocate (WTWNA)* behavior in L2 emphasizes immediate lower-level visibility and no allocation on write misses.

The ordering of read-miss repair and writeback events can also matter. Different servicing orders change replacement-state evolution and therefore alter downstream hit, miss, and writeback counts. If a lower level is disabled in the model, it is often best understood as an always-miss transit layer whose contribution to AAT is only the cost of passing requests to the next active level.

5.7 Prefetching Concepts

+1 *prefetching* predicts the next sequential block ($X + 1$) after a demand miss on block X. *Markov prefetching* instead models miss-stream transitions as a finite-state table, associates each source miss block with a bounded set of likely successors and their frequencies, and selects a prediction from the highest-frequency successor under a deterministic tie-break rule. *Hybrid prefetching* combines correlation-based prediction with sequential fallback when correlation information is absent.

To interpret prefetch behavior correctly, the simulator must also distinguish demand-filled lines from prefetched lines. A prefetch metadata bit provides that distinction. The first demand use of a prefetched line contributes to prefetch-hit interpretation, while eviction before first demand use contributes to prefetch-miss interpretation.

Worked prefetch-counter interpretation Suppose a simulator reports 120 prefetched lines, 70 prefetch hits, and 25 prefetched lines evicted before first use. The 70 prefetch hits show that many speculative fills were in fact useful: later demand accesses avoided ordinary misses because of earlier prefetching. The 25 unused evictions show that some prefetched lines consumed bandwidth and space without helping demand accesses. The remaining prefetched lines are ambiguous without more context, because some may still be resident and not yet demanded. This is why prefetch usefulness is not judged by one counter alone; both benefit and waste must be read together.

5.8 Statistics and AAT Interpretation

Typical hierarchy statistics include access counts, hit/miss counts, hit/miss ratios, writeback counts, and prefetch effectiveness counters. In two-level analysis (Eq. (5.11)):

$$AAT_{L1} = HT_{L1} + MR_{L1} \cdot MP_{L1}, \qquad MP_{L1} = AAT_{L2}. \tag{5.11}$$

$$AAT_{L2} = HT_{L2} + MR_{L2,read} \cdot DRAM_TIME. \tag{5.12}$$

$$DRAM_TIME = DRAM_AT + DRAM_AT_PER_WORD \cdot WORDS_PER_BLOCK. \tag{5.13}$$

This form emphasizes that higher-level penalty is the lower-level average access behavior, not a fixed constant.

5.9 Conceptual Tradeoffs Across Parameters

Increasing C generally reduces capacity misses but tends to raise hit time. Increasing S reduces conflict misses but increases lookup complexity. Increasing B can exploit spatial locality but may also increase pollution and miss penalty. More aggressive prefetching can reduce compulsory misses yet increase bandwidth pressure and useless fills.

5.10 Chapter 5 Summary

Chapter 5 draws together the earlier performance, pipeline, and memory ideas into one simulation-oriented view. Pipeline timing and cache behavior are not separate topics; they are two views of the same broader question of where cycles are spent and why. The simulation concepts in this chapter provide a concrete way to observe those tradeoffs in controlled workloads. That same simulator-oriented reasoning extends naturally to branch-prediction accuracy, fetched-versus-retired instruction counts, queue pressure, and reorder-buffer pressure, which are developed further in Chapter 6. A grouped formula index for the full text is provided in Appendix B.

5.11 Practice Questions

The following questions reinforce the integrated ideas of Chapter 5 and are intended for self-checking after the summary.

1. A hierarchy reports $HT_{L1} = 1$, $MR_{L1} = 0.10$, $HT_{L2} = 7$, $MR_{L2,read} = 0.20$, and $DRAM_TIME = 80$. Compute AAT_{L2} and AAT_{L1}. Then explain why the L1 miss penalty is modeled as the lower-level *average* access time rather than as a fixed constant.

2. A simulator reports 120 prefetched lines, 70 prefetch hits, and 25 prefetched lines evicted before first use. What do these three counters say about prefetch usefulness and prefetch waste? What additional information would you want before concluding that the prefetcher is a net win?

3. In a hierarchy that uses write-through, write-no-allocate behavior at the lower level, why is it often more meaningful to report the lower-level miss ratio over *reads only* rather than over all accesses?

4. Two cache configurations have the same capacity and associativity. Configuration A has lower hit time but higher miss ratio. Configuration B has higher hit time but lower miss ratio. Explain why the better design cannot be chosen correctly without computing or estimating average access time.

5. Explain why the servicing order of read-miss repair and writeback traffic can change later hierarchy statistics even when the high-level policy choices are unchanged.

6. A scalar core has base $CPI = 1$. Loads are 30% of all instructions, the L1 data-cache miss ratio on loads is 5%, and each such miss adds 20 stall cycles. Branches are 18% of all instructions, the branch misprediction rate is 10%, and each misprediction adds 2 cycles. Estimate the total CPI and identify which source of stalls contributes more.

Chapter 6

Dynamic ILP: Superscalar Scheduling and Out-of-Order Execution

6.1 Why Cycles per Instruction Can Go Below One

In basic single-issue pipelines, designers often treat cycles per instruction (CPI) near 1 as a practical lower bound. Superscalar execution challenges that assumption by allowing multiple instructions to make progress in the same cycle. This broader goal is usually described as instruction-level parallelism (ILP). In this context, instructions per cycle (IPC) is the natural companion metric, as shown in Eq. (6.1):

$$IPC = \frac{1}{CPI}. \tag{6.1}$$

So achieving $CPI < 1$ is exactly equivalent to achieving $IPC > 1$. A useful way to view design limits is that average retire throughput cannot exceed the average throughput of the stages before it (Eq. (6.2)):

$$IPC_{retire} \leq \min(IPC_{fetch},\ IPC_{dispatch},\ IPC_{execute},\ IPC_{state\ update})\,. \tag{6.2}$$

If any major stage can only handle one instruction per cycle on average, then the whole machine cannot retire two instructions per cycle on average, even if another stage is wide enough to do so. In other words, the minimum term in Eq. (6.2) is the throughput bottleneck.

Each term corresponds to a different possible limit. IPC_{fetch} asks how many useful instructions the front end can supply per cycle. $IPC_{dispatch}$ asks how many decoded instructions can be prepared and sent onward. $IPC_{execute}$ asks how many instructions the function units (FUs) and memory system can actually complete. $IPC_{state\ update}$ asks how quickly the machine can accept completed work into precise architectural state. The retire rate cannot exceed the weakest of these rates, because retirement is the final visible outcome of all the earlier pipeline activity.

6.2 Historical Motivation

Historically, supercomputer scalar units showed that out-of-order mechanisms could deliver $IPC > 1$ long before that became mainstream in commodity processors. The central lesson is that high throughput comes from combining several mechanisms rather than relying on any one of them in isolation. Stalls are reduced through coordinated fetch, scheduling, execution, and retirement policies. Modern superscalar designs extend the same idea by combining hazard management with wider issue and stronger speculation.

6.3 Superscalar Pipeline View

A practical superscalar model can be described with five main logical units plus a final retirement or commit action. Figure 6.1 provides a compact visual summary of how these units interact in the overall datapath. In that figure, the boxed labels denote the main structures or stages, while the smaller labels on the right identify the dominant action on the adjacent arrow bundle: fetch, dispatch, fire or issue, complete, and retire or commit. The front end is drawn as separate Branch Handling and Instruction Fetch boxes, but together they still represent the single logical instruction fetch (IF) unit described below.

1. Branch Handling and Instruction Fetch (IF) This unit performs instruction fetch (IF) while consulting branch-handling structures such as direction predictors, branch-target mechanisms, and return-address prediction. Its goal is to produce multiple candidate instructions per cycle.

2. Dispatch and Decode (DISP) This unit decodes incoming instructions in program order and dispatches them toward the scheduling structures. Dispatch is where program order is still explicit, which matters for precise recovery.

3. Scheduling Scheduling assigns ready instructions to available FUs in time. This is where execution can diverge from program order to exploit parallelism.

4. Execute (EX) FUs perform arithmetic, memory, and control operations with operation-specific latencies. Multiple FUs allow parallel progress.

5. State Update (SU) State update gathers completed results, handles squash or recovery, and prepares instructions for precise architectural completion. This unit supports squash/recovery when speculation is wrong. The final outgoing flow from this unit is the retirement or commit action, where results become architecturally visible in program order. In Fig. 6.1, retirement is represented by the final outgoing arrows below State Update rather than by a separate boxed stage.

6.4 Dependency Types and Why They Matter

Superscalar scheduling must respect three dependency classes.

- *read-after-write (RAW)* is a true flow dependency. A later instruction needs a value produced by an earlier one, so executing them in the wrong order would use stale data and produce the wrong result.
- *write-after-read (WAR)* is an anti-dependency. A later instruction must not overwrite a register before an earlier instruction has finished reading it, even though the two instructions do not share a true data value.
- *write-after-write (WAW)* is an output dependency. Two instructions write the same destination, so their writes must appear in the correct order or the final architectural value will be wrong.

Only RAW is fundamentally unavoidable. WAR and WAW are name dependencies that arise from register reuse and can be reduced or removed by renaming the register.

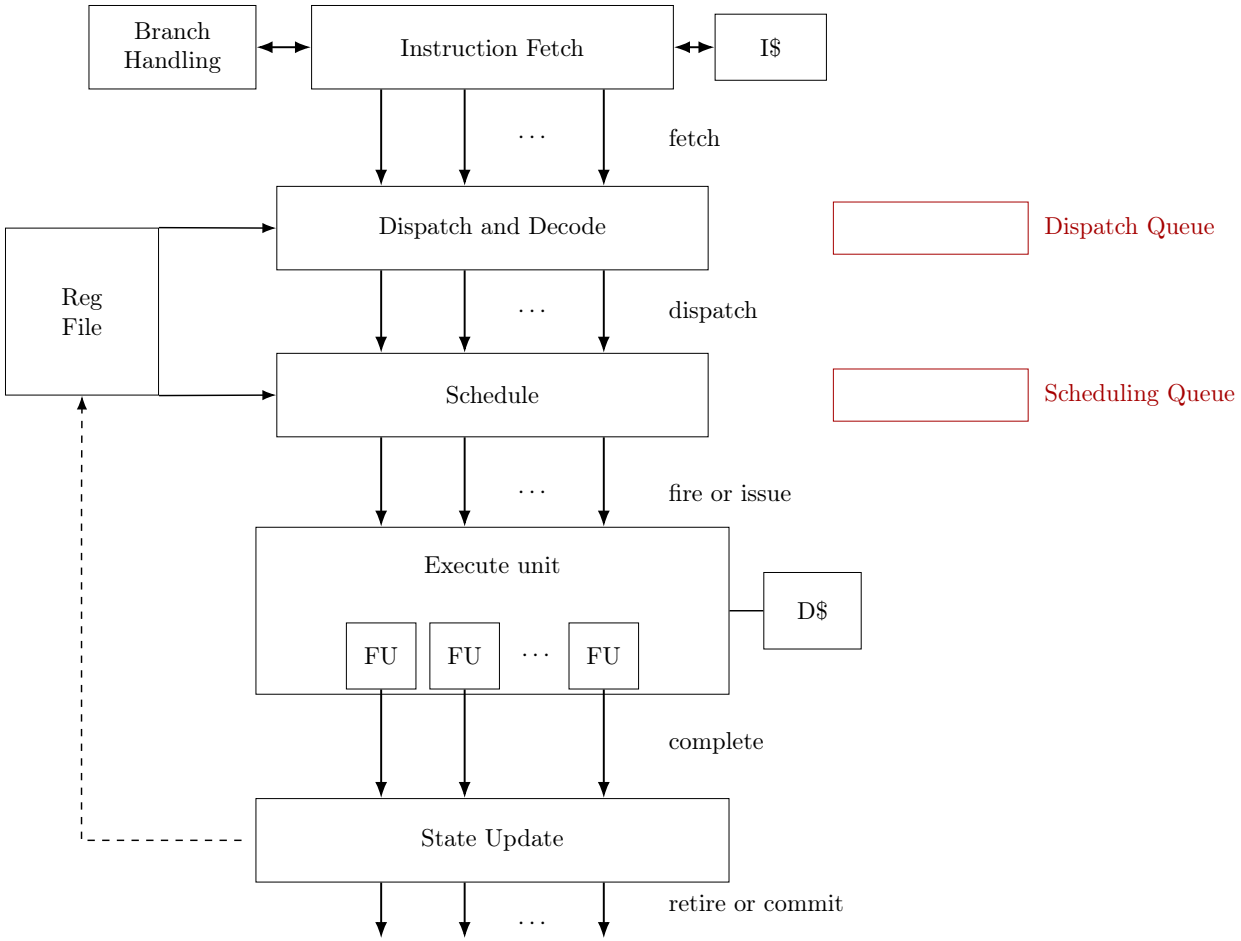


Figure 6.1: Logical superscalar pipeline view illustrating one representative organization. The figure is not specific to FICI, FICO, or FOCO; those classifications depend on the firing and completion policy. Boxed elements denote major structures or stages, while the right-side annotations denote the adjacent flow actions.

6.5 Fire/Complete Taxonomy

A compact way to classify designs is by fire order and complete order.

- *fire in order, complete in order (FICI)* is the classic simple-pipeline behavior. Instructions begin execution in program order and also finish in program order, which simplifies control but limits how much parallelism the machine can exploit.

- *fire in order, complete out of order (FICO)* is a simpler superscalar step. Instructions still begin execution in order, but different latencies allow later instructions to finish first, so some parallel speedup is possible without fully out-of-order scheduling.

- *fire out of order, complete out of order (FOCO)* is more aggressive and offers higher potential ILP. The scheduler may start and finish instructions whenever operands and hardware are ready, which exposes more ILP but requires more machinery to preserve correct architectural behavior.

Type	Typical latency	Why
Floating-point (FP) multiply	3	Denormalize, multiply, renormalize.
FP add	2	Denormalize and add, then renormalize.
Load	2	Typical cache pipeline behavior.
Store	1	Write buffering, or a load/store buffer, can hide much of the visible latency.
Divide	Variable	Often implemented with unpipelined iterative methods such as Newton iteration, and many simplified models treat divide as too rare to emphasize.
ALU	1	Most common operation, so short latency is especially valuable.

Table 6.1: Typical simplified function-unit latencies used in superscalar examples.

6.6 Scoreboarding Intuition (FICO Style)

A scoreboard tracks whether source operands, destinations, and required function units are currently available. A simplified FICO-style fire condition is shown in Eq. (6.3).

$$\text{Fire if } \neg busy(src_1) \ \wedge \ \neg busy(src_2) \ \wedge \ \neg busy(dest) \ \wedge \ \neg busy(FU_{needed}). \tag{6.3}$$

When an instruction fires, destination and function-unit busy state are set. When the instruction completes, those busy states are cleared according to the implementation timing rules. The result is correct execution with moderate hardware complexity, but some parallel opportunities remain unused compared with more aggressive out-of-order firing.

6.7 Function Units and Latency

Superscalar benefit depends on the latency mix and on the number of available units. In many designs, arithmetic logic unit (ALU) operations have short latency, multiply operations take more cycles, and memory accesses vary widely depending on cache behavior. The key point is that latency and throughput are related but not identical. An operation may have a long latency and still support good throughput if the corresponding FU is pipelined and can accept a new instruction every cycle. By contrast, a long-latency operation on a scarce or non-pipelined FU can create backpressure quickly, because younger ready instructions may have no place to execute. Memory operations are especially important here. A cache hit may behave like a modest fixed-latency operation, while a cache miss can hold resources for many extra cycles and expose whether the machine can continue doing useful independent work during the delay.

Unit count matters for the same reason. If the machine has only one ALU, one multiplier, and one load/store path, the scheduler has few choices. Adding more execution resources, including additional FUs such as ALUs, multipliers, or more load/store capacity, gives the scheduler more flexibility to overlap different instruction classes and reduce structural conflicts. Designers therefore optimize both unit availability and scheduling policy, not just raw clock frequency. The practical goal is not merely to make individual operations fast, but to ensure that the mix of latencies, FU counts, and pipeline structure supports high average IPC on realistic workloads. A common simplified latency model is shown in Table 6.1. In this simplified performance model, the exact

Instr.	Operation	0	1	2	3	4	5	6	7	8	9
A	ADD $R1, R2, R3$	IF	DISP	EX	SU						
B	MUL $R5, R1, R4$	IF	DISP	DISP	EX	EX	EX	SU			
C	ADD $R3, R1, R2$		IF	DISP	EX	SU					
D	ADD $R1, R5, R4$		IF	DISP	DISP	DISP	DISP	EX	SU		
E	MUL $R6, R7, R8$			IF	DISP	DISP	DISP	EX	EX	EX	SU
F	ADD $R1, R3, R3$			IF	DISP	DISP	DISP	DISP	EX	SU	
G	ADD $R4, R3, R5$				IF	DISP	DISP	DISP	DISP	EX	SU

Table 6.2: Timing table for a simplified FICO schedule. Repeated DISP entries indicate that an instruction remains blocked in the in-order dispatch-and-fire path while waiting for data or hardware to become available.

cycle count for divide is not emphasized because divide is used relatively rarely. In real designs, however, divide often has high latency and is less deeply pipelined than common ALU operations.

6.8 Worked FICO Timing Example

The following seven-instruction sequence illustrates the scoreboard idea in a concrete schedule:

$$
\begin{aligned}
A &: \texttt{ADD}\ R1, R2, R3 \\
B &: \texttt{MUL}\ R5, R1, R4 \\
C &: \texttt{ADD}\ R3, R1, R2 \\
D &: \texttt{ADD}\ R1, R5, R4 \\
E &: \texttt{MUL}\ R6, R7, R8 \\
F &: \texttt{ADD}\ R1, R3, R3 \\
G &: \texttt{ADD}\ R4, R3, R5
\end{aligned}
$$

Assume a fetch rate of two instructions per cycle, one ALU with latency one cycle, one multiplier with latency three cycles, and a simplified *state update (SU)* stage used to make completion visible. Table 6.2 shows one compact timing schedule for this example. The stage labels are *instruction fetch (IF)*, *dispatch (DISP)*, *execute (EX)*, and *state update (SU)*. Repeated DISP entries indicate that the instruction is blocked in the in-order fire logic, not that it is being redispatched from scratch.

Walkthrough Cycles 0–3 establish the basic pattern. A and B are fetched together, but B cannot leave DISP immediately because it needs the value of $R1$ from A. Once A executes, B can move forward, and C can also reach EX shortly afterward because it depends on the same produced value. This is why B spends an extra cycle in DISP in Table 6.2: the machine is still firing instructions in program order, so the scoreboard must wait until A has released the needed state.

Cycles 4–6 expose the main weakness of FICO. D is genuinely blocked because it needs $R5$ from B, so its repeated DISP entries reflect a real RAW dependence. E, however, is independent of D and could use the multiplier, yet it still cannot fire early because the machine will not let younger instructions pass an older blocked instruction in the dispatch path. That is the key limitation of fire-in-order scheduling: some stalls come from true dependencies, but others come from the policy itself.

Cycles 7–9 show both overlap and continued restrictions. E eventually occupies the multiplier while F and G reach the ALU later, so the machine does achieve some parallel execution and some out-of-order completion. At the same time, the single ALU and single multiplier keep the scheduler's choices narrow, and the in-order fire rule prevents the machine from exploiting all available independence. The example therefore illustrates why FICO is attractive as a first superscalar step—it is much simpler than full FOCO—but also why more aggressive out-of-order firing and renaming are needed to capture more ILP.

6.9 Register Renaming as the Next Step

Register renaming is the introduction of additional physical register identities so later writes do not falsely block earlier reads or writes. Conceptually, renaming transforms many WAR/WAW hazards into non-hazards while preserving program meaning. The dependency pressure that remains is primarily RAW, which reflects true data flow. In a typical implementation, the registers named by the instruction set architecture (ISA) are the *architected registers*, while the larger hidden pool used internally by the machine is the *physical register file*. The architected registers are the names the programmer sees in assembly language and that the ISA defines as part of the visible machine state. The physical register file contains additional hidden storage used internally by the processor to hold renamed versions of those values.

This distinction matters because several in-flight instructions may all refer to the same architected destination at different moments in time. Rather than repeatedly reusing one architectural name and creating false WAR/WAW conflicts, the machine can assign a different physical register to each new write while still preserving the illusion of a small fixed architected register set. Older instructions can continue reading the older physical version, while younger instructions are redirected to the newer one. A *register alias table (RAT)* maps each architected register name to the physical register that currently holds its most recent value. When dispatch renames a destination, it allocates a fresh physical register and updates the RAT so younger instructions will refer to the new location rather than the older one. Ready bits in the physical register file then indicate whether each renamed value has actually been produced yet. Some architectures describe the same underlying status with *busy bits* instead. The terminology varies, but the idea is the same: one convention marks a value as ready to use, while the other marks it as still busy or unavailable.

6.10 Tomasulo's Algorithm

Tomasulo's algorithm is a classic out-of-order execution method that implements *FOCO* by allowing instructions to fire when their operands are ready rather than strictly in program order. Its central idea is to combine renaming with *reservation stations*, which are scheduling entries that hold an instruction until the required operands and execution resources are available. Historically, this approach is associated with Bob Tomasulo's work on the IBM System/360 Model 91, and it remains the conceptual foundation behind many modern out-of-order schedulers.

Main structures A simplified Tomasulo-style machine contains a dispatch unit, a set of reservation stations, one or more FUs, a result-broadcast network, and a renamed register file. Dispatch still runs in program order, but once instructions occupy reservation stations they may wake up and fire out of order. Each reservation-station entry records the required FU type, the destination tag or physical register, and for each source either a ready value or a tag identifying which older instruction will eventually produce that value.

Instruction / event	**0**	**1**	**2**	**3**	**4**	**5**	**6**
A	**DISP** Dispatch	**EX** Fire	**EX** Miss wait	**EX** Miss wait	**EX** Miss wait	**EX** Miss wait	**CDB** Complete
B	**DISP** Dispatch	**RS** Wait on A	**RS** Wait on A	**RS** Wait on A	**RS** Wait on A	**RS** Wait on A	**RS** Ready after A
C	**DISP** Dispatch	**EX** Fire	**EX** Hit wait	**EX** Hit wait	**CDB** Complete		
D	**DISP** Dispatch	**RS** Wait on C	**RS** Wait on C	**RS** Wait on C	**RS** Wake on C	**EX** Fire	**CDB** Complete after A
CDB					**CDB** C		**CDB** A then D

Figure 6.2: Compact cycle table for the four-instruction Tomasulo trace. The table emphasizes logical ordering rather than exact micro-timing details.

Dispatch, fire, and complete At dispatch time, each source operand is examined. If the needed value is already ready, the value is copied into the reservation station. If it is not ready, the reservation station records the producing tag instead. The destination is renamed so younger instructions refer to the new producer rather than to the older architectural name. Later, when both sources are ready and a matching FU is available, the instruction can fire even if an older unrelated instruction is still waiting somewhere else in the machine. Completion is a separate event from firing: when an instruction completes, it broadcasts its tag and value on the common data bus (CDB) so dependent instructions can wake up and the renamed register file can be updated.

Illustrative four-instruction trace The following short trace captures the key behavior. Instruction A is a load that misses in the cache. Instruction B is an add that depends on A. Instruction C is an independent load that hits in the cache. Instruction D is an add that depends on C. Because the machine is out of order and the cache is non-blocking, C and then D can make progress while A is still waiting on the miss. Figure 6.2 gives a compact cycle-by-cycle reference for the same sequence before the text walks through it in words. Each cell shows the dominant structure label first—dispatch (DISP), reservation station (RS), execute (EX), or common data bus (CDB)—followed by the local action.

Cycle 0. A, B, C, and D dispatch into reservation stations. Sources that are already available are copied in as values; missing sources are recorded as tags. Destinations are renamed, and the corresponding destination entries are marked not ready.

Cycle 1. A fires, and independent instruction C can also fire because it does not depend on A. This is the key gain from out-of-order scheduling: an older miss does not automatically freeze younger independent work.

Cycles 2–3. Nothing new completes. B is still waiting for A, and D is still waiting for C.

Cycle 4. C completes first and broadcasts its tag and value on the common data bus (CDB). Reservation stations compare that tag against all waiting operands, so D wakes up and becomes ready to fire.

Cycle 5. D fires and begins its add using the value from C.

Cycle 6. A finally completes its miss and D also wants to complete. With only one CDB, both cannot broadcast at once, so the usual rule is to let the older instruction win. A therefore broadcasts first, and B immediately becomes ready. D then broadcasts its own result, and B can fire as soon as the machine allows post-broadcast wakeup.

This trace shows why it is important to distinguish *fire* from *complete*. Firing sends operands into an FU and begins execution. Completing publishes the result so the rest of the machine can observe it.

Non-blocking caches This Tomasulo-style example also motivates a *non-blocking cache*. Such a cache is often described as supporting *hit under miss*. If A misses in the cache, the machine does not necessarily want every younger memory access to stop. Instead, independent cache accesses may continue to probe the cache pipeline. That allows C to hit and complete while A is still waiting on a lower-level response. That improves throughput, but it also makes control more complex because the cache and scheduler must now track multiple outstanding requests correctly.

6.11 Result Buses and Physical Register Files

Basic Tomasulo has an obvious bottleneck: if all completed instructions must use one common data bus (CDB), then only one completion can be broadcast in a cycle. That limitation can be summarized as Eq. (6.4).

$$IPC_{complete} \leq N_{result\ buses}. \tag{6.4}$$

Adding more result buses raises the completion ceiling, but it also increases wakeup complexity because each reservation station must compare against more simultaneous broadcasts.

One historical way to manage this cost is to redesign Tomasulo around an explicit physical register file and a RAT rather than storing all transient values only in reservation-station entries. With physical registers, the scheduler can often replace some associative tag searches with direct lookup by physical register number. That is faster and usually lower power than asking every waiting instruction to compare against every possible producer all the time.

The tradeoff is a heavier register file. If many reservation-station entries are checked each cycle, the physical register file may need substantial read bandwidth. A common first-order estimate is shown in Eq. (6.5).

$$N_{read\ ports} \approx 2N_{sched}, \qquad N_{write\ ports} \approx N_{result\ buses}. \tag{6.5}$$

The factor of two on the read side comes from the common case of two source operands per instruction. Real designs use banking, clustering, bypass paths, operand caching, and other tricks to reduce this raw multi-port cost, but the estimate is a useful baseline for understanding why renamed out-of-order cores are expensive.

6.12 RATs, Ready Bits, and Dispatch in Renamed Machines

Once the machine uses an explicit physical register file, dispatch must translate instruction set architecture (ISA) register names into physical register numbers. That basic renaming step is summarized by Eq. (6.6).

$$P_{src1} = RAT[A_{src1}], \qquad P_{src2} = RAT[A_{src2}], \qquad RAT[A_{dest}] \leftarrow P_{free}. \tag{6.6}$$

The first two mappings identify the current physical locations of the source values. The third mapping allocates a fresh physical destination so younger instructions depend on the new producer rather than on an older architectural name.

In this style of design, the fire rule is usually expressed with *ready bits* on physical registers rather than with *busy bits* on architected registers. The same logical condition is shown in Eq. (6.7).

$$fire_i \Leftarrow ready(P_{src1}) \wedge ready(P_{src2}) \wedge free(FU_i). \tag{6.7}$$

Each scheduling-queue entry is a reservation station that records the FU type, the destination physical register or tag, and either ready source values or the physical-register identities that will eventually supply them. When a producing instruction completes, the result bus updates the physical register file and wakes any waiting entries that need that value.

The speculative physical-register state may be called the *future file* or the *messy register file*. In either case, the term refers to a speculative state that may be ahead of the last coherent architectural state. The key idea is the same: the speculative state can move ahead aggressively, while the committed state must remain precise.

6.13 Precise State and Recovery

Aggressive out-of-order execution creates a second problem beyond scheduling: the machine may complete instructions in an order that does not match the sequential state the programmer expects. If an interrupt, page fault, or other exception occurs at the wrong moment, software still needs a coherent restart point. That requirement is usually stated as the *precise exception* property in Eq. (6.8).

$$\begin{aligned} \text{precise exception at } i \Rightarrow\ & \text{all instructions older than } i \text{ are committed} \\ & \wedge\ \text{no instruction younger than } i \text{ is architecturally visible} \end{aligned} \tag{6.8}$$

This requirement was not always imposed in the same way. In Tomasulo's original floating-point setting on the IBM System/360 Model 91, the out-of-order machinery covered a narrower part of the processor, and the machine could tolerate *imprecise exceptions* in situations that modern general-purpose cores cannot. Once out-of-order execution extended to ordinary integer, load/store, and control-flow instructions under virtual memory, recovery to an exact sequential state became mandatory rather than optional.

6.14 Reorder Buffers and Retirement

The standard answer to precise-state recovery is the *reorder buffer (ROB)*. Dispatch allocates ROB entries in program order. Execution may finish out of order, but retirement occurs in order from the ROB head, which creates a clean boundary between speculative work and architecturally committed state.

A typical ROB entry stores destination information, a result field, a ready bit, an exception flag, and enough control information to recover the correct program counter (PC). In simple form, retirement follows Eq. (6.9).

$$retire \Leftarrow ready(ROB_{head}) \wedge \neg exception(ROB_{head}). \tag{6.9}$$

If the machine can retire at most W_{retire} instructions per cycle, then one cycle may commit up to the oldest W_{retire} consecutive ready non-excepting entries from the ROB head, but it may not skip

Instr.	Dest reg/tag	Src #1	Src #2	Queue status
A	$R1$/A	$R3$ value ready	n/a	ready to fire
B	$R3$/B	$R2$ value ready	tag A not ready	waiting on A
C	$R1$/C	tag B not ready	$R2$ value ready	waiting on B

ROB entry	Dest reg	Ready	Value source
A	$R1$	0	pending load result
B	$R3$	0	pending add result
C	$R1$	0	pending add result

Table 6.3: Tag-based Tomasulo plus ROB bypass after dispatch of A, B, and C.

over an older unready or excepting entry. Retire width therefore limits architecturally visible IPC even when execution hardware has already completed more work internally. If the instruction at the ROB head is marked as an exception, the machine instead flushes younger speculative work and transfers control to the retry or handler PC, as shown in Eq. (6.10).

$$\begin{aligned} exception(ROB_{head}) \Rightarrow\ & \text{flush younger instructions} \\ & \wedge\ \text{transfer control to the retry or handler PC} \end{aligned} \tag{6.10}$$

For example, a page-faulting load is usually retried after the operating system resolves the fault, whereas a trap-like event may resume at the following instruction.

ROB with bypass or forwarding Some designs let dispatch search the ROB from youngest to oldest for the most recent in-flight producer of a source register. If the matching entry is ready, dispatch can copy the value directly. If it exists but is not ready, dispatch copies the tag or ROB identity and waits. If no matching entry exists, the value must already have retired, so dispatch reads it from the architected register file. This is often described as a ROB with bypass or forwarding. It is effective, but the search itself is associative and can become part of the critical timing path.

Worked tag-based ROB-bypass trace The following tag-based example makes the lookup rule explicit. Assume the architected register file initially holds committed ready values for $R1$, $R2$, and $R3$, and that dispatch allocates ROB tags in program order. Consider:

$$\begin{aligned} A&: \quad \text{LW } R1, 0(R3) \\ B&: \quad \text{ADD } R3, R2, R1 \\ C&: \quad \text{ADD } R1, R3, R2 \end{aligned}$$

At dispatch, the source resolution is:

Instr.	Dest reg/tag	Src #1 at dispatch	Src #2 at dispatch
A	$R1$/A	$R3$ value from architected RF	n/a
B	$R3$/B	$R2$ value from architected RF	waits on tag A from ROB entry A
C	$R1$/C	waits on tag B from ROB entry B	$R2$ value from architected RF

The key rule is that dispatch always takes the newest in-flight producer. So B does not read the old committed $R1$ value; it waits on tag A. Likewise, C does not read the old committed $R3$ value; it waits on tag B. After all three instructions have dispatched, the in-flight structures look like Table 6.3.

From this point onward, execution follows the dependence chain directly. A completes first and marks ROB entry A ready, which wakes B. B then completes and marks ROB entry B ready, which wakes C. C completes last. Retirement still occurs strictly in order A, then B, then C, so after the last retirement both the scheduling queue and the ROB are empty. After retirement, the architected register file holds the retired results, with $R3$ containing the value produced by B and $R1$ containing the younger value produced by C. For any such trace, once the total elapsed cycles are known, fragment IPC is retired instructions divided by elapsed cycles over the chosen measurement interval.

Freeing physical registers Renamed machines also need a rule for reclaiming old physical registers. The exact policy differs across implementations, but the underlying principle is simple: an old physical register cannot return to the free list until no older in-flight instruction can still need it. Common implementations describe this with use counters, pending-free bits, or retirement-based reclamation. For example, if an architected register was formerly mapped to physical register $P23$ and rename later redirects that architected name to $P27$, the old $P23$ mapping cannot be recycled immediately. It becomes reclaimable only after no older instruction can still read it and the newer mapping has safely become the committed architectural version. The details vary, but all correct schemes delay reuse until the older mapping is no longer architecturally or speculatively required.

When architected registers are actually written Completion and retirement are not the same event. Completion usually writes the speculative destination into the physical register file and marks the matching ROB entry ready. The architected register file is updated only when that ROB entry reaches the head and retires in order. This separation is what allows the machine to execute aggressively while still presenting precise sequential state to software.

Worked PRegs/RAT/ROB end-state trace The following short trace makes the bookkeeping easier to follow. Assume the initial mappings are $R0 \to P0$, $R1 \to P1$, $R2 \to P2$, and $R3 \to P3$, with $P4$ through $P7$ free. Assume two-wide fetch and dispatch, two-wide retirement, two 1-cycle ALUs, one 1-cycle LSU, and a back-to-front cycle evaluation so that Execute can mark a destination physical register ready before Schedule runs later in the same cycle. Consider the following sequence:

$$\begin{array}{ll} A: & \text{LW } R3, 0(R1) \\ B: & \text{ADD } R1, R3, R2 \\ C: & \text{ADD } R2, R2, R1 \end{array}$$

One legal cycle-by-cycle interpretation is:

Cycle 1. Fetch A and B.

Cycle 2. Dispatch A and B. A allocates $P4$ as the new destination for $R3$ and records old mapping $P3$ in its ROB entry. B therefore reads $P4$ as its first source, allocates $P5$ as the new destination for $R1$, and records old mapping $P1$ in its ROB entry. Fetch C.

Cycle 3. A fires on the LSU. Dispatch C, which now reads $P2$ and $P5$, allocates $P6$ as the new destination for $R2$, and records old mapping $P2$ in its ROB entry.

Cycle 4. A completes, marking $P4$ ready and its ROB entry ready. Because that completion is visible before scheduling in the same cycle, B can now fire.

Cycle 5. A retires, so the architected copy of $R3$ now comes from $P4$ and old mapping $P3$ can be freed. B completes, marking $P5$ ready, and C can then fire.

Cycle 6. B retires, so the architected copy of $R1$ now comes from $P5$ and old mapping $P1$ can be freed. C completes, marking $P6$ ready.

Cycle 7. C retires, so the architected copy of $R2$ now comes from $P6$ and old mapping $P2$ can be freed.

The exact cycle numbers can shift by one under a different but internally consistent simulator ordering, but the final renamed state does not change.

Instr.	FU	Dest physical reg.	Src #1	Src #2	Final status
A	LSU	$P4$	$P1$	n/a	completed and retired
B	ALU	$P5$	$P4$	$P2$	completed and retired
C	ALU	$P6$	$P2$	$P5$	completed and retired

ROB entry	Dest architected reg.	Prev physical reg.	Final status
A	$R3$	$P3$	retired
B	$R1$	$P1$	retired
C	$R2$	$P2$	retired

Architected reg.	Final physical reg.
$R0$	$P0$
$R1$	$P5$
$R2$	$P6$
$R3$	$P4$

Physical reg.	Ready	Free
$P0$	1	0
$P1$	1	1
$P2$	1	1
$P3$	1	1
$P4$	1	0
$P5$	1	0
$P6$	1	0
$P7$	0	1

Three rules determine the outcome. Dispatch allocates the next free physical destination and records the previous architectural mapping in the ROB. Completion makes the speculative destination ready for younger consumers. Retirement is the event that both updates the architected view and frees the previous physical mapping.

6.15 Future Files and Checkpoint/Repair

One useful way to describe renamed out-of-order state is to separate it into two layers. The *future file* holds the most recent speculative values that execution is building toward. The *architected register file* holds only committed state. Execution and wakeup consult the speculative state, while retirement copies correct in-order results into the architected state. This explains how the machine can behave aggressively internally while still presenting a sequential interface to software. In a physical-register implementation, this often means retirement copies the committed value from the current physical register into the architected register file, while exception recovery can restore a precise view by redirecting the register alias table (RAT) back to the committed architectural mappings.

Comparison with tag-only Tomasulo A physical-register plus RAT design does not automatically increase the amount of ILP that the scheduler can discover. If a tag-based Tomasulo machine and a physical-register machine have the same function units, the same wakeup/select policy, the

same retire width, and enough result buses that completion bandwidth is not the bottleneck, then they can expose the same execution schedule and the same IPC on a given trace. The main differences are in bookkeeping cost, lookup style, recovery organization, and physical-register reuse rather than in the dependence constraints themselves. Performance diverges only when one design changes a practical limit such as result-bus count, port bandwidth, wakeup timing, or retirement bandwidth.

The ROB is not the only recovery method. *Checkpoint/repair* schemes periodically capture a coherent snapshot of register state and use that snapshot as the recovery point. One common description uses three conceptual views of the state: a messy current copy, a working copy being built toward coherence, and a golden copy representing the last known coherent sequential state.

The key control concept is the *instruction boundary*. Dispatch chooses a dynamic boundary, often denoted by markers such as IB1 and IB2. Instructions that complete before the current boundary update both the messy state and the working copy. Instructions that complete after the boundary update only the messy state. Once every instruction older than the boundary has completed, the working copy has become coherent and may be promoted to the new golden copy. The machine then chooses a new boundary and repeats the process.

If an exception occurs, the machine restores the golden copy and continues from that last coherent point. It may then re-execute younger instructions sequentially or under tighter control until it reaches the faulting instruction and handles the event precisely. This can reduce some of the timing pressure associated with aggressive ROB-bypass logic, though it introduces its own bookkeeping and copying costs.

6.16 Speculation Without and With Prediction

Superscalar pipelines use speculation to keep FUs busy. Some designs can speculate without high-quality prediction (for example, eager execution ideas), but the useful work fraction is then limited. High-accuracy branch handling usually gives better effective IPC because fewer speculative instructions are squashed.

6.17 Security Implications of Speculative Execution

Precise exceptions solve the architectural recovery problem, but they do not automatically solve the microarchitectural side-effect problem. An instruction that is later squashed may still have changed caches, predictors, or other timing-visible structures before the machine discovered that the instruction should not retire. That is the core issue behind speculative side channels.

The best-known examples are Spectre- and Meltdown-style attacks. At a high level, the attacker causes the processor to execute work that will later be squashed or faulted. Even though the architectural state rolls back correctly, the speculative work may still have fetched a cache line or trained a predictor. The attacker then measures timing differences to infer information that should never have become visible through the ISA.

Shared-core execution can make this worse. With *hyperthreading*, or more generally simultaneous multithreading, two threads share parts of one physical core. That increases the opportunity for one thread to observe side effects caused by another thread's speculative activity. The security lesson is broader than any single named vulnerability: microarchitectural cleanup matters, not just architectural rollback.

6.18 Implementation Pressure

Once out-of-order execution moved from isolated examples into full commercial processors, these mechanisms had to satisfy real timing, area, power, and verification constraints. That practical pressure is one reason the field explored multiple recovery styles rather than converging immediately on a single universal design.

The architectural lesson is that elegant scheduling ideas are not sufficient by themselves. ROB bypass logic, associative searches, speculative-state tracking, and recovery control all stress timing closure and verification. As out-of-order windows grow wider and deeper, state management and recovery logic become first-order design problems rather than secondary implementation details.

6.19 Interpreting Out-of-Order Processor Statistics

Trace-driven superscalar simulators often report more than cycles and IPC. Those counters are useful only when their meanings and denominators are clear.

Fetched versus retired instructions The architecturally meaningful IPC uses retired instructions, not fetched instructions. *Trace instructions* count the dynamic instructions in the workload itself. *Fetched instructions* count every instruction the front end brought in, including wrong-path work fetched before a branch misprediction was repaired. *Retired instructions* count only the instructions that reached precise architectural state. In a correct full run, retired instructions usually match the trace instruction count, while fetched instructions may be much larger when prediction accuracy is poor.

Miss and misprediction counters Branch-misprediction counts measure control-flow repair events. Instruction-cache misses reduce front-end supply, while data-cache misses usually delay load or store completion and can starve dependent instructions. These counters therefore explain why IPC may fall even when the machine nominally has ample fetch width and many function units.

Queue-pressure counters Maximum and average dispatch-queue, scheduling-queue, and ROB occupancy show where pressure accumulates. If ROB usage remains near its limit and dispatch is frequently blocked because the ROB is full, then commit bandwidth, long-latency instructions, or recovery delays are throttling forward progress. If scheduling queues remain lightly used but *no-fire cycles* are high, then the machine is more likely waiting on ready operands or memory completion than on queue capacity.

Isolating control and memory effects Many simulators also offer a no-miss or no-misprediction mode. Such a configuration does not model real execution, but it is analytically useful because it isolates pure width, scheduling, and dependence limits from penalties caused by cache misses and poor control-flow prediction.

6.20 Chapter 6 Summary

Chapter 6 develops dynamic instruction-level parallelism as a concrete microarchitectural path to $CPI < 1$. Superscalar design achieves this by widening the pipeline's effective throughput and managing dependencies, hazards, and speculative state carefully. FICO-style scoreboarding illustrates the basic mechanism, while renaming, Tomasulo-style wakeup and result broadcast, and stronger

out-of-order policies push performance further. Once the machine executes and completes work out of order, however, scheduling is no longer enough by itself. The core must also maintain precise architectural state through mechanisms such as physical-register renaming, reorder buffers, future files, or checkpoint/repair schemes. Modern processors therefore combine ILP techniques with careful recovery, retirement, and microarchitectural side-effect control. Simulator-visible counters such as fetched-versus-retired instructions, misprediction counts, miss counts, queue occupancy, and ROB pressure then provide a practical way to diagnose which limits are dominating a given design point. These ideas together form the bridge from classic five-stage thinking to modern high-throughput scalar cores.

6.21 Practice Questions

The following questions reinforce the main ideas of Chapter 6 and are intended for self-checking after the summary.

1. Explain why a superscalar machine can achieve $CPI < 1$ without violating the meaning of a clock cycle. Then state the corresponding condition on instructions per cycle (IPC).

2. In a FICO-style machine with one ALU and one load/store unit, consider the sequence `LW R1,0(R2)` followed by `ADD R3,R1,R4`. Describe the earliest cycle in which the add can fire if the load takes two cycles to complete and there is no forwarding of the load result before it is ready.

3. Consider Tomasulo with tags and a ROB with bypass for the sequence `LW R1,0(R3)`; `ADD R3,R2,R1`; `ADD R1,R3,R2`. When the second and third instructions dispatch, which of their source operands come from the register file and which come from the youngest older ROB producer?

4. In a renamed machine, the current mapping of architected register `R1` points to physical register `P10`. A new instruction `ADD R1,R2,R3` dispatches and receives destination physical register `P24`. What mapping change occurs immediately at dispatch, and why can `P10` not be freed at that same moment?

5. Distinguish the event of broadcasting a result on the common data bus from the event of updating precise architected state at retirement. Why is this distinction essential once execution and completion occur out of order?

6. A simulator reports 1,600 fetched instructions, 1,200 retired instructions, 1,000 cycles, 220 no-fire cycles, and 90 ROB-blocked dispatch cycles. Compute IPC using the architecturally meaningful instruction count. Then explain what the gap between fetched and retired instructions suggests and what the ROB-blocked count suggests.

7. Explain why register renaming removes false dependences but does not remove true dependences. Give one short example of each case.

8. Compare reorder-buffer recovery with checkpoint/repair recovery. What state must each approach preserve so that the machine can return to a precise point after a misprediction or exception?

Chapter 7

Compiler-Scheduled ILP: VLIW and Software Pipelining

7.1 VLIW as a Compiler-Scheduled Alternative

One response to dynamic-scheduling complexity is to shift more of the scheduling work into the compiler. As superscalar windows grow, the hardware would like a larger scheduling queue so it can search farther ahead for independent work, but larger queues also make wakeup, select, and operand-readiness checks harder to close at a short cycle time. This tradeoff motivates the *very long instruction word (VLIW)* approach, historically associated with Josh Fisher. In a VLIW machine, one instruction word encodes several operations intended to execute in parallel, so the compiler determines the schedule at compile time rather than leaving that work to large runtime scheduling hardware.

In this context, an individual encoded action is an *op*. A set of ops intended to execute in the same cycle is a *multi-op*. VLIW reduces runtime hardware complexity because the machine no longer needs a large associative scheduling queue, aggressive wakeup logic, or much of the dynamic issue machinery used in FOCO-style processors. The tradeoff is that the compiler must do much more work up front, and the resulting code is less adaptable to runtime variation such as cache misses or changing branch behavior. This tradeoff is attractive in domains such as *digital signal processors (DSPs)*, where workloads are regular and low-power execution is important.

Compiler design must balance three competing goals: code quality, compilation time, and maintainability. In practice, that usually rules out algorithms much worse than quadratic for large compiler passes, and even cubic methods are used cautiously. Because the optimal schedule for a dependence-constrained block is exponential in block size, practical VLIW compilers rely on heuristics rather than exact search. List scheduling is one of the standard heuristics for this job. The broader performance point is that widening fetch and scheduling machinery yields diminishing returns: useful IPC usually grows sublinearly with front-end width, while the cost of dynamic scheduling hardware grows quickly.

7.2 Dependence Graphs and List Scheduling

The compiler begins with a *data dependence graph*. True flow edges capture read-after-write dependencies, while anti-dependence and output-dependence edges capture false constraints caused by register reuse. Pure flow dependence always takes priority over false dependence because it represents real data movement. If the code contains loops, this graph can contain cycles, so it is

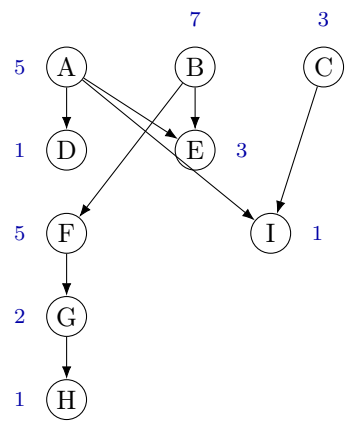
Instr.	Operation	True dependences	Latency
A	LW $R2, X$	None	2
B	LW $R3, Y$	None	2
C	LW $R4, Z$	None	2
D	ADD $R5, R1, R2$	A	1
E	MUL $R6, R2, R3$	A, B	3
F	MUL $R7, R1, R3$	B	3
G	ADD $R8, R7, \#1$	F	1
H	SW $R8, X$	G	1
I	ADD $R9, R2, R4$	A, C	1

Table 7.1: Illustrative basic block for VLIW list scheduling.

Original operations

A: LW $R2, X$
B: LW $R3, Y$
C: LW $R4, Z$
D: ADD $R5, R1, R2$
E: MUL $R6, R2, R3$
F: MUL $R7, R1, R3$
G: ADD $R8, R7, \#1$
H: SW $R8, X$
I: ADD $R9, R2, R4$

Data Dependence Graph



List ordered by dependence height

B (7)
A (5)
F (5)
C (3)
E (3)
G (2)
D (1)
I (1)
H (1)

Figure 7.1: Original operations, dependence graph, and one priority ordering for the VLIW list-scheduling example. The small numeric annotations next to nodes indicate dependence height. Ties may be broken arbitrarily.

not necessarily a directed acyclic graph. Constructing the graph by hand is mechanical. Draw one node per op, then add all true flow edges from each definition to the later uses that require that value. Next add anti-dependence edges from a read to a later write of the same architectural name, and output-dependence edges between two writes of the same name. If two operations are already ordered by a true flow dependence, that true edge is the one that matters for scheduling because it already enforces the required order.

Table 7.1 gives a compact block that will be used for the scheduling example. Assume one pipelined load/store unit, one pipelined arithmetic logic unit (ALU), and one pipelined multiplier. Each unit can accept a new op every cycle; the listed latencies describe when results become available, not how long the unit remains occupied. Also assume the compiler has already assigned registers so that the remaining constraints shown in the table are dominated by true dependences rather than by avoidable name conflicts. Figure 7.1 shows the same block in a more visual form: the original operations, the resulting data dependence graph, and one priority list ordered by dependence height.

One common priority heuristic is *dependence height*, which estimates how much latency remains below a node on the longest dependent path. If the heuristic includes the latency of the current

Cycle	Load/Store Unit	ALU	Multiplier
1	A		
2	B		
3	C	D	
4			F
5		I	E
6			
7		G	
8	H		

Table 7.2: One legal VLIW-style schedule for the block under fixed functional-unit limits.

instruction itself, one convenient definition is Eq. (7.1).

$$height(i) = \begin{cases} lat(i), & succ(i) = \emptyset \\ lat(i) + \max\limits_{j \in succ(i)} height(j), & \text{otherwise} \end{cases} \tag{7.1}$$

Higher-height instructions receive higher priority because delaying them is more likely to lengthen the critical path. To annotate a graph by hand, start from the sinks and assign each sink its own latency. Then sweep backward, adding the current node latency to the maximum height among its successors. The ready list is obtained by sorting nodes from highest height to lowest and breaking ties either arbitrarily or by an explicit secondary rule such as program order. For the block in Table 7.1, the longest true-dependence chain is $B \to F \to G \to H$, so the heuristic tends to prioritize F ahead of E once both become candidates for the single multiplier.

List scheduling then places the highest-priority ready instructions into each cycle, subject to both dependence timing and resource availability. A simple legality condition is shown in Eq. (7.2).

$$cycle(i) = \min\{c \mid \forall j \in pred(i),\ c \geq cycle(j) + lat(j),\ \text{and a compatible FU is free in } c\}\,. \tag{7.2}$$

This is not an optimal algorithm, but it is practical and captures the central VLIW idea: schedule as much parallel work as possible before runtime while still respecting dependencies and resource limits.

Table 7.2 shows one possible compiler-produced schedule for the block running on a machine with one arithmetic logic unit (ALU), one multiplier, and one load/store unit. The point is not that this schedule is unique, but that the compiler can choose the cycle packing in advance and encode it directly. For illustration, this schedule keeps the three initial loads in source order even though a greedier dependence-height policy could start B before A and shorten the block by one cycle. That choice leaves one explicit empty cycle later, which makes the pause-field encoding example concrete. The schedule highlights two points. First, the compiler only packs operations together when both the dependence constraints and the functional-unit constraints allow it. Second, the absence of an operation in a slot does not require the hardware to infer what to do dynamically; the compiler has already encoded the intended schedule. In particular, cycle 6 is intentionally empty because G cannot start until the three-cycle multiply F has finished producing $R7$. Table 7.3 shows the same schedule in a field-level packing format closer to an actual multi-op layout. This table defines which operands occupy each unit's fields in each cycle; it does not, by itself, define the serialized op order in memory.

Cycle	IALU (1 cycle)			MUL (3 cycles)			Load (2 cycles)		Store (1 cycle)	
	d	s_1	s_2	d	s_1	s_2	d	addr	addr	s
1							$R2$	X		
2							$R3$	Y		
3	$R5$	$R1$	$R2$				$R4$	Z		
4				$R7$	$R1$	$R3$				
5	$R9$	$R2$	$R4$	$R6$	$R2$	$R3$				
6										
7	$R8$	$R7$	#1							
8									X	$R8$

Table 7.3: Field-level packing of the schedule from Table 7.2. Rows 3 and 5 are multi-ops because they contain more than one op in the same cycle.

7.3 Multi-Op Encoding, Tail Bits, and Pause Fields

VLIW raises an encoding question: how should a machine represent the boundaries of a multi-op when not every cycle uses every functional unit? One simple approach is to fill every unused slot with an explicit no-op, but that wastes code space. An alternative is to encode only real ops and add a small amount of grouping information. Each encoded op still carries its ordinary fields such as opcode, destination register, and source registers; the additional VLIW-specific fields only describe grouping and delay.

One useful scheme adds a *tail bit* to each encoded op. If the tail bit is 0, the current op is not the end of the current multi-op. If the tail bit is 1, the current op ends the current multi-op. The following op then begins a new multi-op. This lets the instruction stream represent a variable number of ops per cycle without requiring every cycle to contain the same number of explicit no-op fillers.

A related field is a *pause field*, which tells the front-end control logic to wait a specified number of cycles before beginning the next multi-op. In the simple scheme used here, the pause value is interpreted only on the op whose tail bit ends the current multi-op; pause values on earlier ops in the same multi-op are ignored. This is a compact way to represent silence between multi-ops while keeping each op's encoding regular. Table 7.4 illustrates the idea using the schedule from Table 7.2. It shows one legal serialized op order for the same multi-ops that appear in Table 7.3. Some implementations instead use simple interlocking at the multi-op level, which can remove the need for an explicit pause field. That saves encoding space, but it also gives up some of the compiler's direct control over exactly when the following multi-op may begin.

7.4 Software Pipelining and Cyclic Scheduling

Loops create another opportunity for compiler-scheduled parallelism. Instead of scheduling only one copy of a loop body, the compiler can overlap operations from different iterations so that a later iteration begins before an earlier one has fully finished. The older name for this idea is *polycyclic scheduling*; the more common modern name is *software pipelining*.

Conceptually, the compiler behaves as if it had unrolled the loop many times, compacted the overlapping schedule, and then retained only the repeating pattern. That repeating pattern is the *kernel*. The code that fills the pipeline before the kernel reaches steady state is the *prologue*. The code that drains the remaining work after loop exit is the *epilogue*. The distance in cycles between

Encoded op	Tail	Pause	Meaning
A	1	0	Single-op multi-op for cycle 1.
B	1	0	Single-op multi-op for cycle 2.
C	0	0	First op of the cycle-3 multi-op.
D	1	0	Ends the cycle-3 multi-op.
F	1	0	Single-op multi-op for cycle 4.
I	0	0	First op of the cycle-5 multi-op.
E	1	1	Ends the cycle-5 multi-op and inserts one pause cycle before the next multi-op.
G	1	0	Single-op multi-op for cycle 7.
H	1	0	Single-op multi-op for cycle 8.

Table 7.4: Illustrative VLIW encoding using tail bits and a pause field rather than explicit no-op padding.

the start of successive iterations is the *initiation interval (II)*. Once the schedule reaches steady state, the loop begins a new iteration every II cycles, so the throughput is $1/II$ iterations per cycle.

Unrolling view before compaction One useful way to understand software pipelining is to look first at a small explicit unrolling. Suppose a loop body consists of five ops A, B, C, D, E, where E is the loop-closing branch. A two-times unrolled form can be sketched conceptually as

$$A,\ B,\ C,\ D,\ E^*,\ A',\ B',\ C',\ D',\ E',$$

where E^* is the inverted early-exit branch that leaves the loop if the second copy should not begin. This makes the scheduling opportunity visible: once the loop is unrolled, the compiler can interleave A' with still-pending work from the first copy instead of waiting for the whole first iteration to drain first. Software pipelining then takes the next step and *compacts* that overlapping pattern into a finite prologue, a repeating kernel, and a finite epilogue rather than materializing a large explicitly unrolled body.

Which dependences matter most in the first sketch In early cyclic-scheduling sketches, it is common to draw only true dependences and the loop-carried recurrences that constrain them. Anti-dependence and output-dependence edges are still real constraints in the full compiler problem, but they are often omitted from the first visualization because software renaming can frequently remove them. The scheduling bottlenecks that determine $RecII$ come from the true cyclic constraints, so those are the ones worth emphasizing first.

7.5 Iterative Modulo Scheduling

A practical way to construct these schedules is *Iterative Modulo Scheduling (IMS)*. The compiler first estimates a lower bound on II, then attempts to build a modulo schedule at that value. If the attempt fails, it increments II and tries again.

The first bound comes from resource pressure. If a loop contains N_t operations of type t and the machine provides U_t compatible units of that type, then

$$ResII = \max_t \left\lceil \frac{N_t}{U_t} \right\rceil \tag{7.3}$$

This is the resource-constrained initiation interval.

The second bound comes from loop-carried recurrences. If a dependence cycle c has total latency $Lat(c)$ and spans $Dist(c)$ loop iterations, then

$$RecII = \max_c \left\lceil \frac{Lat(c)}{Dist(c)} \right\rceil \tag{7.4}$$

This is the recurrence-constrained initiation interval. If the dependence graph has no loop-carried cycles, then the practical lower bound is $RecII = 1$.

The schedule must satisfy both limits, so the compiler begins from

$$MII = \max(ResII, RecII), \qquad II \geq MII. \tag{7.5}$$

Many texts call this lower bound the *minimum initiation interval (MII)*. IMS follows a simple pattern:

1. Start with $II = MII$.
2. Attempt a legal modulo schedule at that value.
3. If the attempt fails, increment II and retry.

A modulo reservation table for a chosen II has exactly II rows. An operation that would occupy cycle x in a longer unrolled schedule is placed into row $x \bmod II$, provided the dependence and resource constraints remain legal.

In most textbook treatments, the underlying machine is modeled with a point-latency abstraction: an op occupies one issue slot, and its result becomes ready after a fixed latency. This keeps the scheduling problem manageable. More detailed models that explicitly schedule register-file ports, result buses, and intermediate pipeline stages are possible, but they are much more expensive to optimize and are usually avoided in introductory treatments.

7.6 Worked Software-Pipelining Example

The following loop illustrates the idea. Assume two arithmetic logic units, two memory units, one-cycle add/branch latency, and two-cycle load latency. The branch still logically closes the loop body even though later iterations will overlap older ones.

The cyclic dependence view of this loop contains the within-iteration edges $A_i \to B_i$, $B_i \to C_i$, and $D_i \to E_i$, plus the loop-carried edges $A_i \to A_{i+1}$ and $D_i \to D_{i+1}$. Those loop-carried edges are exactly the recurrences used when computing $RecII$.

For this loop, the resource bound is

$$ResII = \max\left(\left\lceil \frac{3 \text{ ALU/branch ops}}{2 \text{ ALUs}} \right\rceil, \left\lceil \frac{2 \text{ memory ops}}{2 \text{ memory units}} \right\rceil \right) = \max(2, 1) = 2.$$

The loop-carried recurrences on $R2$ and $R10$ each have latency 1 and distance 1, so $RecII = 1$. Therefore the first useful attempt is $II = 2$.

Instr.	Operation	Key dependences	Latency
A	ADD $R2, R2, \#1$	loop-carried on $R2$	1
B	LW $R3, 0(R2)$	A	2
C	SW $R0, 0(R3)$	B	1
D	ADD $R10, R10, \#1$	loop-carried on $R10$	1
E	BRGTZ $R10$, LOOP	D; branch must close the loop body	1

Table 7.5: Loop used to illustrate software pipelining and modulo scheduling.

Modulo slot	ALU 0	ALU 1	MEM 0	MEM 1
0	A_i	D_i		
1	E_i		B_i	C_{i-1}

Table 7.6: One modulo reservation table for the loop in Table 7.5 with $II = 2$.

One legal modulo reservation table appears in Table 7.6. The first row begins iteration i. The second row finishes the current iteration's control work while also performing the store from iteration $i - 1$. That overlap is the essence of software pipelining.

The emitted code must still handle loop entry and loop exit correctly. Table 7.7 separates the schedule into prologue, kernel, and epilogue form. The early-exit branch E^* is the inverted form of the loop branch; it lets the machine leave the prologue cleanly if the loop should terminate before the steady-state kernel is entered.

This example also shows why software pipelining is attractive for VLIW machines. The compiler exposes the repeating parallel structure explicitly, so the hardware can execute the steady-state kernel with very little dynamic scheduling logic. At the same time, general-purpose superscalar machines still benefit from dynamic scheduling because real loads do not always return in a fixed number of cycles and because binaries may need to run across several microarchitectural variants.

7.7 Memory Disambiguation

Loads and stores create a special scheduling problem. If a compiler moves a load ahead of an older store to the same address, the load may read the wrong value. The compiler therefore needs *memory disambiguation*: it must prove that two memory references do not overlap, or else preserve the original order or insert explicit synchronization.

This is one of the limits of VLIW compared with dynamic scheduling hardware. An out-of-order processor may speculate and recover if it guesses wrong. A VLIW compiler must usually be more conservative unless it has strong alias information. That is one reason VLIW works best when memory behavior is regular enough for compile-time reasoning to be reliable.

7.8 Chapter 7 Summary

Chapter 7 presents compiler-scheduled ILP as an alternative to increasingly aggressive dynamic superscalar scheduling. Instead of building increasingly complex runtime issue logic, a VLIW machine relies on the compiler to expose parallel ops ahead of time and encode them explicitly. Data dependence graphs, dependence-height ordering, and list scheduling provide the basic machinery for building legal multi-ops. Tail-bit and pause-field encoding then translate that schedule into a

Phase	Operations
Prologue	A_1, D_1 then B_1, E_1^*
Kernel	repeat every $II = 2$ cycles: A_i, D_i then B_i, C_{i-1}, E_i
Epilogue	C_{last}

Table 7.7: Prologue, kernel, and epilogue for the software-pipelined loop.

compact instruction stream without filling unused slots with explicit no-ops. Software pipelining extends the same idea across loop iterations by constructing a prologue, steady-state kernel, and epilogue around a chosen initiation interval. Memory disambiguation remains the key correctness limit: compile-time scheduling can move loads and stores only when aliasing is understood well enough to preserve program behavior. Together, these techniques show how compiler scheduling can approach the performance of aggressive ILP hardware while trading hardware complexity for compile-time analysis.

7.9 Practice Questions

The following questions reinforce the main ideas of Chapter 7 and are intended for self-checking after the summary.

1. For the sequence `LW R1,X`; `ADD R2,R1,R3`; `ADD R1,R4,R5`; `MUL R6,R2,R1`, draw the data dependence graph and label the pure, anti-, and output dependences.

2. Using appropriate latencies for load, add, and multiply, compute the dependence height of each instruction in the previous question. Then list the instructions in descending order of dependence height.

3. Assume a VLIW machine with at most one ALU op, one MUL op, and one memory op per multi-op. Produce one legal list schedule for the instruction sequence in Question 1.

4. A VLIW schedule contains the following multi-ops: row 1 has a single load; row 2 has an add and a multiply; row 3 is empty because of a dependence delay; row 4 has a store. Explain how a tail-bit plus pause-field encoding can represent this schedule without inserting explicit no-op instructions into every unused slot.

5. A loop contains 3 ALU/branch operations, 1 multiply, and 3 memory operations per iteration. The machine has 1 ALU, 1 MUL, and 1 memory unit. Compute $ResII$. Then explain how a loop-carried recurrence could still force MII above that resource bound.

6. For the loop in Question 5, suppose the strongest loop-carried recurrence has total latency 4 and distance 1. Compute $RecII$ and MII.

7. Explain why a modulo reservation table for initiation interval II has exactly II rows, and explain the role of the mapping from cycle x in the unrolled schedule to row $x \bmod II$.

8. Under what conditions may a compiler legally move a load ahead of an older store? Explain why this question is fundamentally a memory-disambiguation problem rather than only a functional-unit scheduling problem.

Chapter 8

Multiprocessor Architecture Fundamentals: Shared Memory, Interconnects, and Cache Coherence

8.1 From Instruction-Level to Thread-Level Parallelism

Once a machine has already exploited as much instruction-level parallelism as is practical, the next major source of speedup is to execute more than one thread of control at the same time. This moves the discussion from single-core scheduling into *multiprocessor* design. At this point the most visible form of parallelism is *thread-level parallelism (TLP)*: the programmer or runtime system must decide how to divide work across multiple processing elements. Unlike instruction-level parallelism, thread-level parallelism is visible to software and therefore directly affects correctness, synchronization, and programmability.

Historically, architecture texts often used the term *processing element (PE)* for one independently executing processor in a larger machine. Modern commercial systems more often use *core*, but the older PE terminology remains useful because it applies equally well to CPUs, accelerator lanes, and other specialized execution agents. Table 8.1 places multiprocessors in the broader grain-size view of parallelism.

At the architecture-classification level, two familiar Flynn terms continue to matter:

- *single instruction, multiple data (SIMD)* machines apply the same operation to multiple data elements at once.
- *multiple instruction, multiple data (MIMD)* machines allow different PEs to execute different instructions on different data at the same time.

Modern shared-memory multiprocessors are usually discussed as MIMD machines. That does not exclude SIMD units inside each core; it only means that the overall machine permits several independently executing instruction streams.

8.2 Memory Organization and System Structure

A multiprocessor can be classified along another axis besides SIMD versus MIMD: how the memory system is organized. The two clean endpoints are *shared memory* and *message passing*.

Parallelism level	Typical mechanism	Visibility to the programmer
Instruction-level	superscalar issue, out-of-order execution, VLIW	usually hidden
Data-level	SIMD vectors, packed operations, vector pipelines	partly visible through data layout or intrinsics
Thread-level	multiple threads or PEs executing concurrently	directly visible

Table 8.1: Common grain sizes of parallelism. Multiprocessor programming is primarily concerned with the thread level.

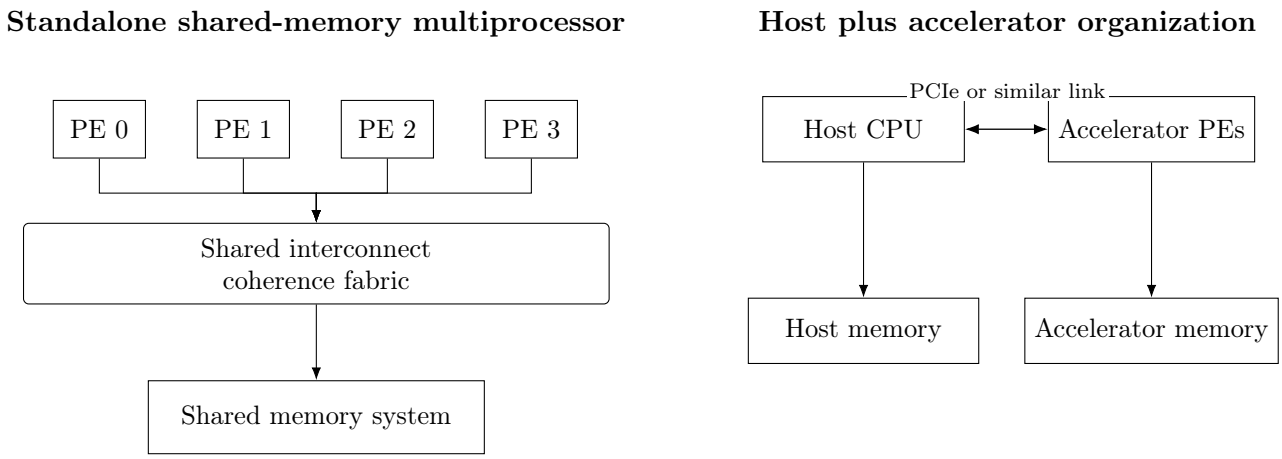


Figure 8.1: Two common multiprocessor organizations. The left machine keeps all PEs in one shared-memory system. The right machine uses an accelerator with distinct memory, so software must explicitly move data across the interconnect.

In a *shared-memory* machine, all PEs conceptually operate in one common address space. A pointer value can identify the same logical object for every PE, and communication occurs when one PE writes data that another PE later reads. This model is attractive because it is easy to think about, but it immediately raises the cache-coherence problem once private caches are introduced.

In a *message-passing* machine, each PE has its own private memory, and data movement occurs through explicit communication operations such as `send()` and `receive()`. This model avoids coherence traffic because data is copied intentionally from one memory to another, but it forces the program to expose that communication directly. Between these extremes lies the *partitioned global address space (PGAS)* idea, where a single address space exists but some portions are substantially more expensive to access than others.

Figure 8.1 contrasts two common organizations. A standalone shared-memory multiprocessor integrates several PEs around a common memory system and runs the operating system and application code directly. An accelerator organization instead separates a host processor from a specialized parallel engine; the host launches work on the accelerator and explicitly copies data to and from accelerator memory.

Accelerators include graphics processors, machine-learning accelerators, and many other specialized engines. Their memories are often kept separate for protection, compatibility, and implementation reasons. The cost of that separation is explicit data motion: the host must copy inputs to the accelerator, launch the parallel work, and copy results back.

Step	Event	Locally visible value of x
0	Initial memory value	memory holds 1
1	P_0, P_1, and P_2 all cache x	each cache now holds 1
2	P_0 writes $x \leftarrow 22$ in its cache	P_0 sees 22
3	P_1 writes $x \leftarrow 44$ in its cache	P_1 sees 44
4	P_2 executes `x++` using its stale cache copy	P_2 sees 2

Table 8.2: An incoherent-cache scenario. Each PE can observe a different value of the same logical variable.

8.3 Interconnects: Shared Buses and Point-to-Point Links

The interconnect between PEs matters just as much as the PE count. In this text, a *bus* means a *shared observable broadcast medium*: all agents attached to the bus can observe the transactions sent across it. That observability is the key property used later by snooping coherence.

A *point-to-point interconnect* behaves differently. A packet is sent from one endpoint to another over a dedicated link, and unrelated agents do not automatically see that traffic. Modern peripheral links such as *PCI Express (PCIe)* fit this model even though informal conversation sometimes calls them buses. PCIe uses differential pairs rather than one shared bundle of broadcast wires, so it scales better than a traditional shared bus but does not naturally support global observation of every memory transaction.

This distinction matters because some coherence mechanisms require every cache to observe every relevant request. Snooping works naturally on a shared observable medium. Directory-based coherence does not require broadcast visibility because a separate directory structure explicitly tracks who has each block.

8.4 The Cache-Coherence Problem

Shared memory becomes difficult as soon as each PE receives a private cache. Without additional coordination, different caches can hold different copies of the same memory location and update them independently. That breaks the programmer's illusion of one shared value.

Consider a memory location x whose initial value is 1. Assume three PEs each read x and therefore bring a copy into their private caches. Now suppose the following operations occur in program order:

1. P_0 writes $x \leftarrow 22$.
2. P_1 writes $x \leftarrow 44$.
3. P_2 executes `x++`.

If the caches do not coordinate, then P_2 may still read its stale local copy of 1, increment it, and write back 2. The program order suggests that the increment should apply to the most recent visible value and therefore produce 45, but an incoherent cache system can easily produce 2 instead. Table 8.2 summarizes the failure.

Simply using write-through caches does not solve the problem. Write-through ensures that memory is updated promptly, but it does not inform the other caches that their local copies have

become stale. The real issue is not whether memory eventually receives the write; it is whether all cached copies remain mutually consistent.

A coherent cache system is normally expected to satisfy three properties:

1. **Read-after-write semantics for one processor.** If a PE writes a value and no one else writes that location in the meantime, a later read by the same PE should return the value it wrote.

2. **Eventual visibility to other processors.** If one PE writes a location and another PE later reads it, then after sufficient time has passed and assuming no intervening writes, the reader must eventually observe the value that was written.

3. **Consistent ordering of writes.** All PEs must observe writes to the same location in a consistent causal order.

The second property is intentionally phrased in terms of *sufficient time* rather than instant propagation. Physical systems cannot propagate signals across large chips or boards instantaneously. Coherence therefore promises eventual agreement and a consistent write story, not magic zero-time updates.

8.5 Coherence Solution Families

Architecturally, there are three standard ways to solve the coherence problem.

The first is a *shared-cache* design in which all PEs communicate through one common cache structure. This avoids inter-cache inconsistency because there is only one place that holds the shared block state. The weakness is scalability: once the shared cache becomes the bottleneck, adding more PEs does not produce linear speedup even for embarrassingly parallel workloads.

The second is *directory-based coherence*. A directory tracks which caches currently hold each block and coordinates invalidations, forwarded data, or ownership transfers. This can scale better than a pure broadcast bus because the system does not require every cache to observe every request.

The third is *snooping coherence*. Here each cache has a *snooper*: logic that knows both what is currently in that cache and what transactions are appearing on the shared interconnect. If the interconnect is a shared observable medium, the snooper can react immediately when another PE requests a block that this cache currently owns or shares. The remainder of this chapter develops the classic snooping approach because it exposes the central state-machine ideas clearly.

8.6 Snooping Coherence and the MSI Protocol

A classic starting point for snooping coherence is the *MSI* protocol. Each cached block is marked as *modified (M)*, *shared (S)*, or *invalid (I)*. The meaning of those states is summarized in Table 8.3.

Two kinds of bus requests are especially important:

- `GetS`: request shared read permission for a block.

- `GetM`: request exclusive access with intent to write.

A dirty eviction produces a `WriteBack` transaction. Snoopers may also perform two high-level actions:

State	My copy	Other caches	Memory copy
M	dirty and writable	no one else has a valid copy	stale
S	clean	others may also hold clean copies	correct
I	not valid in this cache	unspecified	unspecified

Table 8.3: Meaning of the three MSI states for one cache block.

Transaction or action	Meaning
`GetS`	Read miss requesting a clean shared copy.
`GetM`	Read-for-ownership requesting exclusive writable access.
`WriteBack`	Dirty block is written back to memory on eviction or ownership handoff.
abort	Requester must wait or retry because another cache is still handling the block.
intervene	A snooping cache supplies the current data value directly.

Table 8.4: Common bus transactions and snooper responses in a snooping MSI system.

- *abort*: tell the requester to wait or retry because another cache still owns the necessary data or permission.
- *intervene*: supply the data directly rather than letting memory respond.

Table 8.4 summarizes the basic bus-level vocabulary.

From the requesting PE's perspective, the stable-state behavior is simple. If a block is invalid and the PE reads it, the cache issues `GetS` and moves to `S` when the data arrives. If a block is invalid and the PE writes it, the cache issues `GetM` and moves to `M` when ownership arrives. If a block is already shared and the PE wants to write, the cache must first upgrade through `GetM` so that all other copies are invalidated. Table 8.5 gives one compact summary.

From the snooper's perspective, the same bus requests mean something different. If a cache in state `S` sees another PE request `GetM`, it must invalidate its own copy and move to `I`. If a cache in state `M` sees `GetS`, it must ensure that the most recent data becomes visible and then downgrade to `S`. If it sees `GetM` while in `M`, it must give up ownership entirely and move to `I`. Table 8.6 summarizes the stable-state snooping behavior.

8.7 Transient States and a Worked Two-Cache Trace

Real buses and coherence fabrics are not instantaneous, so caches often need *transient states* while waiting for a request to complete. A common notation concatenates the path of the transition. For example, `IM` means that the cache started in `I`, issued a request whose destination is `M`, and is still waiting for that request to complete. Likewise, `SM` means a shared block has requested an upgrade to modified state but has not yet received final ownership.

Table 8.7 walks through a short two-cache scenario that mirrors the classic classroom introduction to MSI:

1. P_1 has a read miss and obtains a shared copy.
2. P_2 then has a read miss and also obtains a shared copy.

Current state	Local event	Bus action	Next state
I	read miss	`GetS`	S
I	write miss	`GetM`	M
S	read hit	none	S
S	write hit	`GetM`	M
M	read hit	none	M
M	write hit	none	M
S	replacement	none	I
M	replacement	`WriteBack`	I

Table 8.5: Stable-state MSI behavior from the requesting cache's perspective.

Current state	Snooped event	Action	Next state
I	sees `GetS` or `GetM`	none	I
S	sees `GetS`	none	S
S	sees `GetM`	invalidate local copy	I
M	sees `GetS`	provide or write back data, downgrade	S
M	sees `GetM`	provide or write back data, invalidate	I

Table 8.6: Stable-state MSI behavior from the snooper's perspective.

3. P_1 writes the block and upgrades from `S` to `M`.
4. P_2 then tries to write the same block, initially receives an abort, enters transient state `IM`, and retries after P_1 writes back or hands off the dirty value.

The important insight is that the coherence protocol is maintaining one simple invariant: a writable dirty copy must be unique. Whenever one cache acquires `M`, all competing copies must either become invalid or be updated into a clean shared form before execution continues.

8.8 Programming Models: Shared Memory and Message Passing

The architecture choices above are reflected directly in parallel programming models. A shared-memory program typically divides the data among threads, lets each thread compute a local partial result, and then uses explicit synchronization to merge those results safely. The following example uses a lock to protect a critical section that updates a global sum:

```
float X[64];
float globalSum = 0;

Thread 0: sum X[0:15] locally, then lock(L); globalSum += local0; unlock(L);
Thread 1: sum X[16:31] locally, then lock(L); globalSum += local1; unlock(L);
Thread 2: sum X[32:47] locally, then lock(L); globalSum += local2; unlock(L);
Thread 3: sum X[48:63] locally, then lock(L); globalSum += local3; unlock(L);
```

The lock prevents a lost update in the critical section. Without it, two threads might read the same old value of `globalSum`, add their local sums independently, and then overwrite one another. That kind of latent synchronization bug is one reason shared memory is powerful but dangerous.

Step	Event	Bus activity	P_1 state	P_2 state
1	P_1 read miss	`GetS`	$I \to S$	$I \to I$
2	P_2 read miss	`GetS`	$S \to S$	$I \to S$
3	P_1 write hit on shared block	`GetM`	$S \to M$	snoops `GetM`: $S \to I$
4	P_2 write miss while P_1 owns block	`GetM`, abort	snoops `GetM`: $M \to I$ with `WriteBack` or intervention	$I \to IM$
5	P_2 retries after ownership transfer	retry `GetM`	$I \to I$	$IM \to M$

Table 8.7: One worked MSI trace showing stable states, a transient `IM` state, and a write-ownership transfer between two caches.

A message-passing version of the same idea avoids shared writable state:

```
Thread 0: compute local0; send(local0, 4);
Thread 1: compute local1; send(local1, 4);
Thread 2: compute local2; send(local2, 4);
Thread 3: compute local3; send(local3, 4);

Thread 4:
  globalSum = 0;
  for (i = 0; i < 4; i++)
    globalSum += receive(i);
```

In this formulation, synchronization is implicit in the blocking `receive()` operation. The collector thread waits until the required messages arrive. The advantage is portability: this model works across independent machines with separate memories. The cost is explicit communication overhead and a more demanding programming model.

8.9 Scaling and Architectural Limits

Amdahl's Law still governs the big picture. If the parallelizable fraction is 1, then the application is often called *embarrassingly parallel*, and the ideal speedup curve is linear in the PE count. Real machines fall short of that ideal once a shared resource saturates. A single shared cache saturates quickly. A shared interconnect can also become a bottleneck. Even a well-designed coherence protocol can lose efficiency if the workload repeatedly bounces writable blocks between PEs.

This is why the interconnect, the memory model, and the programming style cannot be separated cleanly. A coherence protocol is not only a correctness mechanism; it also determines how much communication traffic the machine must generate in order to preserve the illusion of one shared memory.

8.10 Chapter 8 Summary

Chapter 8 moves from instruction-level techniques to multiprocessor architecture. The central classifications are SIMD versus MIMD, shared memory versus message passing, and shared observable

buses versus point-to-point links. Accelerator systems introduce another common organization by separating host and accelerator memory and therefore requiring explicit data movement. Private caches make shared memory fast enough to be attractive, but they also create the cache-coherence problem. Coherence requires local read-after-write correctness, eventual visibility to other processors, and a consistent ordering of writes. Shared caches, directory protocols, and snooping protocols are the three standard solution families. The MSI snooping protocol demonstrates the core state-machine ideas: blocks move among modified, shared, and invalid states in response to local requests and snooped bus traffic, with transient states used while ownership transfers are still in flight. Finally, the programming model seen by software reflects these architectural choices directly: shared memory relies on synchronization such as locks, while message passing relies on explicit communication and blocking receives.

8.11 Practice Questions

The following questions reinforce the main ideas of Chapter 8 and are intended for self-checking after the summary.

1. Explain the difference between SIMD and MIMD. Then explain why a machine with multiple independent cores and a SIMD unit inside each core is still classified overall as MIMD.

2. Compare a shared-memory multiprocessor with a message-passing multiprocessor. What does the programmer gain and lose in each model?

3. Why is a traditional shared bus a natural fit for snooping coherence, while a point-to-point link such as PCIe is not naturally a broadcast snooping medium?

4. In the incoherent-cache example with initial value $x = 1$, explain why the sequence "P_0 writes 22; P_1 writes 44; P_2 executes `x++`" can produce 2 instead of 45.

5. State the three standard coherence properties in words and explain why one of them must be phrased in terms of "sufficient time" rather than instant visibility.

6. For one cache block in an MSI protocol, explain what it means for that block to be in states `M`, `S`, and `I`. In particular, explain what memory does or does not know in each case.

7. Starting from both caches in state `S` for the same block, show the state changes when one cache performs a write hit. Which bus transaction occurs, and what must the other cache do?

8. What does the transient state `IM` mean? Give a short scenario in which a cache would enter `IM` rather than moving immediately from `I` to `M`.

9. Why does a single shared-cache design fail to scale linearly even for a highly parallel workload?

10. In the array-sum examples, explain why the shared-memory version needs a lock around the update of `globalSum`, while the message-passing version does not require the same kind of critical section.

Appendix A

Pipeline Timing-Diagram Interpretation Checklist

A five-stage timing diagram is best read as a checklist rather than as raw symbols. Move cycle by cycle (column by column), then apply the prompts below.

A.1 How to Use This Checklist

Quick Reading Order

1. Mark repeated stage labels (stall behavior).
2. Mark killed instructions (squashes, often shown as dashes).
3. Identify the event that caused each stall/squash (dependence, branch, resource conflict, or memory latency).
4. Compute measured CPI from the traced window.

1. Structural: Resource Organization

1.1 IF and MEM overlap

- **Ask:** Can instruction fetch occur in the same cycle as a load/store data access?
- **Cue:** One cycle shows IF for one instruction while another instruction is in MEM with a real data access.
- **Meaning:** Sustained overlap indicates concurrent instruction/data paths. Repeated IF stalls when load/store reaches MEM suggests a shared bottleneck.

1.2 Single-issue vs superscalar

- **Ask:** Is the machine *single-issue* or *superscalar* (*multi-issue*)?
- **Cue:** In simple single-issue diagrams, each stage has at most one instruction per cycle.
- **Meaning:** Two instructions shown in one stage in one cycle (for example two EX uses) implies multi-issue and/or multiple functional units.

1.3 Multi-cycle execute units

- **Ask:** Do some operations occupy EX for multiple cycles?
- **Cue:** One instruction appears in EX across consecutive cycles (EX, EX, EX, ...).
- **Meaning:** That operation is using a multi-cycle execute unit and can back up younger instructions.

2. Data: Dependence and Forwarding

2.1 RAW hazard location and cost

- **Ask:** Where are *RAW* hazards, and how many bubbles do they add?
- **Cue:** Consumer instruction repeats a stage (usually ID, sometimes IF) while waiting for producer timing.
- **Meaning:** Number of repeated cells is the local dependence penalty.

2.2 Forwarding strength

- **Ask:** Is forwarding present, and from where?
- **Cue:** Back-to-back ALU dependencies proceed without stalling even though producer WB occurs later.
- **Meaning:** This usually implies EX/MEM→EX and MEM/WB→EX forwarding. If simple ALU→ALU pairs stall, forwarding is limited.

2.3 Load-use behavior

- **Ask:** What is the load-use penalty?
- **Cue:** Load immediately followed by consumer, where consumer stalls one cycle.
- **Meaning:** Classic 5-stage with forwarding still needs a one-cycle load-use stall because load data appears at end of MEM.

2.4 WB/ID same-cycle timing

- **Ask:** Can ID read a register in the same cycle an older instruction writes it during the WB stage?
- **Cue:** Consumer ID aligns with producer WB, with no extra bubble.
- **Meaning:** Implies write-before-read register timing and/or WB→ID bypass behavior.

3. Control: Branch and Redirection Policy

3.1 Resolution stage and penalty

- **Ask:** Where is branch outcome resolved, and what is the visible penalty?
- **Cue:** Find where target/direction becomes known; then count wrong-path squashes or inserted bubbles before correct-path flow stabilizes.
- **Meaning:** Later resolution generally increases misfetch cost.

3.2 Delay slot vs squash slot

- **Ask:** Does the post-branch slot always execute, or can it be killed?
- **Cue:** Slot instruction either always completes (*delay slot*) or is conditionally killed (*squash slot*/branch-likely style).
- **Meaning:** Always-executed slot indicates delay-slot semantics; conditional kill indicates squash-style semantics.

3.3 Taken/not-taken inference (branch-likely style)

- **Ask:** Which branch outcome occurred?
- **Cue:** Whether the slot instruction executes or is squashed.
- **Meaning:** In this text's branch-likely examples, a squashed slot corresponds to not taken.

3.4 Prediction vs no prediction

- **Ask:** Does fetch speculate, and is speculation often correct?
- **Cue:** Continued fetching past unresolved branches followed by occasional squashes (speculation), versus front-end waiting for resolution (no speculation).
- **Meaning:** Wrong-path fetches later killed imply predictive/fixed speculation; repeated front-end pauses imply non-speculative handling.

4. Memory: Miss Effects in the Timeline

4.1 Miss penalty visibility

- **Ask:** Are long memory latencies visible?
- **Cue:** Load/store occupies MEM for multiple cycles (M, M, M, . . .), with younger instructions backing up.
- **Meaning:** Indicates cache-miss (or similar long-latency memory) behavior.

4.2 Blocking vs non-blocking cache behavior

- **Ask:** Is cache behavior *blocking*, or can it do *hit-under-miss* (*non-blocking*)?
- **Cue:** Either all younger work freezes behind a miss, or independent work continues.
- **Meaning:** Full freeze implies blocking behavior (common in simple in-order models). Continued independent progress suggests non-blocking support and/or more advanced scheduling.

5. Performance from the Diagram

5.1 Measured CPI of a trace window If a window spans C cycles and retires N_{ret} non-squashed instructions (Eq. (A.1)):

$$\mathrm{CPI}_{measured} = \frac{C}{N_{ret}}. \tag{A.1}$$

Squashed instructions consume cycles but do not count as retired instructions.

5.2 Stall-cycle attribution by cause Assign repeated-stage cells and squashes to causes:

- dependence waiting (RAW/load-use),
- control redirection (branch squashes/bubbles),
- structural conflicts (resource contention),
- memory-latency extension (repeated MEM).

This gives a practical CPI breakdown from one timing chart.

6. Common Gotchas

- ALU instructions may still appear in MEM in pedagogical 5-stage diagrams; this alone does not prove a data-memory access.
- IF stalls can be secondary backpressure from a downstream stall (often ID), not a fetch-unit limitation by itself.
- Squash and stall are different: squashed instructions are killed; stalled instructions are preserved in place with repeated stage labels.

Appendix B

Formula Compendium (Grouped and Labeled)

This appendix gathers the core formulas used across Chapters 1–8.

Chapter 1 Formula Set

Chapter 1-F1 (Vector Serial vs Pipelined Cycles)

$$\text{serial cycles} = nk, \qquad \text{pipeline cycles} \approx k + n - 1. \tag{B.1}$$

Chapter 2 Formula Set

Chapter 2-F1 (CPU Time Decomposition)

$$\text{CPU time} = \text{IC} \times \text{CPI} \times \text{CT}. \tag{B.2}$$

Chapter 2-F2 (Frequency Form)

$$\text{CPU time} = \frac{\text{IC} \times \text{CPI}}{f}. \tag{B.3}$$

Chapter 2-F3 (Instruction-Mix Total Cycles)

$$\text{Total cycles} = \sum_i \left(\text{IC}_i \times \text{CPI}_i\right). \tag{B.4}$$

Chapter 2-F4 (Average CPI)

$$\text{CPI}_{avg} = \frac{\text{Total cycles}}{\text{Total instructions}}. \tag{B.5}$$

Chapter 2-F5 (Speedup Definition)

$$S = \frac{T_{old}}{T_{new}}. \tag{B.6}$$

Chapter 2-F6 (Amdahl's Law)

$$S_{Amdahl} = \frac{1}{(1-f) + \frac{f}{s}}. \tag{B.7}$$

Chapter 2-F7 (Amdahl Upper Bound)

$$S_{\text{upper}} = \frac{1}{1-f}. \tag{B.8}$$

Chapter 2-F8 (Gustafson's Law)

$$S_{Gustafson} = \alpha + (1-\alpha)N. \tag{B.9}$$

Chapter 2-F9 (Arithmetic Mean for Time)

$$\bar{T}_{arith} = \frac{1}{n}\sum_{i=1}^{n} T_i. \tag{B.10}$$

Chapter 2-F10 (Harmonic Mean for Rate)

$$\bar{R}_{harm} = \frac{n}{\sum_{i=1}^{n} \frac{1}{R_i}}. \tag{B.11}$$

Chapter 3 Formula Set

Chapter 3-F1 ((C, B, S) Definitions)

$$\text{capacity} = 2^C \text{ bytes}, \qquad \text{block size} = 2^B \text{ bytes}, \qquad \text{associativity} = 2^S \text{ ways}. \tag{B.12}$$

Chapter 3-F2 (Total Blocks and Sets)

$$\text{blocks} = 2^{C-B}, \qquad \text{sets} = \frac{2^C}{2^B \cdot 2^S} = 2^{C-B-S}. \tag{B.13}$$

$$\text{lines} = \text{sets} \cdot \text{ways} = 2^{C-B-S} \cdot 2^S = 2^{C-B}. \tag{B.14}$$

Direct-mapped corollary: if $S = 0$, then ways $= 1$ and lines $=$ sets. Set-associative corollary: if $S > 0$, then lines $>$ sets.

Chapter 3-F3 (Address-Field Widths, General AW)

$$|\text{Offset}| = B, \qquad |\text{Index}| = C - B - S, \qquad |\text{Tag}| = AW - (C - S). \tag{B.15}$$

$$\text{Address} = [\text{Tag} \mid \text{Index} \mid \text{Offset}]. \tag{B.16}$$

Chapter 3-F4 (Address-Field Widths, 32-bit Address)

$$|\text{Offset}| = B, \qquad |\text{Index}| = C - B - S, \qquad |\text{Tag}| = 32 - (C - S) = 32 - C + S. \tag{B.17}$$

Chapter 3-F5 (Way Selection Relationship)

$$\text{ways} = 2^S, \qquad \log_2(\text{ways}) = S. \tag{B.18}$$

Chapter 3-F6 (Single-Level Average Access Time)

$$AAT = HT + MR \cdot MP. \tag{B.19}$$

Chapter 3-F7 (Bit Overhead from Valid/Dirty)

$$\text{overhead fraction} = \frac{n_v + n_d}{2^{B+3}}, \qquad n_v = 1, \quad n_d \in \{0, 1\}. \tag{B.20}$$

Chapter 3-F8 (Full-Block Miss Penalty Model)

$$MP_{full} = T_1 + (N_{sb} - 1)T_f, \qquad N_{sb} = 2^B/W. \tag{B.21}$$

Chapter 3-F9 (Early-Restart Approximation)

$$MP_{ER} \approx T_1 + \frac{N_{sb} - 1}{2} T_f. \tag{B.22}$$

Chapter 3-F10 (Recursive Two-Level Form)

$$AAT_{L1} = HT_{L1} + MR_{L1} \cdot AAT_{L2}. \tag{B.23}$$

Chapter 3-F11 (VIPT Constraint)

$$C - S \leq P. \tag{B.24}$$

Chapter 4 Formula Set

Chapter 4-F1 (Pipelined Clock Approximation)

$$T_{clk,pipe} \approx T_s + T_{reg}. \tag{B.25}$$

Chapter 4-F2 (Ideal Long-Run Pipeline Speedup)

$$S_{ideal} \approx \frac{NT_s}{T_s + T_{reg}}. \tag{B.26}$$

Chapter 4-F3 (Finite-Length Pipelined Time)

$$T_{pipe} \approx (N + M - 1)T_{clk,pipe}. \tag{B.27}$$

Chapter 4-F4 (Unpipelined Time for M Instructions)

$$T_{unpiped} \approx MNT_s. \tag{B.28}$$

Chapter 4-F5 (Practical CPI Model)

$$CPI = 1 + \text{stalls}_{avg}. \tag{B.29}$$

Chapter 4-F6 (Branch-Stall CPI Contribution)

$$CPI \approx 1 + f_{branch} \cdot \text{stall}_{branch,avg}. \tag{B.30}$$

Chapter 5 and Simulation Formula Set

Chapter 5-F1 (Two-Level AAT with Explicit L1 Penalty)

$$AAT_{L1} = HT_{L1} + MR_{L1} \cdot MP_{L1}, \qquad MP_{L1} = AAT_{L2}. \tag{B.31}$$

Chapter 5-F2 (L2 AAT Model)

$$AAT_{L2} = HT_{L2} + MR_{L2,read} \cdot DRAM_TIME. \tag{B.32}$$

Chapter 5-F3 (DRAM Block Service Model)

$$DRAM_TIME = DRAM_AT + DRAM_AT_PER_WORD \cdot WORDS_PER_BLOCK. \tag{B.33}$$

Chapter 6 Formula Set (Dynamic ILP, Superscalar, and Out-of-Order Execution)

Chapter 6-F1 (IPC and CPI Relationship)

$$IPC = \frac{1}{CPI}. \tag{B.34}$$

Chapter 6-F2 (CPI/IPC Threshold Equivalence)

$$CPI < 1 \iff IPC > 1. \tag{B.35}$$

Chapter 6-F3 (Retire-Throughput Bound)

$$IPC_{retire} \leq \min(IPC_{fetch},\ IPC_{dispatch},\ IPC_{execute},\ IPC_{state\ update}). \tag{B.36}$$

Chapter 6-F4 (Simplified Scoreboard Fire Condition)

$$\text{Fire if } \neg busy(src_1) \wedge \neg busy(src_2) \wedge \neg busy(dest) \wedge \neg busy(FU_{needed}). \tag{B.37}$$

Chapter 6-F5 (Result-Bus Completion Bound)

$$IPC_{complete} \leq N_{result\ buses}. \tag{B.38}$$

Chapter 6-F6 (Physical Register File Port Estimate)

$$N_{read\ ports} \approx 2N_{sched}, \qquad N_{write\ ports} \approx N_{result\ buses}. \tag{B.39}$$

Chapter 6-F7 (Precise Exception Property)

$$\begin{aligned}\text{precise exception at } i \Rightarrow\ & \text{all instructions older than } i \text{ are committed} \\ & \wedge\ \text{no instruction younger than } i \text{ is architecturally visible}\end{aligned} \tag{B.40}$$

Chapter 6-F8 (RAT Rename Mapping)

$$P_{src1} = RAT[A_{src1}], \qquad P_{src2} = RAT[A_{src2}], \qquad RAT[A_{dest}] \leftarrow P_{free}. \tag{B.41}$$

Chapter 6-F9 (Ready-Based Fire Condition)

$$fire_i \Leftarrow ready(P_{src1}) \wedge ready(P_{src2}) \wedge free(FU_i). \tag{B.42}$$

Chapter 6-F10 (ROB Retirement Condition)

$$retire \Leftarrow ready(ROB_{head}) \wedge \neg exception(ROB_{head}). \tag{B.43}$$

Chapter 6-F11 (ROB Exception Rule)

$$\begin{aligned}exception(ROB_{head}) \Rightarrow\ & \text{flush younger instructions} \\ & \wedge\ \text{transfer control to the retry or handler PC}\end{aligned} \tag{B.44}$$

Chapter 7 Formula Set (VLIW and Software Pipelining)

Chapter 7-F1 (Dependence Height)

$$height(i) = \begin{cases} lat(i), & succ(i) = \emptyset \\ lat(i) + \max\limits_{j \in succ(i)} height(j), & \text{otherwise} \end{cases} \tag{B.45}$$

Chapter 7-F2 (Earliest Legal List-Scheduling Cycle)

$$cycle(i) = \min\{c \mid \forall j \in pred(i),\ c \geq cycle(j) + lat(j),\ \text{and a compatible FU is free in } c\}. \tag{B.46}$$

Chapter 7-F3 (Resource-Constrained Initiation Interval)

$$ResII = \max_t \left\lceil \frac{N_t}{U_t} \right\rceil \tag{B.47}$$

Chapter 7-F4 (Recurrence-Constrained Initiation Interval)

$$RecII = \max_c \left\lceil \frac{Lat(c)}{Dist(c)} \right\rceil \tag{B.48}$$

Chapter 7-F5 (Minimum Initiation Interval Bound)

$$MII = \max(ResII, RecII), \qquad II \geq MII. \tag{B.49}$$

Chapter 8 Formula Set (Multiprocessors and Coherence)

Chapter 8-F1 (Ideal Embarrassingly Parallel Scaling)

$$S(N) = N \qquad \text{when } f = 1 \text{ and no shared resource becomes the bottleneck.} \tag{B.50}$$

Appendix C

Glossary of Core Definitions

This appendix collects first-pass definitions used throughout Chapters 1–8.

Parallelism Doing multiple useful operations concurrently to increase throughput.

Speculation Executing work before all required conditions are confirmed, with recovery if assumptions are wrong.

Prediction Estimating uncertain future behavior to guide speculative execution.

Instruction-Level Parallelism (ILP) Overlap of independent instructions within one thread.

Very Long Instruction Word (VLIW) Instruction encoding style that packs several compiler-scheduled ops into one instruction word.

Polycyclic Scheduling Early name for overlapping multiple loop iterations in one compact repeating schedule.

Software Pipelining Loop-scheduling technique that overlaps operations from different iterations and emits a prologue, steady-state kernel, and epilogue.

Thread-Level Parallelism (TLP) Concurrent execution of multiple threads on one or more cores.

Processing Element (PE) One independently executing processor, core, or execution agent within a multiprocessor system.

Digital Signal Processor (DSP) Processor class optimized for regular numeric workloads such as filtering, radio, and image-processing kernels.

SIMD One instruction stream operating on multiple data elements in lockstep.

MIMD Multiple instruction streams operating on multiple data streams independently.

Shared Memory Multiprocessor organization in which all participating processors conceptually operate in one common address space.

Message Passing Multiprocessor organization in which processors communicate by explicitly sending data between private memories.

Partitioned Global Address Space (PGAS) Memory model with a global address space in which some regions are much more expensive to access than others.

Accelerator Specialized processing engine, often attached to a host CPU through an external link and using distinct memory.

Bus Shared observable broadcast medium across which all attached agents can see the same transactions.

Point-to-Point Interconnect Link organization in which packets move between specific endpoints rather than across one shared observable medium.

Cache Coherence Property that keeps multiple cached copies of the same location mutually consistent enough to preserve shared-memory semantics.

Snooping Coherence Coherence method in which each cache monitors shared interconnect traffic and reacts when another processor requests a block it holds.

Directory-Based Coherence Coherence method in which a directory tracks which caches currently hold a block and coordinates invalidations or forwarding.

MSI Coherence protocol whose stable states are modified, shared, and invalid.

GetS Coherence request for a shared readable copy of a block.

GetM Coherence request for exclusive writable ownership of a block.

WriteBack Propagation of a dirty cache block to lower memory.

Intervene Snooper action in which a cache supplies data directly in response to another processor's request.

Abort Snooper response instructing a requester to wait or retry because ownership transfer is not yet complete.

Transient State Intermediate coherence state used while a block is waiting for a request or ownership transfer to complete.

Critical Section Region of code that must not be executed concurrently by multiple threads if shared data is to remain correct.

Lock Synchronization primitive that grants exclusive access to a critical section.

Data Race Incorrect unsynchronized overlap of conflicting accesses to shared state.

Embarrassingly Parallel Workload whose useful work is effectively all parallelizable and therefore scales nearly ideally if resources permit.

Pipelining Temporal parallelism where different stages process different instructions simultaneously.

Superscalar (Multi-Issue) Pipeline organization that can issue more than one instruction per cycle.

Register Renaming Introduction of additional physical register identities so later writes do not falsely block earlier reads or writes.

Op One encoded operation within a VLIW instruction stream.

Multi-Op Group of ops intended to execute in the same cycle in a VLIW design.

Tail Bit Per-op field marking whether the current op ends the current multi-op.

Pause Field Encoded delay field that inserts a specified number of idle cycles before the next multi-op begins.

Iterative Modulo Scheduling (IMS) Practical software-pipelining algorithm that searches for a legal modulo schedule by trying increasing initiation intervals.

Initiation Interval (II) Number of cycles between the start of successive loop iterations in a software-pipelined schedule.

Minimum Initiation Interval (MII) Lower bound on II, equal to $\max(ResII, RecII)$.

Resource-Constrained Initiation Interval (ResII) Lower bound on II imposed by the number of available functional units of each type.

Recurrence-Constrained Initiation Interval (RecII) Lower bound on II imposed by loop-carried dependence cycles.

Prologue Non-repeating prefix of a software-pipelined loop that fills the pipeline before steady state.

Kernel Repeating steady-state portion of a software-pipelined loop.

Epilogue Non-repeating suffix of a software-pipelined loop that drains remaining work after loop exit.

Point Latency Model Scheduling model that treats an op as occupying one issue point and becoming ready after a fixed latency.

Tomasulo's Algorithm Out-of-order scheduling method that combines renaming, reservation stations, and result broadcast to wake dependent instructions when operands become ready.

Reservation Station Scheduling entry that holds an instruction, its operand state, and dependency tags until it can execute.

Common Data Bus (CDB) Shared result-broadcast path used to distribute a completed value and its producer tag to waiting instructions.

Result Bus Generalized form of the CDB; modern designs often use more than one such broadcast path.

Architected Register Register name defined by the ISA and visible to software.

Architected Register File Register file containing the committed architectural state visible to software.

Physical Register Internal hardware register used to hold renamed values beyond the architected register set.

Register Alias Table (RAT) Mapping table from architected register names to the current physical register locations.

Ready Bit Status bit indicating that a register or operand value has already been produced and may be consumed.

Reorder Buffer (ROB) In-order retirement structure that records speculative results and enables precise exceptions.

Future File Register organization that holds the most recent speculative values while a separate architected register file holds committed state.

Messy Register File Informal name for the current speculative register state that may contain out-of-order completions not yet reflected in a coherent sequential state.

Precise Exception Exception model in which all older instructions appear committed and all younger instructions appear not to have changed architectural state.

Imprecise Exception Exception model that reports an error without guaranteeing an exact sequential recovery point.

Checkpoint/Repair Recovery method that periodically captures coherent register state and restores it when an exception or mis-speculation occurs.

Instruction Boundary Dynamic marker used to denote a coherence point or recovery boundary in checkpoint-based schemes.

Hyperthreading (SMT) Simultaneous multithreading in which multiple threads share one core's execution resources.

Speculative Side Channel Information leak caused when squashed speculative work still changes microarchitectural state such as caches or predictors.

Runtime Time to complete one job (response/turnaround time).

Throughput Completed work per unit time.

Amdahl's Law Speedup limit for fixed-size workloads when only a fraction is accelerated.

Gustafson's Law Scaled-speedup model where workload size grows with processor count.

Average Access Time (AAT) Expected memory-access latency, typically modeled as hit time plus miss rate times miss penalty.

SRAM Fast, low-latency, higher-area memory used for on-chip caches.

DRAM Dense, lower-cost, higher-latency memory used for main memory capacity.

Hit Time (HT) Latency to serve an access from the current cache level.

Miss Ratio (MR) Fraction of accesses at a level that miss.

Miss Penalty (MP) Extra latency to service a miss from lower levels.

Compulsory Miss First-reference miss for data not previously loaded.

Capacity Miss Miss caused by working set exceeding effective cache capacity.

Conflict Miss Miss caused by mapping/placement constraints despite sufficient total capacity.

Write-Back Write policy where modified data is propagated on eviction.

Write-Through Write policy where lower level is updated on each write hit.

Write-Allocate Write-miss policy that fetches and installs the block before write.

Write-No-Allocate Write-miss policy that bypasses install and writes lower level directly.

PIPT Physically indexed, physically tagged cache organization in which TLB translation typically precedes L1 indexing/tag lookup.

VIPT Virtually indexed, physically tagged cache organization that overlaps indexing with translation.

TLB Translation lookaside buffer storing recent virtual-to-physical translations.

Structural Hazard Stall risk from insufficient hardware resources.

Data Hazard Stall risk from operand read/write ordering conflicts.

Control Hazard Stall risk from unresolved branch/jump control flow.

RAW Read-after-write dependence hazard.

WAR Write-after-read dependence hazard.

WAW Write-after-write dependence hazard.

Data Dependence Graph Graph representation of instruction ordering constraints induced by true and false dependences.

Dependence Height Priority heuristic measuring the remaining latency on the longest dependent path below an instruction.

List Scheduling Heuristic compiler scheduling method that repeatedly places the highest-priority ready instructions into the earliest legal cycle.

Memory Disambiguation Compile-time or runtime reasoning used to determine whether two memory references may access the same location.

Forwarding (Bypassing) Sending a produced value directly to a consumer before register write-back.

Branch Target Buffer (BTB) Predictor structure that supplies likely branch targets early in fetch.

Aliasing Predictor or cache entry interference from unrelated addresses mapping to the same slot.

Return-Address Stack (RAS) Small stack predictor for call/return target addresses.

Direction Prediction Branch predictor decision for taken versus not taken.

One-Bit Predictor Dynamic branch predictor that records only the last observed branch outcome.

Smith Predictor Saturating-counter branch predictor, commonly implemented with a two-bit counter per entry.

GShare Global-history predictor that uses an exclusive-or of low PC bits and global history to index the prediction table.

GSelect Global-history predictor that forms its index by concatenating selected PC bits and global-history bits.

Yeh/Patt Predictor Two-level branch predictor that uses recorded branch history to index a table of direction counters.

Perceptron Predictor Branch predictor that computes a weighted sum over branch-history bits and predicts from the sign of that sum.

Hybrid Predictor Predictor that chooses among multiple underlying predictors, often through a tournament-style selector.

BTFNT Backward taken, forward not taken static branch heuristic.

Delay Slot ISA semantic in which the instruction after a branch executes before control transfer takes effect.

Squash Slot Post-branch instruction position that may be conditionally killed based on branch outcome/policy.

WBWA Write-back, write-allocate pairing.

WTWNA Write-through, write-no-allocate pairing.

Blocking Cache Cache behavior where an outstanding miss prevents servicing younger accesses.

Non-Blocking Cache Cache behavior allowing selected younger accesses to proceed during a miss.

Hit-Under-Miss Ability to serve cache hits while a miss is still outstanding.

Acronym List

AAT Average Access Time

ALU Arithmetic Logic Unit

AVX Advanced Vector Extensions

AW Address Width

BTFNT Backward Taken, Forward Not Taken

BTB Branch Target Buffer

CDB Common Data Bus

CISC Complex Instruction Set Computer

CPI Cycles Per Instruction

CPU Central Processing Unit

CT Cycle Time

DDG Data Dependence Graph

DSP Digital Signal Processor

DISP Dispatch and Decode

DRAM Dynamic Random-Access Memory

EX Execute

FICI Fire In Order, Complete In Order

FIFO First In, First Out

FICO Fire In Order, Complete Out of Order

FOCO Fire Out of Order, Complete Out of Order

FP Floating Point

FU Functional Unit

GPU Graphics Processing Unit

HT Hit Time

IC Instruction Count

ID Instruction Decode

IF Instruction Fetch

II Initiation Interval

ILP Instruction-Level Parallelism

IMS Iterative Modulo Scheduling

IPC Instructions Per Cycle

ISA Instruction Set Architecture

LFU Least Frequently Used

LIP Least Recently Used Insertion Policy

LRU Least Recently Used

MEM Memory

MII Minimum Initiation Interval

MIMD Multiple Instruction, Multiple Data

MIP Most Recently Used Insertion Policy

MISD Multiple Instruction, Single Data

MP Miss Penalty

MR Miss Ratio

MRU Most Recently Used

MSI Modified, Shared, Invalid

OS Operating System

PC Program Counter

PCIe Peripheral Component Interconnect Express

PE Processing Element

PFN Physical Frame Number

PGAS Partitioned Global Address Space

PIPT Physically Indexed, Physically Tagged

RAS Return-Address Stack

RAT Register Alias Table

RAW Read After Write

RecII Recurrence-Constrained Initiation Interval

ResII Resource-Constrained Initiation Interval

RISC Reduced Instruction Set Computer

ROB Reorder Buffer

RS Reservation Station

SIMD Single Instruction, Multiple Data

SIMT Single Instruction, Multiple Threads

SISD Single Instruction, Single Data

SMT Simultaneous Multithreading

SRAM Static Random-Access Memory

SSE Streaming SIMD Extensions

SSD Solid-State Drive

SU State Update

TLB Translation Lookaside Buffer

TLP Thread-Level Parallelism

VIPT Virtually Indexed, Physically Tagged

VLIW Very Long Instruction Word

VPN Virtual Page Number

WAR Write After Read

WAW Write After Write

WB Write Back

WBWA Write-Back, Write-Allocate

WTWNA Write-Through, Write-No-Allocate